Quantum Monte Carlo

Lecture notes

Guglielmo Mazzola

SISSA
Trieste
Italy

Abstract

These notes cover Monte Carlo and several flavors of quantum Monte Carlo. Unlike many other sources, we treat lattice and continuous-space models together to highlight the foundations of the idea, which are model-independent. We focus on variational and projective QMC methods.

There are exercises that mostly require a numerical solution. Exercises are marked with a (*), (**) or more, to signal their difficulty level.

Contents

1 Many-Body Quantum Models **1**

1.1 A bit of complexity theory . . . 1
1.1.1 Input size, runtime, and verification . . . 1
1.1.2 Hardness, completeness, and special cases . . . 2
1.1.3 Approximation and the requested accuracy . . . 3
1.1.4 Quantum computation and ground-state energies . . . 3
1.2 Two Many-Body Hamiltonians . . . 4
1.2.1 Quantum spins: the transverse-field Ising model . . . 5
1.2.2 Fermions: the Hubbard model . . . 5
1.3 Exact Diagonalization and Its Scaling . . . 6
1.3.1 Constructing the TFIM matrix . . . 7
1.4 The Variational Principle and Compact Ansatze . . . 8
1.4.1 A spin correlator . . . 8
1.4.2 One correlator in place of a determinant expansion . . . 9
1.5 Ground-State Projection . . . 10
1.6 The Plan for the Course . . . 11

2 Classical Monte Carlo Sampling **13**

2.1 Probability and Central Limit Theorem . . . 13
2.1.1 Mean and variance . . . 14
2.1.2 Chebyshev's inequality . . . 15
2.1.3 Law of Large Numbers . . . 15
2.1.4 Central Limit Theorem . . . 17
2.2 Markov Chains . . . 18
2.2.1 Example: Convergence to the Stationary Distribution . . . 21
2.2.2 Detailed Balance . . . 22
2.2.3 Spectral Convergence and Autocorrelation Time . . . 22
2.3 The Metropolis Algorithm . . . 26
2.4 Error Estimation and Binning . . . 28
2.5 Importance Sampling . . . 29
2.5.1 General importance sampling and its variance . . . 31
2.5.2 Expectation values and unknown normalization: reweighting. . . . 33
2.6 Cluster Updates for Spin Systems . . . 34
2.7 Metropolis Sampling in Continuous Space . . . 38

2.7.1 Simple Metropolis Proposals . . . 38
2.7.2 Langevin Dynamics . . . 39
2.7.3 From the Langevin Dynamics to Fokker-Planck equation . . . 41
2.7.4 From the Fokker-Planck to the Schrödinger Equations . . . 42
2.8 Exponential Divergence of Autocorrelation Time in Disordered Systems . . . 45

3 Variational Monte Carlo 48
3.1 Local Energy and Zero Variance Property . . . 48
3.1.1 Lattice Models . . . 50
3.1.2 Continuous Systems . . . 54
3.1.3 Electronic systems: Jastrow and Cusp Conditions . . . 55
3.2 Optimization of the Wavefunction . . . 57
3.2.1 Correlated Sampling . . . 58
3.2.2 Gradients . . . 58
3.2.3 Natural Gradients and Stochastic Reconfiguration . . . 59
3.3 Restricted Boltzmann Machines . . . 65
3.4 Quantum State Tomography . . . 68
3.4.1 Simple case: reconstructing a positive-valued state . . . 68
3.4.2 KL divergence and maximum likelihood . . . 68
3.4.3 An RBM probability model and its gradients . . . 69
3.4.4 Block Gibbs sampling and contrastive divergence . . . 70
3.4.5 Reconstructing observables . . . 71

4 Diffusion and Green's Function Monte Carlo 73
4.1 Imaginary-time projection . . . 73
4.2 Trotter Formulas . . . 75
4.3 Implementing the Projection Stochastically: Simple Algorithm . . . 76
4.3.1 Single Walker DMC . . . 76
4.3.2 Multi-Walker DMC . . . 77
4.4 Sign-Problem in Fermionic DMC and Fixed Node Approximation . . . 78
4.4.1 1D Example . . . 78
4.4.2 Fixing the Nodal Structure . . . 80
4.5 Importance Sampling DMC . . . 80
4.5.1 Computing the energy . . . 82
4.5.2 Single-Walker Implementation . . . 82
4.6 Green's Function Monte Carlo . . . 84

5 Path Integral Monte Carlo 86
5.1 Finite Temperature Formalism . . . 86
5.2 Path Integrals in Continuous Variable Models . . . 87
5.2.1 Ring-polymer isomorphism . . . 88
5.2.2 A Simple Energy Estimator . . . 91
5.2.3 Monte Carlo Sampling and Path Updates . . . 91
5.2.4 Path Integral Ground State . . . 93
5.2.5 Conceptual Difference Between DMC and PIMC . . . 94

5.3 Implementing Symmetries in Real-Space PIMC: Bosons and Permutations . . 94
5.3.1 The Sign Problem in Continuous Space PIMC 96
5.4 Path Integrals in Spin Models 98
5.4.1 A Single Quantum Spin 98
5.4.2 Evaluation of the off-diagonal operator 100
5.4.3 Multidimensional Generalization 103
5.5 The Loop Algorithm 104
5.5.1 The XXZ Model 104
5.5.2 Graph representation and the loop break-ups 107
5.5.3 The Kandel–Domany Graph Framework 110
5.5.4 Complexity of the Sign Problem in Lattice Models 111
5.6 Real-Time Dynamics vs PIMC Pseudodynamics 114
5.6.1 Coherent and incoherent tunneling 114
5.6.2 Instantons and tunneling in path space 114

A Representations, Discretization, and Error Estimates 118
A.1 Coordinates, Spin, and Exchange Symmetry 118
A.2 From First to Second Quantization 119
A.2.1 Occupation states and operator ordering 119
A.2.2 Field operators and Hamiltonian matrix elements 120
A.3 A Minimal Molecular Model and Its Approximations 121
A.4 Discretizing a Continuous Coordinate 122
A.4.1 The harmonic oscillator as a controlled test 122
A.4.2 The sinc kinetic-energy matrix 123
A.4.3 Optional application: a Morse model for deuterium vibration 123
A.5 Energy, Fidelity, and Variance 125

B Details of the Fokker–Planck calculations 126

C Cusp Conditions 130

D Details of Quantum State Tomography 134
D.1 From the KL divergence to a data likelihood 134
D.2 Marginalizing a binary RBM 135
D.3 Differentiating the partition function and the loss 135
D.4 The conditional distributions and block Gibbs chain 136
D.5 What contrastive divergence approximates 137
D.6 Fisher geometry for the positive-state model 138
D.7 Likelihood gradients for a general complex wavefunction 138

E Derivation of Importance Sampling DMC 140

F Supplementary Calculations for PIMC 142
F.1 Derivation of the Centroid-Virial Estimator 142
F.2 Two-State Relaxation with Dephasing 143
F.3 An Explicit Quartic Instanton 144

Chapter 1

Many-Body Quantum Models

1.1 A bit of complexity theory

Before discussing quantum many-body systems, it is useful to ask what makes a computational problem difficult. A short equation can define a calculation whose cost grows rapidly with system size. This happens already in classical physics. Quantum mechanics introduces further challenges, but computational difficulty is not exclusively quantum.

We will mostly study ground states. The relevant questions are therefore: how does the cost of finding a low-energy state grow with the system size, and what accuracy can we obtain with the available resources? Complexity theory gives a language for discussing these questions. The overview below is informal; the distinctions between solving, verifying, and approximating will be the most useful ones.

1.1.1 Input size, runtime, and verification

Let n be the number of bits needed to specify an instance of a problem. For a lattice Hamiltonian, this includes the lattice, the couplings, and their numerical precision. We often discuss scaling with the number of sites L, assuming that the input description has size polynomial in L. The distinction matters: a compact list of interactions and an explicitly stored many-body matrix are very different inputs.

An algorithm has polynomial runtime if its worst-case cost is bounded by $O(n^k)$ for some fixed k. This is the conventional meaning of *efficient*. The exponent and prefactor still matter in practice. Exponential costs, such as $O(2^L)$, are qualitatively different: adding a few degrees of freedom can multiply the required resources.

The familiar classes P and NP concern **decision problems**, whose answer is yes or no:

- A problem is in P if a deterministic classical algorithm answers it in polynomial time for every input.
- A problem is in NP if every yes-instance has a certificate of polynomial length that a deterministic classical algorithm can verify in polynomial time. For a no-instance, no certificate may pass the verification.

Thus $\mathsf{P} \subseteq \mathsf{NP}$. Whether the two classes are equal is unknown. In particular, membership in NP does not mean that no efficient algorithm is known: all problems in P also belong to NP.

The name NP means *nondeterministic polynomial time*. One can picture a computation that guesses a certificate and accepts if some guess passes the check. This is a mathematical definition, not an available machine that searches exponentially many possibilities for free. It is also not the computational model of a quantum computer.

A classical ground-state problem. Consider Ising variables $s_i = \pm 1$ on a graph with edge set $\mathcal{E}$:

$$E(s) = -\sum_{(i,j)\in\mathcal{E}} J_{ij} s_i s_j. \tag{1.1}$$

Each edge is counted once. The optimization problem asks for $\min_s E(s)$. Its decision version asks

Does there exist a configuration s such that $E(s) \leq E_c$?

A configuration is a certificate: it contains L signs, and its energy can be evaluated by summing over the edges. With polynomially specified rational couplings, this verification takes polynomial time. Merely reporting a number below E_c is not a certificate; one must also supply a configuration that attains it.

1.1.2 Hardness, completeness, and special cases

To compare two problems, we use a *reduction*: a polynomial-time transformation of an instance of problem A into an instance of problem B that preserves the yes/no answer. An efficient solver for B would then give an efficient solver for A.

- A problem is NP-hard if every problem in NP can be reduced to it in this way.
- A decision problem is NP-complete if it is both in NP and NP-hard.

An NP-hard problem need not belong to NP. Optimization problems, for example, have a different output type; they can be called NP-hard through suitable reductions, but are not themselves members of the decision class NP. Verifying the energy of a candidate and certifying its global optimality are different tasks.

For general graphs and couplings, the Ising threshold problem above is NP-complete. Its connection with a familiar combinatorial problem is immediate: set $J_{ij} = -1$ on every edge and let $C(s)$ count edges joining opposite spins. Then

$$E(s) = |\mathcal{E}| - 2C(s). \tag{1.2}$$

Minimizing the energy is equivalent to maximizing the size of a cut of the graph, the Max-Cut problem. By contrast, when all $J_{ij} \geq 0$, an aligned configuration minimizes every bond simultaneously. A difficult problem family can contain easy, physically relevant subclasses.

An efficient deterministic algorithm for any NP-complete problem would imply P = NP. However, NP-hardness is not a proof of an exponential runtime lower bound. Even assuming P ≠ NP excludes polynomial algorithms, not every possible subexponential algorithm. These are worst-case statements, rather than predictions of the runtime on every instance.[1]

[1] For a systematic account, see S. Arora and B. Barak, *Computational Complexity: A Modern Approach*, Cambridge University Press (2009).

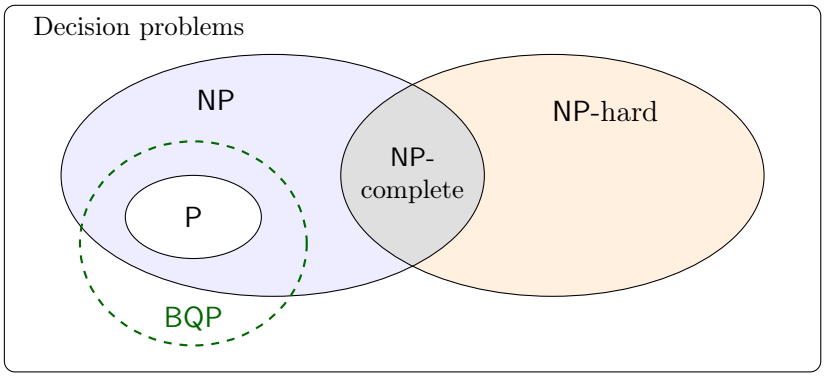


Figure 1.1: Schematic relations among decision problems, drawn under the assumption $\mathsf{P} \neq \mathsf{NP}$. The shaded intersection is $\mathsf{NP} \cap \mathsf{NP}$-hard. The dashed BQP contour shows a conjectured placement: $\mathsf{P} \subseteq \mathsf{BQP}$ is known, but neither $\mathsf{BQP} \subseteq \mathsf{NP}$ nor $\mathsf{NP} \subseteq \mathsf{BQP}$ is known. Excluding NP-complete problems from BQP is also conjectural. Factoring is in NP and in BQP.

1.1.3 Approximation and the requested accuracy

Often the exact optimum is unnecessary. An *approximation algorithm* provides a quantified guarantee on the quality of its answer; a useful heuristic may work well without such a guarantee.

For a minimization problem with positive objective value, a constant-factor guarantee has the form

$$f_{\mathrm{opt}} \leq f_{\mathrm{alg}} \leq \rho f_{\mathrm{opt}}, \qquad \rho \geq 1, \tag{1.3}$$

where ρ is independent of the input size. For maximization, the corresponding guarantee is $f_{\mathrm{alg}} \geq f_{\mathrm{opt}}/\rho$. The class APX consists of NP optimization problems admitting polynomial-time constant-factor approximation algorithms. Here an NP optimization problem has polynomial-size candidate solutions, efficiently checkable feasibility, and an efficiently evaluable objective.

For energies, a multiplicative ratio can be misleading: the zero of energy is arbitrary, and adding a constant changes the ratio without changing the physics. An additive error is often more natural,

$$|E_{\mathrm{alg}} - E_0| \leq \epsilon, \qquad \text{or} \qquad \left| \frac{E_{\mathrm{alg}} - E_0}{L} \right| \leq \epsilon_e. \tag{1.4}$$

Fixed accuracy in total energy and fixed accuracy per site are different requirements. The computational cost must be discussed together with the requested accuracy. Approximation can help, but some problems remain hard even at specified nonzero tolerances.

1.1.4 Quantum computation and ground-state energies

Two further distinctions are useful. BPP describes decision problems solvable in polynomial time by a randomized classical algorithm with bounded error; BQP is the analogous class for a quantum computer. The error probability can be reduced by repetition. Quantum interference can give computational advantages, but no polynomial-time quantum algorithm is known for all NP-complete problems.

The class QMA, or Quantum Merlin–Arthur, concerns verification: a yes-instance has a polynomial-size quantum witness that a quantum verifier accepts with high probability, while all witnesses are rejected with high probability on no-instances. In this discussion the quantum classes also allow *promise problems*, for which inputs are guaranteed to lie in specified yes or no sets. The useful containments are

$$\mathsf{P} \subseteq \mathsf{BPP} \subseteq \mathsf{BQP} \subseteq \mathsf{QMA}, \qquad \mathsf{P} \subseteq \mathsf{NP} \subseteq \mathsf{QMA}. \tag{1.5}$$

These inclusions are not asserted to be strict; the general relation between NP and BQP is unknown.

For a Hamiltonian $H = \sum_\ell h_\ell$ on L qubits, suppose there are polynomially many terms, each acting on at most a fixed number k of qubits, with bounded norms and polynomially specified coefficients. The *local Hamiltonian problem* asks whether

$$E_0 \le a \qquad \text{or} \qquad E_0 \ge b, \qquad b - a \ge \frac{1}{\mathrm{poly}(L)}, \tag{1.6}$$

under the promise that one of the two alternatives holds. In the standard bounded-term formulation this problem is QMA-complete in general, already for $k = 2$. The word "local" here means few-body, and need not mean nearest-neighbor in space.[2]

An efficient quantum algorithm for every such instance would imply $\mathsf{BQP} = \mathsf{QMA}$, which is generally considered unlikely but has not been ruled out. This result concerns a family of problems at a specified accuracy. It does not say that every physical Hamiltonian is hard, or that useful approximate calculations are impossible.

For this course, the practical lesson is to identify *which structure an algorithm uses*, *which resources it requires*, and *which errors it controls*. A compact Hamiltonian is only the starting point.

1.2 Two Many-Body Hamiltonians

We now introduce two examples that will recur throughout the chapter: the transverse-field Ising model and the Hubbard model. Both contain competing terms whose separate limits are easy to understand. Their competition produces states that generally require a calculation.

We focus on the ground-state problem

$$H\,|\psi_0\rangle = E_0\,|\psi_0\rangle\,, \qquad E_0 = \min_{\psi \neq 0} \frac{\langle\psi|\,H\,|\psi\rangle}{\langle\psi|\psi\rangle}. \tag{1.7}$$

In addition to E_0, we want physical observables such as correlations and double occupancy. Analytical limits give intuition and benchmarks; numerical methods let us examine the intermediate regimes and more general geometries.

[2] J. Kempe, A. Kitaev, and O. Regev, *The complexity of the local Hamiltonian problem*, SIAM J. Comput. **35**, 1070–1097 (2006), arXiv:quant-ph/0406180.

1.2.1 Quantum spins: the transverse-field Ising model

A spin-1/2 degree of freedom has two basis states, $|\uparrow\rangle$ and $|\downarrow\rangle$. We use Pauli operators, with $\sigma^z|\uparrow\rangle = |\uparrow\rangle$, $\sigma^z|\downarrow\rangle = -|\downarrow\rangle$, and σ^x exchanging the two states. Physical spin operators are $S^\alpha = \hbar\sigma^\alpha/2$.

On a graph, the transverse-field Ising model (TFIM) is

$$H = -\sum_{(i,j)\in\mathcal{E}} J_{ij}\sigma_i^z\sigma_j^z - \Gamma\sum_{i=1}^{L}\sigma_i^x. \tag{1.8}$$

Each bond is counted once. We mainly consider ferromagnetic nearest-neighbor couplings $J > 0$ and $\Gamma \geq 0$.

At $\Gamma = 0$, an aligned configuration minimizes every ferromagnetic bond. At $J = 0$ and $\Gamma > 0$, the ground state is $|+x\rangle^{\otimes L}$, where $|+x\rangle = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$. The two terms favor alignment along different axes and do not commute. At intermediate coupling, the ground state contains correlated spin fluctuations.

The uniform one-dimensional nearest-neighbor chain is analytically solvable and has a zero-temperature transition at $\Gamma/J = 1$ in the thermodynamic limit.[3] We use this special model as a benchmark; the same numerical construction applies when extra couplings or a different geometry remove that solvability. A finite chain has a rounded crossover rather than a thermodynamic singularity.

Configurations and matrix elements. In the σ^z basis,

$$|s\rangle = |s_1\rangle \otimes \cdots \otimes |s_L\rangle, \qquad s_i = \pm 1, \qquad |\psi\rangle = \sum_s \psi(s)\,|s\rangle. \tag{1.9}$$

The Ising interaction is diagonal, while each transverse-field term connects two configurations differing by one spin flip. For two sites joined by one bond, use the ordered basis

$$\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}.$$

The Hamiltonian is

$$H = \begin{pmatrix} -J & -\Gamma & -\Gamma & 0 \\ -\Gamma & J & 0 & -\Gamma \\ -\Gamma & 0 & J & -\Gamma \\ 0 & -\Gamma & -\Gamma & -J \end{pmatrix}. \tag{1.10}$$

This small matrix already displays the basic numerical operation: enumerate configurations and apply the terms of H.

1.2.2 Fermions: the Hubbard model

For electrons in a localized orbital basis, a configuration specifies which spin-orbitals are occupied. The operator $c_{i\sigma}^\dagger$ creates an electron with spin σ on site i, and $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$. Fermionic operators obey

$$\{c_{i\sigma}, c_{j\sigma'}^\dagger\} = \delta_{ij}\delta_{\sigma\sigma'}, \qquad \{c_{i\sigma}, c_{j\sigma'}\} = 0. \tag{1.11}$$

[3] P. Pfeuty, *The one-dimensional Ising model with a transverse field*, Ann. Phys. **57**, 79–90 (1970).

These relations encode exchange antisymmetry. The connection with coordinate-space wavefunctions and the required ordering convention are developed in Appendix A.

The Hubbard Hamiltonian is

$$H = -t \sum_{\langle i,j\rangle,\sigma} \left(c^\dagger_{i\sigma} c_{j\sigma} + c^\dagger_{j\sigma} c_{i\sigma} \right) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \tag{1.12}$$

with $t > 0$, $U \geq 0$, and each nearest-neighbor bond included once. Hopping favors delocalization; repulsion penalizes two electrons occupying the same site. A site has four states: empty, spin up, spin down, or doubly occupied. At half filling the electron number equals the number of sites.

The two-site problem. For two electrons with $S^z_{\text{tot}} = 0$, use the orthonormal basis

$$|1\rangle = c^\dagger_{1\uparrow} c^\dagger_{1\downarrow} |\varnothing\rangle , \qquad |2\rangle = c^\dagger_{1\uparrow} c^\dagger_{2\downarrow} |\varnothing\rangle , \tag{1.13}$$

$$|3\rangle = c^\dagger_{2\uparrow} c^\dagger_{1\downarrow} |\varnothing\rangle , \qquad |4\rangle = c^\dagger_{2\uparrow} c^\dagger_{2\downarrow} |\varnothing\rangle . \tag{1.14}$$

The displayed operator order defines the phases of these states. The vacuum $|\varnothing\rangle$ is a normalized zero-particle state, not the zero vector. In this basis,

$$H = \begin{pmatrix} U & -t & -t & 0 \\ -t & 0 & 0 & -t \\ -t & 0 & 0 & -t \\ 0 & -t & -t & U \end{pmatrix}. \tag{1.15}$$

This model captures the competition between covalent and ionic configurations in a minimal description of a hydrogen molecule. It is a simplified model, not the full projection of the molecular Coulomb Hamiltonian; the additional approximations are explained in Appendix A.3.

Define the normalized covalent and symmetric ionic states

$$|C\rangle = \frac{|2\rangle + |3\rangle}{\sqrt{2}}, \qquad |I\rangle = \frac{|1\rangle + |4\rangle}{\sqrt{2}}. \tag{1.16}$$

Both are spin singlets with the operator convention above. In their subspace,

$$H_{CI} = \begin{pmatrix} 0 & -2t \\ -2t & U \end{pmatrix}, \qquad E_0 = \frac{U - \sqrt{U^2 + 16t^2}}{2}. \tag{1.17}$$

The two remaining states in the $S^z_{\text{tot}} = 0$ sector have energies 0 and U. Thus the lower eigenvalue above is the ground-state energy for $t > 0$, $U \geq 0$. This exact result will let us judge a correlated ansatz without relying on intuition alone.

1.3 Exact Diagonalization and Its Scaling

For a chosen finite basis, exact diagonalization determines eigenvalues and eigenvectors to numerical precision. It provides a benchmark for small systems. The word "exact" refers to the specified finite Hamiltonian; it does not remove errors from a truncated orbital basis, a finite simulation cell, or an approximate physical model.

For L spins, the dimension is $D = 2^L$. For L Hubbard sites the full Fock-space dimension is 4^L, while fixed particle numbers give

$$D_{N_\uparrow, N_\downarrow} = \binom{L}{N_\uparrow}\binom{L}{N_\downarrow}. \tag{1.18}$$

Symmetries can reduce the calculation substantially, although these sectors still grow exponentially at fixed nonzero filling fractions away from full occupation.

Object or operation	Scaling for $D = 2^L$
Store a state vector	$O(D) = O(2^L)$
Store a dense Hamiltonian	$O(D^2) = O(4^L)$
Dense full diagonalization	$O(D^3) = O(8^L)$
Apply a local TFIM Hamiltonian	$O(LD) = O(L2^L)$

The last line assumes $O(L)$ bonds and matrix-free application to a dense vector. It is much cheaper than dense matrix multiplication, but still requires a vector of length D. Fifty spins already require 16×2^{50} bytes, or 16 PiB (approximately 18 PB), for a full complex-double state vector.

This counting describes the cost of an explicit representation. It is not a proof that every state or observable requires exponential work. A product state has a compact description despite having 2^L components in a generic basis. The useful question is whether the states and observables of interest admit a representation and a calculation that exploit their structure.

1.3.1 Constructing the TFIM matrix

For a periodic chain with $L \geq 3$,

$$H = -J\sum_{i=1}^{L}\sigma_i^z\sigma_{i+1}^z - \Gamma\sum_{i=1}^{L}\sigma_i^x, \qquad \sigma_{L+1}^z = \sigma_1^z. \tag{1.19}$$

Encode a configuration by bits $b_i = (1 - s_i)/2$, and assign the integer $n = \sum_{i=1}^{L} b_i 2^{i-1}$. Then

$$H_{nn} = -J\sum_{i=1}^{L} s_i s_{i+1}, \qquad H_{n\,\mathrm{xor}\,2^{i-1},n} = -\Gamma. \tag{1.20}$$

Each column has at most $L+1$ nonzero entries. One can generate the connected configurations directly, without testing all pairs or storing a dense matrix.

Exercise 1.3.1: Diagonalization of TFIM

Use periodic boundary conditions and nearest-neighbor coupling $J = 1$.

1. Construct the Hamiltonian for $L = 6$ and check its Hermiticity. Verify the ground-state energies at $\Gamma = 0$ and $J = 0$ analytically.
2. Calculate E_0/L as a function of Γ/J, then repeat for several sizes. Compare the memory cost of dense storage and matrix-free application.
3. Define $m_z = L^{-1} \sum_i \sigma_i^z$. Study $\langle m_z^2 \rangle$ as the size increases. Explain why a finite parity eigenstate has $\langle m_z \rangle = 0$, even in the parameter regime that becomes ferromagnetic in the thermodynamic limit.

1.4 The Variational Principle and Compact Ansatze

An *ansatz* is a family of trial states $|\psi_\theta\rangle$ specified by parameters θ. Its usefulness depends on whether it captures the relevant physics with manageable resources. The variational principle gives an objective for choosing the parameters:

$$E(\theta) = \frac{\langle \psi_\theta | H | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle} \geq E_0. \tag{1.21}$$

Indeed, a normalized state expanded in energy eigenstates, $|\psi\rangle = \sum_n a_n |\phi_n\rangle$, has

$$\langle \psi | H | \psi \rangle = \sum_n |a_n|^2 E_n \geq E_0 \sum_n |a_n|^2 = E_0. \tag{1.22}$$

Equality holds precisely when the state lies in the ground-state eigenspace. Enlarging a variational family cannot increase its globally minimized energy; actually finding that minimum may still be difficult.

For a nondegenerate ground state with gap $\Delta = E_1 - E_0 > 0$,

$$1 - |\langle \phi_0 | \psi \rangle|^2 \leq \frac{\langle \psi | H | \psi \rangle - E_0}{\Delta}. \tag{1.23}$$

Thus an energy error small compared with the gap is sufficient to guarantee high ground-state fidelity. A small energy error per site alone does not provide that guarantee. Further error and variance identities are collected in Appendix A.5.

1.4.1 A spin correlator

The TFIM suggests a simple construction. The reference state $|+x\rangle^{\otimes L}$ assigns equal amplitudes to all σ^z configurations. A diagonal correlator can favor aligned neighbors:

$$|\psi_K\rangle \propto \exp\left(K \sum_{\langle i,j \rangle} \sigma_i^z \sigma_j^z \right) |+x\rangle^{\otimes L}, \qquad \psi_K(s) \propto \exp\left(K \sum_{\langle i,j \rangle} s_i s_j \right). \tag{1.24}$$

For $K = 0$ this is the exact $J = 0$ ground state. For $K > 0$, each domain wall reduces the amplitude relative to an aligned bond by a factor e^{-2K}. On a connected ferromagnetic graph, $K \to \infty$ selects the symmetric combination of the two fully aligned configurations, a ground state at $\Gamma = 0$.

This one-parameter Jastrow ansatz interpolates between two correct limits. It is not generally exact at intermediate coupling. Its quality can be checked against exact diagonalization. A more flexible pair correlator replaces K by W_{ij}:

$$\psi_W(s) \propto \exp\left(\sum_{i<j} W_{ij} s_i s_j\right). \tag{1.25}$$

Translation invariance and a distance cutoff can reduce the parameter count. Real parameters give positive amplitudes, a suitable starting point for the finite TFIM with $\Gamma > 0$, whose connected configuration graph and nonpositive off-diagonal matrix elements permit a strictly positive ground state. Other Hamiltonians may require a nontrivial sign or phase structure.

1.4.2 One correlator in place of a determinant expansion

Return to the Hubbard dimer. Define bonding and antibonding orbitals through

$$d^\dagger_{b\sigma} = \frac{c^\dagger_{1\sigma} + c^\dagger_{2\sigma}}{\sqrt{2}}, \qquad d^\dagger_{a\sigma} = \frac{c^\dagger_{1\sigma} - c^\dagger_{2\sigma}}{\sqrt{2}}. \tag{1.26}$$

Putting both electrons in the bonding orbital gives the single determinant

$$|B\rangle = d^\dagger_{b\uparrow} d^\dagger_{b\downarrow} |\varnothing\rangle = \frac{|1\rangle + |2\rangle + |3\rangle + |4\rangle}{2} = \frac{|I\rangle + |C\rangle}{\sqrt{2}}. \tag{1.27}$$

It is exact at $U = 0$. However, it assigns total probability $1/2$ to double occupancy, independently of U/t. The same determinant therefore becomes a poor description at large repulsion.

Let $\hat{D} = n_{1\uparrow} n_{1\downarrow} + n_{2\uparrow} n_{2\downarrow}$ count doubly occupied sites. Multiplying by a correlator gives

$$|\psi_g\rangle \propto e^{-g\hat{D}} |B\rangle, \qquad g \geq 0. \tag{1.28}$$

Writing $\lambda = e^{-g}$, the normalized state is

$$|\psi_\lambda\rangle = \frac{\lambda |1\rangle + |2\rangle + |3\rangle + \lambda |4\rangle}{\sqrt{2(1+\lambda^2)}} = \frac{|C\rangle + \lambda |I\rangle}{\sqrt{1+\lambda^2}}. \tag{1.29}$$

The parameter has a direct physical meaning: it reduces the amplitude of ionic configurations relative to covalent ones. In particular,

$$\langle \hat{D} \rangle_\lambda = \frac{\lambda^2}{1+\lambda^2}, \qquad E(\lambda) = \frac{U\lambda^2 - 4t\lambda}{1+\lambda^2}. \tag{1.30}$$

Minimization gives

$$\lambda_* = \frac{4t}{\sqrt{U^2 + 16t^2} + U}, \qquad E(\lambda_*) = E_0. \tag{1.31}$$

Thus **one correlator parameter reproduces the exact ground state of the symmetric repulsive Hubbard dimer**. At $U = 0$, $\lambda_* = 1$. At large U/t, $\lambda_* \simeq 2t/U$ and the state approaches the covalent singlet $|C\rangle$, the analogue of the Heitler–London state. The small ionic contribution at finite U remains physically important: it produces the energy lowering $E_0 \simeq -4t^2/U$.

The same state can also be written using two molecular-orbital determinants. With

$$|A\rangle = d^\dagger_{a\uparrow} d^\dagger_{a\downarrow} |\varnothing\rangle = \frac{|I\rangle - |C\rangle}{\sqrt{2}}, \tag{1.32}$$

we obtain

$$|\psi_\lambda\rangle = \frac{(1+\lambda)\,|B\rangle + (\lambda - 1)\,|A\rangle}{\sqrt{2(1+\lambda^2)}}. \tag{1.33}$$

Equations 1.29 and 1.33 are two representations of the same state. A determinant expansion describes correlation through coefficients of several configurations; a correlator imposes a physically motivated relation among amplitudes directly.

For a larger Hubbard system the corresponding Gutzwiller family is

$$|\psi_g\rangle \propto \exp\left(-g \sum_i n_{i\uparrow} n_{i\downarrow}\right) |\Phi_0\rangle\,, \tag{1.34}$$

where $|\Phi_0\rangle$ is a reference Slater determinant. One parameter modifies many amplitudes simultaneously. This is a compact description of correlation, although the onsite factor alone is not generally sufficient for an accurate many-body ground state. Longer-range correlators or a better reference state may be needed. A positive correlator also preserves the signs and zeros of the reference amplitudes, which limits the states it can represent.

Exercise 1.4.1: Determinants and a correlator

For the repulsive Hubbard dimer with $t = 1$:

1. Derive the matrix in Eq. 1.15 using the fermionic anticommutation relations.
2. Compare the energies of $|B\rangle$, $|C\rangle$, and the optimized $|\psi_\lambda\rangle$ as functions of U/t. Plot their double occupancies.
3. Verify Eq. 1.33. Explain why the correlator is exact for this dimer and why the same conclusion does not automatically extend to a larger lattice.

1.5 Ground-State Projection

A second strategy improves a trial state by filtering its energy components. Let $H\,|\phi_n\rangle = E_n\,|\phi_n\rangle$, with $E_0 < E_1 \leq \cdots$, and assume $|\psi_T\rangle = \sum_n a_n\,|\phi_n\rangle$ has $a_0 \neq 0$. Imaginary-time evolution gives

$$e^{-\tau(H-E_0)}\,|\psi_T\rangle = a_0\,|\phi_0\rangle + \sum_{n>0} a_n e^{-\tau(E_n-E_0)}\,|\phi_n\rangle\,. \tag{1.35}$$

After normalization the state approaches the ground state, up to an overall phase. The energy shift only changes the normalization, so knowing E_0 is not required to use the filtering idea. The gap and the initial overlap determine the projection time needed for a given accuracy.

For a finite-dimensional Hamiltonian, the *power method* uses $G = \Lambda I - H$, with $\Lambda > E_{\max}$:

$$\left|\psi^{(m+1)}\right\rangle = \frac{G\left|\psi^{(m)}\right\rangle}{\|G\left|\psi^{(m)}\right\rangle\|}. \tag{1.36}$$

The ratio of an excited-state amplitude to the ground-state amplitude is multiplied at each step by $(\Lambda - E_n)/(\Lambda - E_0)$. The slowest possible ratio is

$$q = \frac{\Lambda - E_1}{\Lambda - E_0} < 1. \tag{1.37}$$

A small gap or an unnecessarily large shift makes convergence slow. Applying H itself instead would select the eigenvalue of largest magnitude, which need not be the ground state. With a degenerate ground space, projection retains the component of the initial state in that space.

Only the action of H on a vector is required. For the TFIM it can be generated from spin flips without storing the matrix. However, the vector still generally contains exponentially many components; an initially sparse vector need not stay sparse. Lanczos and related Krylov methods use successive matrix-vector products more effectively, but do not by themselves remove this state-storage bottleneck.

Exercise 1.5.1: Projection and convergence

For a small TFIM chain or Hubbard dimer, compare the power method with exact diagonalization.

1. Choose Λ from a rigorous upper bound on $\|H\|$. For the TFIM, $\|H\| \leq \sum_{(i,j)\in\mathcal{E}} |J_{ij}| + L|\Gamma|$ is sufficient.
2. Start from a generic vector with nonzero ground-state overlap and iterate Eq. 1.36. Plot the energy and residual $\|H\psi - E\psi\|$ against iteration number.
3. Implement the TFIM action without a stored matrix. Compare convergence for different shifts and gaps, and track the growth of the support of an initially sparse vector.

1.6 The Plan for the Course

We now have two routes toward a ground state: choose a compact ansatz and optimize it, or filter an initial state by projection. Neither route is automatically efficient. A short expression for a wavefunction does not guarantee cheap expectation values, easy optimization, or rapid projection. These are separate computational questions.

We will learn how to perform such calculations while keeping track of the relevant errors. A model approximation changes the Hamiltonian; basis and grid truncations change its representation; finite size changes the system; an ansatz restricts the state; and an iterative solver

has a convergence tolerance. Statistical uncertainty will enter when we introduce sampling. Each source of error needs its own check.

The questions guiding the next chapters are:

- How can we evaluate expectation values for a compact many-body ansatz?
- How can we optimize its parameters and assess its limitations?
- How can we implement ground-state projection without explicitly storing every amplitude?
- Which properties of the model determine the cost and reliability of these methods?

Our small systems and solvable limits will remain useful throughout: they provide controlled tests of each new numerical idea.

Chapter 2

Classical Monte Carlo Sampling

Monte Carlo (MC) methods play a crucial role in computational physics, but also in technology, finance and so on. Learning MC opens therefore a lot of "doors" for several, seemingly disconnected areas. MC is essential for evaluating high-dimensional integrals. In our context, these arise when dealing with many-body wavefunctions that are not simple Slater determinants. In this chapter, we introduce the mathematical foundation of Monte Carlo, explain how random sampling techniques are used to compute physical observables, and explore their application in both classical and quantum systems. We also discuss practical implementation considerations and statistical tools for estimating uncertainty.

One of the primary motivations for using Monte Carlo techniques arises in quantum systems involving many interacting particles. For example, the Hartree-Fock approximation reduces the computation of the total energy of N electrons to the evaluation of one- and two-body integrals. This simplification brings down the dimensionality of the problem from $3N$ to at most six dimensions, making it computationally tractable.

However, when we introduce electron correlation effects through more accurate (trial) wavefunctions $|\psi_T\rangle$ e.g., by multiplying the Slater determinant with a Jastrow factor,this simplification is lost. The energy expectation value:

$$E_T = \frac{\langle\psi_T| H |\psi_T\rangle}{\langle\psi_T|\psi_T\rangle} \tag{2.1}$$

requires integration over all $3N$ coordinates, which is computationally infeasible using deterministic methods for large N. Other large-dimensional integrals appear also when computing thermodynamical properties. Monte Carlo methods provide a way to handle these high-dimensional integrals efficiently by sampling configurations probabilistically instead of deterministically computing integrals over all variables.

2.1 Probability and Central Limit Theorem

Here we give a brief reminder of some concept of probability theory. A random variable X is a function from a sample space to the outcome space (could be a real number, or binary etc). The function itself is fixed; its value is random because the outcome in the sample space is random. For instance, the event i, produce and outcome x_i and occurs with probability p_i.

The probabilities satisfy $p_i \geq 0$ and are normalized

$$\sum_i p_i = 1. \tag{2.2}$$

When dealing with two random variables, x and y, we define the *joint probability* $p(x, y)$ as the probability that both x and y take specific values simultaneously. The *marginal probability* of one variable, e.g. $p(x)$, is obtained by summing (or integrating) over the other:

$$p(x) = \sum_y p(x, y), \qquad \text{or} \qquad p(x) = \int dy\, p(x, y) \tag{2.3}$$

in the continuous case, where $p(x, y)$ and $p(x)$ denote probability densities, rather than probabilities at a single point. The *conditional probability* $p(x|y)$ quantifies the probability of x given that y has occurred, and is defined as

$$p(x|y) = \frac{p(x, y)}{p(y)}. \tag{2.4}$$

The conditional probability is defined where $p(y) > 0$. These quantities are related by the fundamental identity $p(x, y) = p(x|y)\, p(y)$.
Two random variables are independent if their joint probability is the product of their marginal probabilities. In this case, the conditional probability $p(x|y)$ is independent on y.

2.1.1 Mean and variance

For discrete random variables x with probabilities p_i, the mean and variance are:

$$\langle x \rangle = \sum_i p_i x_i, \tag{2.5}$$

$$\mathrm{var}(x) = \sum_i p_i (x_i - \langle x \rangle)^2 = \langle x^2 \rangle - \langle x \rangle^2 \tag{2.6}$$

The standard deviation $\sigma = \sqrt{\mathrm{var}(x)}$ measures the spread of individual values of x. It is distinct from the statistical error on an estimated mean. In the continuous case, using a probability density $\rho(x)$:

$$\langle x \rangle = \int dx\, \rho(x) x, \tag{2.7}$$

$$\mathrm{var}(x) = \int dx\, \rho(x) (x - \langle x \rangle)^2. \tag{2.8}$$

In continuos case the normalization reads: $\int dx\, \rho(x) = 1$.

2.1.2 Chebyshev's inequality

Let X be a random variable with finite mean $\mu = \mathbb{E}[X]$ and finite variance $\mathrm{var}(X) = \sigma^2$. For any $\varepsilon > 0$,

$$\mathbb{P}\big(|X - \mu| \geq \varepsilon\big) \leq \frac{\mathrm{var}(X)}{\varepsilon^2}. \tag{2.9}$$

For $\sigma > 0$, rewriting with $k > 0$ gives

$$\mathbb{P}\big(|X - \mu| \geq k\sigma\big) \leq \frac{1}{k^2}. \tag{2.10}$$

Proof (discrete case). Let the possible values of X be x_i with probabilities p_i. By the definition of variance,

$$\mathrm{var}(X) = \sum_i p_i (x_i - \mu)^2.$$

Split the sum into those indices with $|x_i - \mu| < \varepsilon$ and those with $|x_i - \mu| \geq \varepsilon$. On the subset with $|x_i - \mu| \geq \varepsilon$ we have $(x_i - \mu)^2 \geq \varepsilon^2$, therefore

$$\begin{aligned} \mathrm{var}(X) &= \sum_{|x_i - \mu| < \varepsilon} p_i (x_i - \mu)^2 + \sum_{|x_i - \mu| \geq \varepsilon} p_i (x_i - \mu)^2 \\ &\geq \sum_{|x_i - \mu| \geq \varepsilon} p_i \, \varepsilon^2 = \varepsilon^2 \sum_{|x_i - \mu| \geq \varepsilon} p_i. \end{aligned}$$

The last sum is exactly $\mathbb{P}(|X - \mu| \geq \varepsilon)$, so rearranging gives (2.9).

2.1.3 Law of Large Numbers

The *Law of Large Numbers* (LLN) describes how the average of many independent random outcomes tends to the expected value as the number of observations increases. Intuitively, it formalizes the idea that random fluctuations average out in the long run.

The Weak Law of Large Numbers states the following: let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables, with mean $\mu = \langle X_i \rangle$ and finite variance $\sigma^2 = \mathrm{var}(X_i)$. Define the sample mean

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i. \tag{2.11}$$

Then, for any $\varepsilon > 0$,

$$\lim_{n \to \infty} \mathbb{P}\big(|\overline{X}_n - \mu| \geq \varepsilon\big) = 0. \tag{2.12}$$

That is, the sample mean $\overline{X}_n$ converges in probability to the true mean μ. The Strong Law states instead that

$$\mathbb{P}\Big(\lim_{n \to \infty} \overline{X}_n = \mu\Big) = 1. \tag{2.13}$$

The word "strong" refers to this stronger mode of convergence: almost every infinite sample sequence has a convergent average. The weak law controls the probability of a deviation at each sufficiently large sample size.

Since the X_i are independent and identically distributed, the mean of $\overline{X}_n$ is

$$\langle \overline{X}_n \rangle = \frac{1}{n} \sum_{i=1}^{n} \langle X_i \rangle = \mu.$$

To find the variance, we evaluate the second moment $\langle \overline{X}_n^2 \rangle$. Averaging $\overline{X}_n^2$ over the joint distribution of the variables, we obtain

$$\langle \overline{X}_n^2 \rangle = \Big\langle \frac{1}{n^2} \sum_{i,j=1}^{n} X_i X_j \Big\rangle = \frac{1}{n^2} \sum_{i,j=1}^{n} \langle X_i X_j \rangle. \tag{2.14}$$

There are two types of terms in the double sum:

- For $i = j$, we have $\langle X_i^2 \rangle$, and since all variables are identically distributed, there are n such terms.
- For $i \neq j$, independence implies $\langle X_i X_j \rangle = \langle X_i \rangle \langle X_j \rangle = \mu^2$, and there are $n(n-1)$ such terms.

Combining both contributions gives

$$\langle \overline{X}_n^2 \rangle = \frac{1}{n^2} \Big[n \langle X^2 \rangle + n(n-1)\mu^2 \Big]. \tag{2.15}$$

Using the definition of the variance, $\mathrm{var}(\overline{X}_n) = \langle \overline{X}_n^2 \rangle - \langle \overline{X}_n \rangle^2$, we find

$$\mathrm{var}(\overline{X}_n) = \frac{1}{n^2} \Big[n \langle X^2 \rangle + n(n-1)\mu^2 \Big] - \mu^2 \tag{2.16}$$

$$= \frac{1}{n^2} \Big[n(\langle X^2 \rangle - \mu^2) \Big] = \frac{\mathrm{var}(X)}{n} = \frac{\sigma^2}{n}. \tag{2.17}$$

Hence, the variance of the sample mean decreases inversely with the number of samples n, meaning that $\overline{X}_n$ becomes sharply peaked around the true mean μ as n increases. This fact basically prove the statement, but if we want to be formal:

Proof (using Chebyshev's inequality). Applying Chebyshev's inequality to $\overline{X}_n$ gives, for any $\varepsilon > 0$,

$$\mathbb{P}\big(|\overline{X}_n - \mu| \geq \varepsilon \big) \leq \frac{\mathrm{var}(\overline{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

As $n \to \infty$, the right-hand side tends to zero:

$$\lim_{n \to \infty} \mathbb{P}\big(|\overline{X}_n - \mu| \geq \varepsilon \big) = 0.$$

Therefore, the sample mean $\overline{X}_n$ converges in probability to μ, proving the weak law of large numbers. □

2.1.4 Central Limit Theorem

The law of large number tells us that the average of random variables converge to the true mean, but it does not tell what is the probability distrubution of this new random variable. An essential result for statistical estimation is the Central Limit Theorem. Let $x_1, \ldots, x_N$ be i.i.d. random variables with mean $\langle x \rangle$ and variance $0 < \sigma^2 < \infty$. The sample mean:

$$\bar{x}_N = \frac{1}{N} \sum_{i=1}^{N} x_i \tag{2.18}$$

has, for large N, the Gaussian approximation:

$$\bar{x}_N \approx \mathcal{N}\left(\langle x \rangle, \frac{\sigma^2}{N} \right). \tag{2.19}$$

Proof (using characteristic functions). Let $X_1, X_2, \ldots$ be independent and identically distributed (i.i.d.) random variables with finite mean $\mu = \langle X_i \rangle$ and finite, nonzero variance $\sigma^2 = \mathrm{var}(X_i)$. Put

$$S_n = \sum_{i=1}^{n} X_i.$$

Introduce the *standardized* sum

$$Z_n \;=\; \frac{S_n - n\mu}{\sigma\sqrt{n}} \;=\; \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{X_i - \mu}{\sigma}.$$

The Central Limit Theorem states that Z_n converges in distribution to the standard normal:

$$Z_n \xrightarrow{d} \mathcal{N}(0,1).$$

Let $\phi_X(t) = \langle e^{itX} \rangle$ denote the characteristic function (CF) of X. This expectation value for a continuos variable is

$$\int dx \; p(x) \; e^{itx},$$

where $p(x)$ is the probability distrubution for the values of the random variable X. Define the standardized single-variable

$$Y := \frac{X - \mu}{\sigma},$$

so that $\langle Y \rangle = 0$ and $\mathrm{var}(Y) = 1$. Let $\phi_Y(t) = \langle e^{itY} \rangle$ be its CF. By the differentiation properties of characteristic functions, [1]

$$\phi_Y(0) = 1, \qquad \phi_Y'(0) = i\langle Y \rangle = 0, \qquad \phi_Y''(0) = i^2 \langle Y^2 \rangle = -1.$$

[1] Assume that $\langle |Y|^n \rangle < \infty$. Only the first two moments are needed for the proof here. Then differentiation of $\phi_Y(t)$ under the sum or integral is valid up to order n, giving:

$$\phi_Y'(t) = \frac{d}{dt} \langle e^{itY} \rangle = \langle iY e^{itY} \rangle, \tag{2.20}$$

$$\phi_Y''(t) = \frac{d^2}{dt^2} \langle e^{itY} \rangle = \langle (iY)^2 e^{itY} \rangle, \tag{2.21}$$

$$\phi_Y^{(n)}(t) = \langle (iY)^n e^{itY} \rangle. \tag{2.22}$$

where each displayed derivative requires the corresponding finite moment. Thus $\phi_Y'(0) = i\,\langle Y \rangle$, $\phi_Y''(0) = -\,\langle Y^2 \rangle$, $\phi_Y'''(0) = -i\,\langle Y^3 \rangle$.

Hence, for $t \to 0$, a second-order Taylor expansion yields

$$\phi_Y(t) = 1 - \tfrac{1}{2}t^2 + o(t^2). \tag{2.23}$$

Now remember that $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i$. Since the characteristic function of a sum independent random variablesis just the product of the CF, we have

$$\phi_{Z_n}(t) \;=\; \Big[\phi_Y\Big(\frac{t}{\sqrt{n}}\Big)\Big]^n = \Big[1 - \frac{t^2}{2n} + o(n^{-1})\Big]^n. \tag{2.24}$$

Taking the limit $n \to \infty$ yields[2]

$$\lim_{n\to\infty} \phi_{Z_n}(t) = e^{-t^2/2},$$

for every fixed real t. The right-hand side $e^{-t^2/2}$ is the characteristic function of the standard normal distribution $\mathcal{N}(0,1)$.

The Gaussian approximation gives a more informative estimate of fluctuations on the scale of the standard deviation of the mean, $\sigma/\sqrt{N}$. For instance, Chebyshev bounds the probability of a deviation of at least four standard deviations by 1/16, whereas a standard Gaussian has $\mathbb{P}(|Z| \geq 4) \simeq 6.33 \times 10^{-5}$.

2.2 Markov Chains

To progress further we need to study random processes. Contrary to before we introduce a time depence on the random variables $x_1, x_2, \cdots$. A Markov chain (MC) is a stochastic sequence x_n generated by:

$$x_{n+1} = F(x_n, \xi_n), \tag{2.25}$$

where ξ_n are i.i.d. random variables and F is a deterministic function. The future depends only on the present, not on the full past, but in a probabilistic way. MC are a special stochastic process where only the most recent information matters in predicting the probability of the next state.

We can think of x_n of the state of a system at discrete time n, and we will use this pseudo-dynamics as **a tool to sample** from a target distribution $\pi(x)$. So the whole field of Monte Carlo is to construct a Markov Chain that enable one to sample from a distribution we are actually interested in, regardless if the "dynamics" is actually a good representation of the real system dynamics.

We can think that the random variable are distributed accordingly to an initial probability density $\rho_0(x)$ The evolution of the marginal distribution $\rho_n(x)$ is governed by the Master equation:

$$\rho_{n+1}(x') = \sum_x P(x'|x)\rho_n(x), \tag{2.26}$$

[2] $\lim_{n\to\infty}(1 + x/n)^n = e^x$.

where $P(x'|x)$ is the transition probability.[3] This gives the probability to transition from state x to x', and its rows are normalized to 1:

$$\sum_{x'} P(x'|x) = 1. \tag{2.27}$$

So the probability of being at state x' is given by the convolution between the transition probability and the probability density at the previous step.
Markov chains can also be depicted visually in the form of a graph, with the state space $\mathcal{S}$ drawn as a collection of nodes and the edges representing the non-zero transition probabilities P_{ij}.

As an example, suppose that our space is given by nodes A, B, C, and D. and the transition probability is

$$P = \begin{pmatrix} 0.6 & 0.2 & 0.1 & 0.1 \\ 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.5 & 0.5 \\ 0.5 & 0.2 & 0.2 & 0.1 \end{pmatrix} \tag{2.28}$$

You can visualize the transitions using the following diagram:

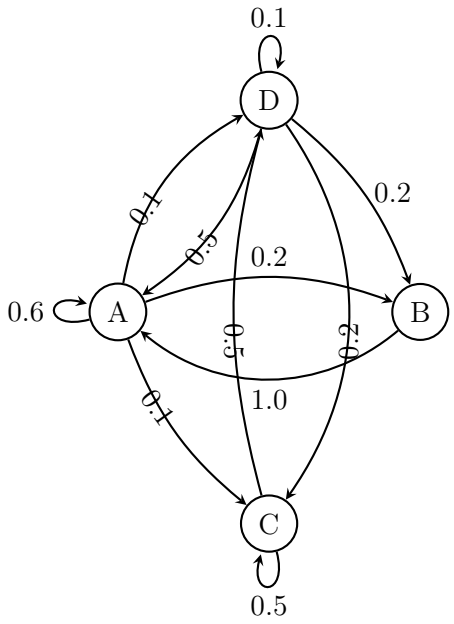


Figure 2.1: Transition graph for the example Markov chain.

In Markov Chain MC one typically starts from a definite state, e.g A at $n = 0$. For this pedagogical example we use explicitely write the probability density as vectors. Then:

$$\vec{\rho}_0^T = (1, 0, 0, 0)$$

[3]Sometimes we will also use the notation $P(x, x'), P(x \to x')$, or $P_{xx'} \equiv P(x'|x)$ (notice the reverse order x, x' compared to the notation $(x'|x)$).

At $n = 1$, the distribution becomes:

$$\vec{\rho}_1^T = \vec{\rho}_0^T P = (0.6, 0.2, 0.1, 0.1)$$

Notice that we are using this 'strange' notation for matrix-vector multiplication, where we apply the matrix from the right to a row vector,[4] If sampling from this vector results in state D, this becomes the current state. Repeating this generates a trajectory, for example:

$$A - D - A - C - D - A - B - A - C - ...$$

To calculate the probability distribution at $n = 2$, we use:

$$\vec{\rho}_2^T = \vec{\rho}_1^T P,$$

recursively. From a trajectory perspective, one needs to evaluate all the 'paths' that connect the inital state A from each of the final states x', given that there is an intermediate step. For example, the probability of being in state D at $n = 2$ is:

$$\begin{aligned}
\Pr(X_2 = D|X_0 = A) &= \sum_{j=1}^{4} \Pr(X_1 = j|X_0 = A) \cdot \Pr(X_2 = D|X_1 = j) \\
&= \sum_{j=1}^{4} P_{1j} \cdot P_{j4} \\
&= P_{11}P_{14} + P_{12}P_{24} + P_{13}P_{34} + P_{14}P_{44} \\
&= 0.6 \cdot 0.1 + 0.2 \cdot 0 + 0.1 \cdot 0.5 + 0.1 \cdot 0.1 \\
&= 0.06 + 0 + 0.05 + 0.01 = 0.12
\end{aligned}$$

From a linear algebra persepctive, we evolve distributions using repeated application or matrix-vectors operations:

$$\vec{\rho}_{n+1}^T = \vec{\rho}_n^T P \quad \text{and} \quad \vec{\rho}_{n+k}^T = \vec{\rho}_n^T P^k$$

A stationary distribution $\vec{\pi}$ satisfies:

$$\vec{\pi}^T = \vec{\pi}^T P, \quad \text{or} \quad \sum_i \pi_i P_{ij} = \pi_j \tag{2.29}$$

which means that a chain initialized with distribution π has this same distribution at every subsequent step. Notice therefore a fundamental distinction between the random variable x and its probability distribution $\rho(x)$: The evolution of the probability distribution is fully deterministic, while the dynamics of the variable x is not. Obviously, not every MC has a limiting distribution.

[4]The resulting vector $\vec{y}^T = \vec{x}^T P$ is given by

$$y_j = \sum_{i=1}^{N} x_i P_{ij}, \quad \text{for } j = 1, \ldots, N,$$

so we are touching all the rows i at a fixed column j. Informally, rows tell us where we can go from a particular state, while columns tell us where we start to reach a particular state.

2.2.1 Example: Convergence to the Stationary Distribution

We now show numerically that the Markov chain defined above converges to its stationary distribution. Consider again the transition matrix of Eq. 2.28. Starting from:

$$\vec{\rho}_0^T = (1, 0, 0, 0)$$

we compute:

$$\begin{aligned}
\vec{\rho}_1^T &= (0.600,\ 0.200,\ 0.100,\ 0.100),\\
\vec{\rho}_2^T &= (0.610,\ 0.140,\ 0.130,\ 0.120),\\
\vec{\rho}_3^T &= (0.566,\ 0.146,\ 0.150,\ 0.138).
\end{aligned}$$

and there seems to be convergence to a stationary distribution.[5]

Indeed, a natural question is asking under which conditions a P evolution leads to a stationary distribution $\bar{\rho}(x)$ such that:

$$\bar{\rho}(x') = \sum_x P(x'|x)\bar{\rho}(x). \tag{2.30}$$

The theory of Markov-Chains is pretty heavy, and we will leave theorems aside. We are going to rely on some physical intuitions. The reason why the above example led to a limiting distribution is because the Markov chain is

- *Irreducible*: It's possible to reach any state from any other state in a finite number of steps.
- *Aperiodic*: The system does not get trapped in cycles, i.e., the chain does not return to states only at fixed multiples of time steps.

For a finite state space, irreducibility guarantees a unique stationary distribution with positive probabilities. Aperiodicity additionally guarantees convergence to it from every initial distribution. Infinite state spaces require further assumptions. For a finite irreducible chain, the time average of an observable also obeys an ergodic law:

$$\frac{1}{N}\sum_{n=1}^{N} f(x_n) \longrightarrow \sum_x \pi(x) f(x) \quad \text{almost surely.} \tag{2.31}$$

[5] To find the stationary distribution analytically, solve $\vec{\pi}^T P = \vec{\pi}^T$ with $\sum_i \pi_i = 1$. This leads to the system:

$$\begin{aligned}
0.6\pi_1 + 1.0\pi_2 + 0.0\pi_3 + 0.5\pi_4 &= \pi_1\\
0.2\pi_1 + 0.0\pi_2 + 0.0\pi_3 + 0.2\pi_4 &= \pi_2\\
0.1\pi_1 + 0.0\pi_2 + 0.5\pi_3 + 0.2\pi_4 &= \pi_3\\
0.1\pi_1 + 0.0\pi_2 + 0.5\pi_3 + 0.1\pi_4 &= \pi_4\\
\pi_1 + \pi_2 + \pi_3 + \pi_4 &= 1
\end{aligned}$$

Solving this yields $\vec{\pi} \approx (0.5385, 0.1385, 0.1692, 0.1538)$.

This time-average statement holds even for a periodic chain. Aperiodicity is needed for convergence of the distribution at a given time, a different statement.

In our example, the graph defined by P allows movement (possibly indirectly) between all states (irreducibility), and contains self-loops (i.e non-zero diagonal elements) breaking periodicity.
It is clear that, if P allows a limiting distribution π, then this is eigenvector with eigenvalue 1.

Finally, let's notice the subtle difference between limiting and stationary distributions. A finite irreducible periodic chain still has a stationary distribution, but generally does not converge to it from a single starting state. If initialized in that stationary distribution, it remains there at all times.

2.2.2 Detailed Balance

The example featured above is quite artificial, but there is a useful condition that allows one to verify that a prescribed distribution is stationary. A sufficient (but not necessary) condition for stationarity is detailed balance:

$$P(x'|x)\pi(x) = P(x|x')\pi(x'). \tag{2.32}$$

This expresses a balance between forward and reverse transitions, analogous to equilibrium in physical systems. Detailed balance implies stationarity, and it can be simply shown. If we sum both sides of Eq. 2.32:

$$\sum_x P(x'|x)\pi(x) = \sum_x P(x|x')\pi(x') \tag{2.33}$$

$$\sum_x P(x'|x)\pi(x) = \pi(x')\sum_x P(x|x') \tag{2.34}$$

$$\sum_x P(x'|x)\pi(x) = \pi(x'), \tag{2.35}$$

which is the definition of stationarity, from the master equation. Convergence also requires the irreducibility and aperiodicity conditions discussed above.

Exercise 2.2.1: Implementation of Markov Chain trajectories

Write a code that implements the Markov chain, and check that the empirical frequencies approach the stationary distribution. Consecutive measurements are correlated; use independent trajectories or a correlation-aware error estimate. You can simply output an histogram of visited states, since the space is just four-dimensional.

Exercise 2.2.2: Violation of Detailed Balance

Show that the example of Eq. 2.28, does not respect the detailed balance property.

2.2.3 Spectral Convergence and Autocorrelation Time

Let P be the transition matrix of a finite irreducible, aperiodic Markov chain, with the row convention $P_{xx'} = P(x'|x)$. Assume first that P is diagonalizable, and order its eigenvalues

by modulus:

$$1 = \lambda_0 > |\lambda_1| \geq |\lambda_2| \geq \cdots . \tag{2.36}$$

The left eigenvector corresponding to $\lambda_0 = 1$ is the stationary distribution π. For a nonreversible chain the other eigenvalues and eigenvectors can be complex.

Suppose the initial distribution is expanded in left eigenvectors v_i^T:

$$\rho_0^T = \sum_i a_i v_i^T, \tag{2.37}$$

with $v_0 = \pi$ and $a_0 = 1$ by normalization. Then after n steps,

$$\rho_n^T = \rho_0^T P^n = \sum_i a_i \lambda_i^n v_i^T. \tag{2.38}$$

As $n \to \infty$, all components with $|\lambda_i| < 1$ decay:

$$\rho_n \longrightarrow a_0 v_0 = \pi. \tag{2.39}$$

Writing $\lambda_* = \max_{i \geq 1} |\lambda_i| = |\lambda_1|$, the expansion gives

$$\|\rho_n - \pi\| \leq C_{\rho_0} \lambda_*^n. \tag{2.40}$$

The slowest rate is present only if the initial distribution has a component in the corresponding eigenspace. The *absolute spectral gap*

$$\delta = 1 - \lambda_* \tag{2.41}$$

therefore controls the slowest relaxation scale. Notice the formal similarity with the power method. Without diagonalizability, Jordan blocks can also produce polynomial factors multiplying the exponential decay.

A direct convergence proof using detailed balance. For a reversible chain, the above expansion and a useful quantitative bound follow by transforming P into a symmetric matrix. This argument applies, for example, to a Metropolis chain satisfying detailed balance. The four-state example above is not reversible, so the symmetric transformation does not apply to that example.

Let D be diagonal with $D_{xx} = \pi(x) > 0$, and define

$$S = D^{1/2} P D^{-1/2}, \qquad S_{xx'} = P_{xx'} \sqrt{\frac{\pi(x)}{\pi(x')}}. \tag{2.42}$$

Detailed balance gives

$$S_{xx'} = P_{x'x} \sqrt{\frac{\pi(x')}{\pi(x)}} = S_{x'x}, \tag{2.43}$$

$$S_{xx'} = \sqrt{P_{xx'} P_{x'x}}. \tag{2.44}$$

Thus S is real symmetric and has the same eigenvalues as P. Its eigenvectors u_i can be chosen orthonormal. The stationary eigenvector is

$$u_0(x) = \sqrt{\pi(x)}, \qquad (Su_0)(x) = \sqrt{\pi(x)} \sum_{x'} P_{xx'} = u_0(x), \qquad \|u_0\|_2^2 = 1. \tag{2.45}$$

Now use column vectors for the density difference and define

$$g_n = D^{-1/2}(\rho_n - \pi), \qquad g_n(x) = \frac{\rho_n(x) - \pi(x)}{\sqrt{\pi(x)}}. \tag{2.46}$$

The original row convention means $\rho_{n+1} = P^T \rho_n$ for these column vectors. Since $P^T D = DP$, we obtain

$$g_{n+1} = D^{-1/2} P^T (\rho_n - \pi) \tag{2.47}$$

$$= D^{-1/2} P^T D^{1/2} g_n = S g_n. \tag{2.48}$$

Normalization removes the stationary component:

$$u_0^T g_n = \sum_x [\rho_n(x) - \pi(x)] = 0. \tag{2.49}$$

Expanding $g_0 = \sum_{i \geq 1} c_i u_i$ therefore gives

$$g_n = S^n g_0 = \sum_{i \geq 1} c_i \lambda_i^n u_i, \tag{2.50}$$

$$\|g_n\|_2^2 = \sum_{i \geq 1} |c_i|^2 |\lambda_i|^{2n} \leq \lambda_*^{2n} \sum_{i \geq 1} |c_i|^2 = \lambda_*^{2n} \|g_0\|_2^2. \tag{2.51}$$

This proves convergence in the weighted norm

$$\left[\sum_x \frac{[\rho_n(x) - \pi(x)]^2}{\pi(x)} \right]^{1/2} \leq \lambda_*^n \left[\sum_x \frac{[\rho_0(x) - \pi(x)]^2}{\pi(x)} \right]^{1/2}. \tag{2.52}$$

To express the result in terms of probabilities, define the total variation distance

$$\|\rho_n - \pi\|_{\mathrm{TV}} = \frac{1}{2} \sum_x |\rho_n(x) - \pi(x)|. \tag{2.53}$$

Cauchy–Schwarz and $\sum_x \pi(x) = 1$ imply

$$\|\rho_n - \pi\|_{\mathrm{TV}} = \frac{1}{2} \sum_x \sqrt{\pi(x)}\, |g_n(x)| \tag{2.54}$$

$$\leq \frac{1}{2} \left(\sum_x \pi(x) \right)^{1/2} \|g_n\|_2 \tag{2.55}$$

$$\leq \frac{1}{2} \lambda_*^n \|g_0\|_2 \leq \frac{1}{2} e^{-n\delta} \|g_0\|_2. \tag{2.56}$$

In the last step we used $1-\delta \le e^{-\delta}$. For a chain started at a definite state x_0,

$$\|g_0\|_2^2 = \frac{[1-\pi(x_0)]^2}{\pi(x_0)} + \sum_{x\neq x_0} \pi(x) = \frac{1}{\pi(x_0)} - 1. \tag{2.57}$$

Thus the absolute gap controls the decay, while the initial state affects the prefactor.

Mixing time and the inverse gap. Define $t_{\mathrm{mix}}(\epsilon)$ as the smallest integer n for which the total variation distance is at most ϵ, for every starting state. For a finite reversible chain and $0<\epsilon<1/2$, a standard pair of bounds is

$$(\delta^{-1}-1)\ln\left(\frac{1}{2\epsilon}\right) \le t_{\mathrm{mix}}(\epsilon) \le \left\lceil \delta^{-1}\ln\left(\frac{1}{\epsilon\pi_{\min}}\right)\right\rceil, \qquad \pi_{\min}=\min_x \pi(x). \tag{2.58}$$

The upper bound follows from the estimate just proved, with a conservative prefactor. For the lower bound, choose a nonconstant right eigenfunction r of P with eigenvalue of modulus λ_* and $\langle r\rangle_\pi = 0$. Start at a state where $|r|$ is maximal. Then

$$2\|\rho_n-\pi\|_{\mathrm{TV}}\|r\|_\infty \ge \left|\sum_x [\rho_n(x)-\pi(x)]r(x)\right| = \lambda_*^n\|r\|_\infty. \tag{2.59}$$

Consequently $n \ge \ln[1/(2\epsilon)]/[-\ln\lambda_*]$ is necessary. The inequality $-\ln\lambda_* \le (1-\lambda_*)/\lambda_*$ yields the stated lower bound when $0<\lambda_*<1$; for $\lambda_*=0$ it is trivial. Mixing therefore involves the inverse gap together with accuracy- and stationary-probability-dependent factors, rather than being exactly $1/\delta$.

For $0<\lambda_*<1$, the slowest exponential relaxation time is

$$\tau_{\exp} = -\frac{1}{\ln\lambda_*} = -\frac{1}{\ln(1-\delta)} \simeq \frac{1}{\delta} \qquad (\delta \ll 1). \tag{2.60}$$

The approximation follows from $-\ln(1-\delta) = \delta + O(\delta^2)$. A negative eigenvalue produces alternating signs, but its envelope decays with its modulus. If desired, a lazy chain $(I+P)/2$ removes negative eigenvalues. For a lazy reversible chain the absolute gap equals the usual gap below the stationary eigenvalue.

Autocorrelation of an observable and statistical error. Constructing P is infeasible in large systems. Instead, correlations are measured along an equilibrated trajectory. For an observable f, define

$$\widetilde{f}(x) = f(x) - \langle f\rangle_\pi, \qquad C_f(n) = \langle f(x_0)f(x_n)\rangle_\pi - \langle f\rangle_\pi^2. \tag{2.61}$$

Here the average includes $x_0\sim\pi$ and subsequent transitions. For a reversible chain, expand $\sqrt{\pi(x)}\,\widetilde{f}(x) = \sum_{i\ge1} b_i u_i(x)$. Then

$$C_f(n) = \sum_{x,x'} \pi(x)\widetilde{f}(x)(P^n)_{xx'}\widetilde{f}(x') \tag{2.62}$$

$$= (D^{1/2}\widetilde{f})^T S^n (D^{1/2}\widetilde{f}) = \sum_{i\ge1} b_i^2\lambda_i^n, \tag{2.63}$$

$$\sigma_f^2 = C_f(0) = \sum_{i\ge1} b_i^2. \tag{2.64}$$

The autocorrelation is generally a sum of modes, rather than a single exponential. An observable with $b_i = 0$ does not see mode i. This is why distributional convergence and observable autocorrelation are distinct questions even though they involve the same spectrum.

For $\sigma_f^2 > 0$, define the integrated autocorrelation time in units of recorded steps by

$$\tau_{\mathrm{int},f} = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{C_f(n)}{C_f(0)}. \tag{2.65}$$

Summing each geometric series gives

$$\tau_{\mathrm{int},f} = \frac{1}{2} + \sum_{i\geq 1} \frac{b_i^2}{\sigma_f^2} \frac{\lambda_i}{1-\lambda_i} \tag{2.66}$$

$$= \sum_{i\geq 1} \frac{b_i^2}{\sigma_f^2} \frac{1+\lambda_i}{2(1-\lambda_i)} \leq \frac{1+\lambda_*}{2(1-\lambda_*)} = \frac{1}{\delta} - \frac{1}{2}. \tag{2.67}$$

Thus $\tau_{\mathrm{int},f}$ is an observable-dependent weighted average of mode contributions, not universally the inverse gap. For one positive slow mode it is approximately $1/\delta$ when δ is small. Negative correlations can reduce it below $1/2$.

To see why this is the time relevant to error bars, let $\bar{f} = N^{-1}\sum_{j=1}^{N} f(x_j)$ for a stationary trajectory. Retaining all covariance terms,

$$\mathrm{var}(\bar{f}) = \frac{1}{N^2} \sum_{i,j=1}^{N} \mathrm{cov}(f(x_i), f(x_j)) \tag{2.68}$$

$$= \frac{1}{N^2} \left[N C_f(0) + 2 \sum_{n=1}^{N-1} (N-n) C_f(n) \right] \tag{2.69}$$

$$= \frac{\sigma_f^2}{N} \left[1 + 2 \sum_{n=1}^{N-1} \left(1 - \frac{n}{N}\right) \frac{C_f(n)}{C_f(0)} \right]. \tag{2.70}$$

For a run long compared with the relevant correlation scales, this becomes

$$\sigma_{\mathrm{eff}}^2 \equiv \mathrm{var}(\bar{f}) \simeq \frac{2\tau_{\mathrm{int},f}\sigma_f^2}{N}, \qquad N_{\mathrm{eff},f} = \frac{N}{2\tau_{\mathrm{int},f}}. \tag{2.71}$$

Independent observations have $\tau_{\mathrm{int},f} = 1/2$, recovering σ_f^2/N. Positive correlations reduce the effective number of samples. The statistical error of the mean is the square root of this variance.

The equilibration period is discarded as burn-in before this stationary analysis. Apparent stabilization of one observable does not prove equilibration: it may fail to detect a slow mode. Runs from different initial conditions and sufficiently long trajectories help diagnose this problem.

2.3 The Metropolis Algorithm

The Metropolis algorithm constructs a Markov chain whose stationary distribution is a given target distribution $\pi(x)$. Notice that the working hypothesis is that we *known* what probability we need to sample from, e.g. $\pi(x) \propto e^{-\beta V(x)}$, but we don't know how to generate

configurations sampled exactly as that.
The Metropolis Markov Chain is defined by a transition probability of the form:

$$P(x'|x) = T(x'|x)A(x'|x), \quad \text{for } x' \neq x \tag{2.72}$$

where:

- $T(x'|x)$ is the **proposal** or *trial* probability, giving the probability of proposing a move from x to x';
- $A(x'|x)$ is the **acceptance** probability, which decides whether the proposed move is accepted.

To ensure that $\pi(x)$ is the stationary distribution, $A(x'|x)$ is chosen to satisfy the *detailed balance condition*:

$$\pi(x)T(x'|x)A(x'|x) = \pi(x')T(x|x')A(x|x'). \tag{2.73}$$

A common choice that satisfies detailed balance is:

$$A(x'|x) = \min\left(1, \frac{\pi(x')T(x|x')}{\pi(x)T(x'|x)}\right) \tag{2.74}$$

If $T(x'|x)$ is symmetric (i.e., $T(x'|x) = T(x|x')$), then the acceptance probability simplifies to:

$$\boxed{A(x'|x) = \min\left(1, \frac{\pi(x')}{\pi(x)}\right)} \tag{2.75}$$

One of the most important aspect of the algorithm is that it does not require the *normalization* of the probability distribution, since only the ratio $\pi(x')/\pi(x)$ enters in the evaluation.
The pseudocode of the Metropolis algorithm with symmetric proposal is:

Input: Initial state $x = x_0$

For loop: $n = 0, \ldots, M_{\text{steps}} - 1$; set $x = x_n$.

- Propose $x' \sim T(x'|x)$
- Compute acceptance ratio $r = \frac{\pi(x')}{\pi(x)}$
- Accept with probability $A = \min(1, r)$
  - if accepted: $x_{n+1} = x'$
  - else: $x_{n+1} = x$
- Record the resulting state and compute $f_{n+1} = f(x_{n+1})$, including repeated states after rejection.

The probability of staying at x includes proposals of x itself and all rejected moves. It is fixed by normalization:

$$P(x|x) = 1 - \sum_{x' \neq x} T(x'|x)A(x'|x). \tag{2.76}$$

Operationally, draw a uniform number z between 0 and 1 and accept if $z < \min(1, r)$. The diagonal probability need not be positive: if every proposed move changes the state and is accepted, it is zero. A chain with single-spin flips can then be periodic; adding a probability $1/2$ of doing nothing guarantees aperiodicity. The acceptance move allows us to move from a configuration with high probability toward ones less probable.

There is a tradeoff in choosing the trial distribution $T(x'|x)$. Proposing a state x' very close to x increases the acceptance probability, since $\pi(x') \approx \pi(x)$, but leads to small changes and thus high autocorrelation in the samples. Conversely, proposing distant moves lowers acceptance but may explore the space more rapidly if accepted.

An acceptance rate around 50% is sometimes a useful tuning heuristic in low dimensions, but there is no universal optimal acceptance rate. The relevant quantity is the statistical error per unit computational cost. Independent exact draws, when available, achieve both unit acceptance and zero autocorrelation; generic local proposals do not. When a move is rejected, the chain stays at x, which still contributes to averages. Rejections increase the number of times the chain remains in the same state, slowing exploration but maintaining correctness.

2.4 Error Estimation and Binning

When the sum of the nonzero-lag autocorrelations is positive, treating MCMC samples as independent underestimates uncertainties. A standard method is binning analysis:

1. Divide data (originally an array of M_{steps} values) into M bins of size n_{bin}.
2. Compute bin averages $\bar{f}_i$.
3. Estimate variance:
$$\sigma^2 = \frac{1}{M-1} \sum_{i=1}^{M} (\bar{f}_i - \bar{f})^2. \tag{2.77}$$
4. The standard error on the average is $\sigma/\sqrt{M}$.

Here σ^2 estimates the variance of the *bin averages*, not that of the original individual measurements. The standard error means the estimated standard deviation of the final sample mean; it is the quantity used for its error bar.

As n_{bin} becomes much larger than the correlation scale, correlations *between bin averages* become small. Correlations inside each bin have not vanished: their contribution is included in the fluctuations of the bin average. We are treating the binned values as approximately independent samples, and there are M of them, not M_{steps}.

Repeat the analysis with increasing bin sizes. A stable error estimate over a range of sizes is evidence that the relevant correlations have been captured, provided many bins remain. If the error keeps increasing, the run may be too short to resolve its slow modes. A plateau obtained from only a few bins is unreliable. Binning estimates sampling fluctuations; it does not remove a bias from incomplete equilibration or from a discretized dynamics with the wrong stationary distribution. For sufficiently many approximately independent, well-behaved bin averages, a Gaussian interval $\bar{f} \pm 1.96\,\sigma/\sqrt{M}$ is an approximate 95% confidence interval.

2.5 Importance Sampling

Now that we have presented the basics of Markov Chain and the Metropolis algorithms, let's take a step back and find a motivation for that. The concepts introduced so far find a natural and powerful application in the context of statistical physics. consider a system of L classical spins, with configuration denoted by

$$x = (\sigma_1, \sigma_2, \ldots, \sigma_L), \tag{2.78}$$

where each spin $\sigma_i \in \{-1, +1\}$. The energy of a configuration is given by a function $E(x)$, for example the Hamiltonian of the classical Ising model:

$$E(x) = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j - h \sum_i \sigma_i, \tag{2.79}$$

where $\langle i, j \rangle$ denotes nearest-neighbor pairs, J is the interaction strength, and h is the external magnetic field.

Our goal is to compute thermal expectation values of observables $f(x)$ under the Boltzmann distribution:

$$\langle f \rangle = \frac{1}{Z} \sum_x f(x) e^{-\beta E(x)}, \tag{2.80}$$

where the partition function is

$$Z = \sum_x e^{-\beta E(x)}, \tag{2.81}$$

and $\beta = 1/(k_B T)$ is the inverse temperature. Direct evaluation of Eq. 2.81 is impractical for large generic systems, since the number of configurations is $\Omega = 2^L$. We reserve N for the number of sampled configurations. On the constrary, computing the energy $E(x)$ of a configuration is efficient because it involves a number of operations that scales with L. We need to escape the direct summation way to compute both integrals.

A naive approach would be to sample configurations x uniformly and compute the average:

$$\langle f \rangle \approx \frac{\Omega}{ZN} \sum_{i=1}^{N} f(x_i) e^{-\beta E(x_i)}, \qquad x_i \sim 1/\Omega \tag{2.82}$$

assuming Z is known. If Z is unknown, one can use a ratio of sampled sums, discussed below. This method is often highly inefficient because most randomly chosen configurations contribute negligibly due to their large energy (if the temperature is low enough). If one is interested to predict the low-temperature phases, there is vanishing probability to pick randomly such configurations, e.g. $x = (1, 1, 1, \cdots)$. This leads to high variance and poor convergence.

A more efficient strategy is *importance sampling*: we sample configurations x according to the target distribution

$$\pi(x) = \frac{e^{-\beta E(x)}}{Z}, \tag{2.83}$$

and estimate the average as

$$\langle f \rangle \approx \frac{1}{N} \sum_{i=1}^{N} f(x_i), \qquad x_i \sim \pi(x). \tag{2.84}$$

Notice that now, the sum over x_i is understood as x_i distributed accordingly to $\pi(x)$. This often greatly reduces the variance for thermodynamic observables, although the improvement depends on the observable and on the correlations of the sampled chain. This motivates our need to use Markov Chain to sample from a target $\pi(x)$.

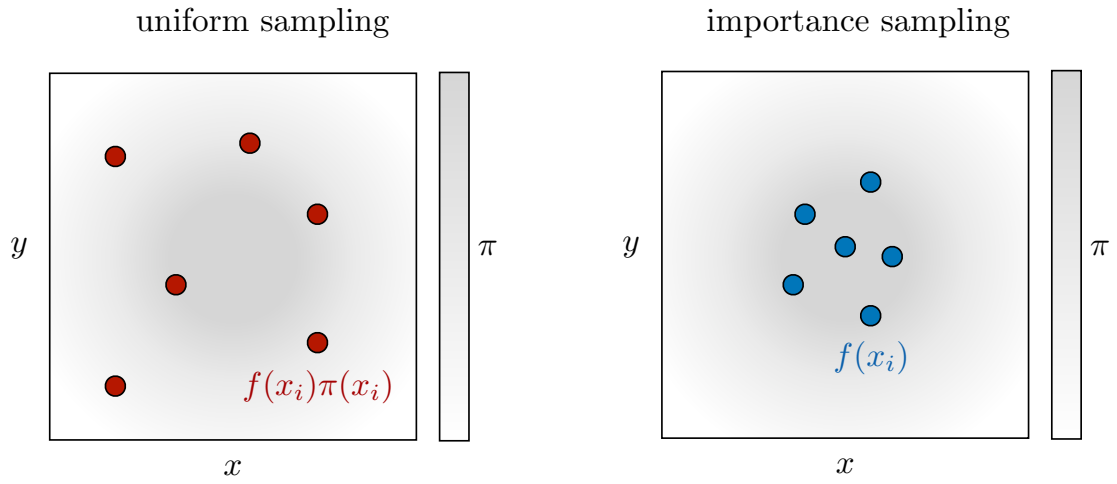


Figure 2.2: Left: we generate samples, x_i, uniformly in the space, each sample contributes to the sum, $\langle f \rangle$, with $\Omega f(x_i) e^{-\beta E(x_i)}/Z$; most of these contribution vanish due to the probability weight. Right: we generate samples distributed already as $\pi(x) = e^{-\beta E(x)}/Z$; now each sample contributes with $f(x_i)$.

The Metropolis algorithm is particularly well-suited for importance sampling in this context. It generates a Markov chain whose stationary distribution is precisely the Boltzmann distribution $\pi(x)$. Crucially, the algorithm **does not** require knowledge of the partition function Z, since it only needs the ratio of probabilities:

$$\frac{\pi(x')}{\pi(x)} = e^{-\beta(E(x')-E(x))}. \tag{2.85}$$

In the Metropolis algorithm, a common choice for the trial[6] distribution $T(x'|x)$ is to propose a new configuration x' by flipping a single spin at a randomly chosen site. This ensures the Markov chain explores the configuration space in small, local steps.

Once the Markov chain reaches equilibrium (after an initial thermalization period), we collect N samples $\{x_1, x_2, \ldots, x_N\}$ from the chain and compute:

$$\langle f \rangle \approx \frac{1}{N} \sum_{i=1}^{N} f(x_i). \tag{2.86}$$

The associated statistical error can be estimated using standard techniques such as binning analysis, which accounts for the autocorrelation between samples. The proposal moves must explore the intended configuration space. They need not preserve every symmetry of the

[6]Do not confuse this T with the T of temperature.

Hamiltonian. For example, a classical Heisenberg simulation can use single-spin rotations. Spin exchanges are appropriate if one chooses an ensemble with fixed total magnetization; a symmetry of the Hamiltonian alone does not impose that constraint.

Exercise 2.5.1: Critical Slowing Down in the 2D Ising Model

Consider the classical Ising model on a two-dimensional square lattice of size $L \times L$ with periodic boundary conditions. The energy of a spin configuration $x = (\sigma_1, \ldots, \sigma_{L^2})$ is given by:

$$E(x) = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j, \tag{2.87}$$

where $\sigma_i \in \{-1, +1\}$ and the sum is over nearest-neighbor pairs $\langle i, j \rangle$. Assume $J = k_B = 1$ and $\beta = 1/T$ is the inverse temperature. Start with small L, i.e. 4×4, then try to increase the size.

1. Implement the Metropolis algorithm using a local spin-flip proposal (i.e., $T(x'|x)$ consists of flipping a randomly chosen spin).
2. Measure the average magnetization $\langle M \rangle$ and energy $\langle E \rangle$ for several values of temperature T, particularly near the infinite-lattice critical temperature $T_c \approx 2.269$. At zero field the exact finite-system mean $\langle M \rangle$ vanishes by spin-inversion symmetry; also measure $\langle |M| \rangle$ and $\langle M^2 \rangle$.
3. Plot the autocorrelation function of the magnetization time series and extract the autocorrelation time τ_{corr} as a function of T.
4. Discuss how the efficiency of the Metropolis algorithm degrades near T_c. This phenomenon is known as *critical slowing down*.
5. **The Art of Debug.** Start small. When in doubt remember that you can exactly compute the expectation values and Z, if 2^{L^2} is still decently small.

2.5.1 General importance sampling and its variance

The preceding Boltzmann example is a special case of a general change of sampling distribution. To make its statistical meaning explicit, consider the integral

$$I = \int_{\mathcal{D}} dx\, h(x). \tag{2.88}$$

Here $h(x)$ denotes a real integrand; it is distinct from the sampling density $q(x)$ and from the constant magnetic field h in the Ising Hamiltonian above. Assume $\int |h(x)|\, dx < \infty$. Choose a normalized density $q(x)$ such that $q(x) > 0$ wherever $h(x) \neq 0$, apart from sets of measure zero. Then

$$I = \int_{\mathcal{D}} dx\, q(x) \frac{h(x)}{q(x)} \tag{2.89}$$

$$= \left\langle \frac{h(X)}{q(X)} \right\rangle_q . \tag{2.90}$$

For independent samples $X_1, \ldots, X_N$ from q, the estimator is

$$\widehat{I}_N = \frac{1}{N}\sum_{i=1}^{N}\frac{h(X_i)}{q(X_i)}, \qquad \langle \widehat{I}_N \rangle_q = I. \tag{2.91}$$

The weight $1/q(X_i)$ compensates for sampling different regions with different frequencies. Changing q changes how often a contribution appears, while leaving the expected answer unchanged.

Derivation of the variance. Write $Y_i = h(X_i)/q(X_i)$. If $\int h(x)^2/q(x)\,dx < \infty$, independence gives

$$\begin{aligned}
\mathrm{var}_q(\widehat{I}_N) &= \frac{1}{N^2}\sum_{i,j=1}^{N}\mathrm{cov}_q(Y_i, Y_j) && (2.92)\\
&= \frac{1}{N^2}\sum_{i=1}^{N}\mathrm{var}_q(Y_i) = \frac{1}{N}\mathrm{var}_q\left(\frac{h(X)}{q(X)}\right) && (2.93)\\
&= \frac{1}{N}\left[\left\langle\left(\frac{h(X)}{q(X)}\right)^2\right\rangle_q - \left\langle\frac{h(X)}{q(X)}\right\rangle_q^2\right] && (2.94)\\
&= \frac{1}{N}\left[\int_{\mathcal{D}} dx\, q(x)\frac{h(x)^2}{q(x)^2} - I^2\right] && (2.95)\\
&= \boxed{\frac{1}{N}\left[\int_{\mathcal{D}} dx\,\frac{h(x)^2}{q(x)} - I^2\right]}. && (2.96)
\end{aligned}$$

The square in the final integrand is on $h(x)$; only one power of $q(x)$ remains after multiplication by the sampling density. Equivalently,

$$\mathrm{var}_q(\widehat{I}_N) = \frac{1}{N}\int_{\mathcal{D}} dx\, q(x)\left[\frac{h(x)}{q(x)} - I\right]^2. \tag{2.97}$$

Importance sampling reduces the variance prefactor by making the sampled contributions $h(X)/q(X)$ more uniform. Regions where $|h|$ is appreciable but q is tiny give large weights and can dominate the error. A poor density can therefore increase the variance, or even make it infinite. Merely choosing a nonuniform density does not guarantee an improvement.

For a domain of finite volume $\mathcal{V}_{\mathcal{D}} = \int_{\mathcal{D}} dx$, uniform sampling has $q = 1/\mathcal{V}_{\mathcal{D}}$ and gives

$$\mathrm{var}_{\mathrm{unif}}(\widehat{I}_N) = \frac{1}{N}\left[\mathcal{V}_{\mathcal{D}}\int_{\mathcal{D}} h(x)^2\,dx - I^2\right]. \tag{2.98}$$

Comparing this expression with the one for another q quantifies the change in error at a fixed number of independent samples. For example, reducing the single-sample variance by a factor of ten requires ten times fewer samples for the same variance of the mean, if the sampling costs are equal.

The variance-minimizing density. Since I does not depend on q, minimizing the variance amounts to minimizing $\int h^2/q$. Cauchy–Schwarz gives

$$\left(\int_{\mathcal{D}} |h(x)|\,dx\right)^2 = \left(\int_{\mathcal{D}} \frac{|h(x)|}{\sqrt{q(x)}}\sqrt{q(x)}\,dx\right)^2 \tag{2.99}$$

$$\leq \left(\int_{\mathcal{D}} \frac{h(x)^2}{q(x)}\,dx\right)\left(\int_{\mathcal{D}} q(x)\,dx\right) \tag{2.100}$$

$$= \int_{\mathcal{D}} \frac{h(x)^2}{q(x)}\,dx. \tag{2.101}$$

Equality holds when $|h|/\sqrt{q}$ is proportional to $\sqrt{q}$. If h is not identically zero, the optimum is therefore

$$q_{\mathrm{opt}}(x) = \frac{|h(x)|}{\int_{\mathcal{D}} |h(y)|\,dy}, \qquad \mathrm{var}_{\mathrm{opt}}(\widehat{I}_N) = \frac{1}{N}\left[\left(\int_{\mathcal{D}} |h(x)|\,dx\right)^2 - I^2\right]. \tag{2.102}$$

For an integrand of constant sign, $h(X)/q_{\mathrm{opt}}(X) = I$ and the variance is zero. Obviously it is not possible in general to generate points distributed accordingly to the integrand itself.

For a simple example, take $h(x) = 2x$ on $[0,1]$, so $I = 1$. Uniform sampling gives

$$\mathrm{var}_{q=1}(\widehat{I}_N) = \frac{1}{N}\left[\int_0^1 4x^2\,dx - 1\right] = \frac{1}{3N}. \tag{2.103}$$

Instead, with $q(x) = 2x$, every sampled contribution is $h(X)/q(X) = 1$. The estimator has zero variance. We will see how the zero variance property comes in the calculation of energies in quantum mechanical systems.

2.5.2 Expectation values and unknown normalization: reweighting.

Suppose that we want to compute expectation values for the Boltzmann distribution associated with the energy $E(x)$, but can more easily sample a model with a simpler energy $E_{\mathrm{model}}(x)$. At the same inverse temperature β, the two distributions are

$$\pi(x) = \frac{e^{-\beta E(x)}}{Z}, \qquad q(x) = \frac{e^{-\beta E_{\mathrm{model}}(x)}}{Z_{\mathrm{model}}}. \tag{2.104}$$

We can use configurations sampled from q to estimate expectation values under π, provided we can evaluate $E(x)$ on those configurations. This is called *reweighting*.

Define the relative weight

$$W(x) = e^{-\beta[E(x) - E_{\mathrm{model}}(x)]}. \tag{2.105}$$

Since $q(x)W(x) = e^{-\beta E(x)}/Z_{\mathrm{model}}$, the desired expectation value can be written as

$$\langle f\rangle_\pi = \frac{\int dx\, f(x) e^{-\beta E(x)}}{\int dx\, e^{-\beta E(x)}} \tag{2.106}$$

$$= \frac{\int dx\, q(x) f(x) W(x)}{\int dx\, q(x) W(x)} \tag{2.107}$$

$$= \frac{\langle fW\rangle_q}{\langle W\rangle_q}. \tag{2.108}$$

Both unknown normalization constants cancel. For discrete configurations, replace the integrals by sums.

Given N configurations x_i sampled from q, we therefore estimate

$$\boxed{\widehat{f}_N = \frac{\sum_{i=1}^{N} W(x_i) f(x_i)}{\sum_{i=1}^{N} W(x_i)}}. \tag{2.109}$$

The weights correct the relative frequency with which the model samples different configurations. If $E_{\text{model}} = E$, all weights are one and we recover the ordinary sample average. An additive constant difference between the energies also cancels from the ratio.

The reweighting identity is exact, but its usefulness depends on the overlap between the two distributions. We require $q(x) > 0$ wherever $\pi(x) > 0$. Moreover, configurations important under π must occur sufficiently often in samples from q. Otherwise, a few very large weights dominate the estimate, producing large statistical uncertainty.

The finite-N ratio estimator is generally biased, although it converges to the correct expectation under the usual integrability and sampling assumptions. Its error must account for fluctuations of both numerator and denominator, which are correlated. For Markov-chain samples, a block jackknife can do this: remove the same block of consecutive samples from both sums, recompute the ratio, and estimate its uncertainty from these repeated calculations. Blocks must be long enough to account for the relevant autocorrelations.

2.6 Cluster Updates for Spin Systems

Local-update Monte Carlo methods such as the Metropolis algorithm suffer from *critical slowing down* near second-order phase transitions. In particular, for models like the 2D Ising model close to the critical temperature T_c, spin configurations develop large correlated domains. A local spin-flip move becomes inefficient because it is unlikley to reverse the magnetization of a large cluster by flipping one spin at time, creating many domain walls. The system evolves slowly, leading to large autocorrelation times. A good thing about MC is that the pseudodynamics we follow does not need to be close to the physical dynamics of the system. If we are clever, we can therefore solve critical slow downs.

One great example of this idea are *cluster algorithms*. These non-local[7] update schemes flip entire clusters of spins simultaneously, allowing the simulation to decorrelate much faster, even near T_c. One of the most well-known cluster algorithms is the *Wolff algorithm*, which we describe below.

For the ferromagnetic Ising model with $J > 0$ and zero external field, the Wolff algorithm (Fig 2.3) constructs a cluster of aligned spins starting from a randomly chosen seed spin and grows the cluster by adding neighboring aligned spins with a certain probability. Once the cluster is built, all spins in the cluster are flipped. This method satisfies detailed balance and samples from the correct Boltzmann distribution $\pi(x) \propto e^{-\beta E(x)}$ without needing to know the partition function.

Let x be a spin configuration, and define a new configuration x' by flipping a cluster C:

[7] The algorithm is however local in the sense that the information to proceed constructing the cluster are local.

- Choose a random site i on the lattice as the *seed* of the cluster.
- Initialize the cluster $C = \{i\}$.
- Process each newly added spin i once, examining its neighbors j:
  - If $\sigma_j = \sigma_i$ and $j \notin C$, add j to C with probability $p_{\text{add}} = 1 - e^{-2\beta J}$.
- Continue until every spin in the cluster has been processed. Do not repeatedly retest a rejected bond.
- Flip all spins in C: $\sigma_k \to -\sigma_k$ for all $k \in C$.

The Wolff algorithm preserves detailed balance and is irreducible at finite positive temperature on a finite lattice, since single-spin clusters have nonzero probability. Aperiodicity must also be checked; a lazy step can ensure it in exceptional periodic cases. The acceptance of a cluster flip is always 1 (since the Bolzmann acceptance probability has been transfered into the proposal step), so the update is rejection-free.
The forward and reverse proposal probabilities are generally different. Their ratio compensates the Boltzmann weight ratio, so the probability fluxes are equal:

$$\pi(x)P(x'|x) = \pi(x')P(x|x'). \tag{2.110}$$

Close to criticality this algorithm can be much more efficient than local updates. The improvement is model- and observable-dependent; there is no universal factor L^2. A fair comparison accounts for the number of spins processed or for elapsed computing time. They allow to observe many full-volume spin-reversal along the same Markov Chain, while the single spin-flip trajectory would be trapped in one of the basins (all-up spins + fluctuations, or the opposite).

Explanation. We now demonstrate why the choice of p works in pratice and allow for unit acceptance. Suppose that, at the end of the cluster formation process, there are still n_s links between *same* (s) type nodes that could have been added but they were not. Each of these links could have been accepted with probability p, but has been rejected. Conversely, there are n_d links that could not be added, since they connect *different* (d) species. So the total probability of forming this exact cluster can be written as $T(x \to x')_{\text{interior}} \times T(x \to x')_{\text{boundary}}$, with $T(x \to x')_{\text{boundary}} = (1-p)^{n_s}$. The boundary contribution to energy of the starting configuration x is

$$E_{\text{boundary}}(x) = (-n_s + n_d)J. \tag{2.111}$$

Now, we need to ask what would be the probability of forming the exact same cluster starting from the new configuration x'. Given that the links that are at the interior are still parallel, the probability of accepting them is the same of before, $T(x \to x')_{\text{interior}} = T(x' \to x)_{\text{interior}}$. What differs is the number of links which are now of the same type. By definition, now the link connecting same type spins are the n_d of before, and the ones that cannot be added are n_s. So the probability of not join to the cluster the former is $T(x' \to x)_{\text{boundary}} = (1-p)^{n_d}$,

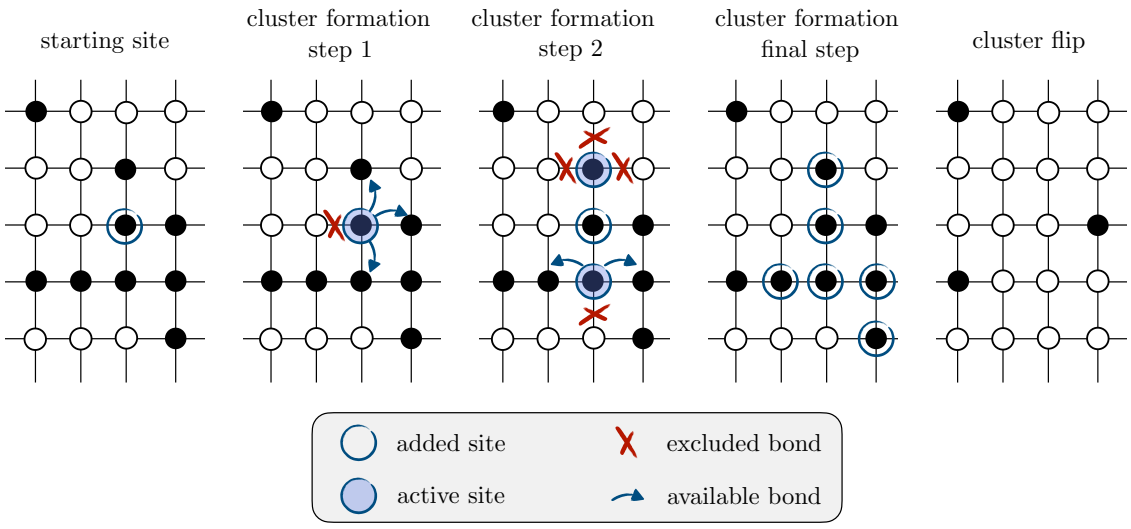


Figure 2.3: Steps of the Wolff algorithm. Notice that at each active site there may be excluded bond to be added (if you try to add to the cluster an opposite spin) or available bonds (linking a parallel spin). Available bonds are added only with a probability. Notice that, at the end of the cluster formation there are left $n_s = 3$ possible links not added due to probability, and $n_d = 8$ links that could not be added (we include here only visible sites). A suitable data structure can simplify the book-keeping of the active sites at each cluster formation step.

while the energy at the boundary of the new configuration is $E_{\text{boundary}}(x') = (-n_d + n_s)J$. We can therefore calculate the acceptance probability that satisfy detailed balance

$$A(x \to x') = \min\left[1,\ \frac{\pi(x')}{\pi(x)}\frac{T(x' \to x)}{T(x \to x')}\right] \tag{2.112}$$

$$= \min\left[1,\ \frac{e^{-\beta J(n_s - n_d)}}{e^{-\beta J(n_d - n_s)}}\frac{(1-p)^{n_d}}{(1-p)^{n_s}}\right] \tag{2.113}$$

$$= \min\left[1,\ \frac{e^{-2\beta J n_s}}{(1-p)^{n_s}}\frac{(1-p)^{n_d}}{e^{-2\beta J n_d}}\right] \tag{2.114}$$

$$= \min\left[1,\ \left(\frac{e^{-2\beta J}}{1-p}\right)^{n_s}\left(\frac{1-p}{e^{-2\beta J}}\right)^{n_d}\right]. \tag{2.115}$$

The acceptance immensele simplifies if we chose precisely $1 - p = e^{-2\beta J}$.

Extension: restricted clusters. In principle, one can restrict the growth of the cluster to a specific subregion of the lattice or to specific directions. For example, in anisotropic systems where the coupling J is much stronger along one direction (e.g., the z-axis) and weaker or disordered in orthogonal directions (e.g., x, y-axis), it may be more efficient to grow clusters only along the strongly interacting direction.

In this modified Wolff algorithm, since the cluster construction does not sample all bonds surrounding the flipped cluster, the update no longer automatically satisfies detailed balance. Therefore, an acceptance step must be reintroduced.
The idea is to compute the energy change ΔE only from the bonds crossing the boundary of the cluster that were not considered during the cluster growth (i.e., those not "tested"). Tested boundary bonds can also change their energy, but their Boltzmann factors are already compensated by the cluster proposal ratio. More precisely, write $E = E_0 + E_1$, where the fixed set of bonds used to construct the cluster defines E_0. The proposal satisfies detailed balance with respect to $e^{-\beta E_0}$, so the remaining acceptance ratio is $e^{-\beta \Delta E_1}$. Arbitrary state-dependent restrictions require a fresh calculation of the proposal ratio.

The acceptance probability is then:

$$A = \min\left(1, e^{-\beta \Delta E_{\text{boundary}}}\right) \tag{2.116}$$

where $\Delta E_{\text{boundary}} = \Delta E_1$ is the change in energy of the excluded interactions (and of any other terms not included in E_0).

This technique is particularly useful when simulating disordered systems. Indeed, in disordered couplings and low temperatures, the clusters still may grow as to include all spins. This is unwanted if the model is disordered, as one never flips single (or a small group of) spins as to accomodate frustrations.

Exercise 2.6.1: Solving the Critical Slowing Down in the 2D Ising Model (*)

Consider the same model of Exercise 2.5.1.

1. Implement the cluster spin-flip proposal $T(x'|x)$ of the Wolff algorithm, with its unit acceptance.
2. Compare the results from Exercise 2.5.1.
3. Compare the autocorrelation times. Plot the magnetization time series as a diagnostic, then estimate its autocorrelation function and integrated autocorrelation time; visual inspection alone is insufficient.
4. Devise a fair way to compare the convergence of the two methods (single vs cluster flips) as a function of the MC iteration. Is the computational speed-up surviving?

2.7 Metropolis Sampling in Continuous Space

In classical statistical mechanics, the canonical ensemble is defined by the Boltzmann distribution:

$$\pi(x) = \frac{1}{Z} e^{-\beta V(x)}, \tag{2.117}$$

where $V(x)$ is the potential energy of the system, $\beta = 1/k_B T$ is the inverse temperature, and $Z = \int dx\, e^{-\beta V(x)}$ is the partition function. Our goal is to generate configurations x distributed according to $\pi(x)$ using a Markov chain with transition kernel $K(x'|x)$ satisfying detailed balance:

$$\pi(x) K(x'|x) = \pi(x') K(x|x'). \tag{2.118}$$

2.7.1 Simple Metropolis Proposals

In continuous spaces, the simplest Metropolis algorithm begins by proposing a new configuration x' via a random displacement from the current position x. For example, one may use:

- **Uniform proposal:** $x' = x + \delta$, with δ drawn uniformly from $[-\epsilon, \epsilon]$.
- **Gaussian proposal:** $x' = x + \eta$, where $\eta \sim \mathcal{N}(0, \sigma^2)$.

The trial distribution $T(x'|x)$ is symmetric in both cases. Here K denotes the full continuous-space transition kernel, while T denotes the proposal, as before. Therefore, the Metropolis acceptance probability reduces to:

$$A(x \to x') = \min\left[1, \frac{\pi(x')}{\pi(x)}\right] = \min\left[1, e^{-\beta[V(x') - V(x)]}\right]. \tag{2.119}$$

The choice of ϵ (for uniform) or σ (for Gaussian) directly affects the acceptance rate. These are hyperparameters tuned to ensure efficient sampling; values that are too small lead to slow exploration, while values that are too large result in low acceptance.

2.7.2 Langevin Dynamics

Another approach uses information about the gradient of the potential. Langevin dynamics introduces a drift term that biases proposals toward lower-energy regions:

$$\dot{x}(t) = f(x) + \eta(t), \tag{2.120}$$

where $f(x) = -\nabla V(x)$ is the deterministic force (with mobility set to one) and $\eta(t)$ is Gaussian white noise with:

$$\langle \eta(t) \rangle = 0, \quad \langle \eta(t)\eta(t') \rangle = 2k_B T \delta(t - t'). \tag{2.121}$$

For a general constant mobility, both the drift and the diffusion coefficient are multiplied by that mobility, as required by the fluctuation–dissipation relation. We keep mobility equal to one here.

To numerically integrate this stochastic differential equation, we discretize time into steps of length Δ. Over one time interval (t_n, t_{n+1}), the increment in position is:

$$x_{n+1} - x_n = \int_{t_n}^{t_{n+1}} dt\, f(x(t)) + \int_{t_n}^{t_{n+1}} dt\, \eta(t). \tag{2.122}$$

The first term is approximated using the Euler method as $\Delta f(x_n)$. For the second term, we use the fact that the integral of white noise over a finite time interval is a Gaussian random variable:

$$\int_{t_n}^{t_{n+1}} dt\, \eta(t) = \sqrt{2\Delta k_B T} z_n, \tag{2.123}$$

where z_n is a standard normal variable: $z_n \sim \mathcal{N}(0, 1)$. Notice that now the amplitude of the random component is proportional to $\sqrt{\Delta}$. To check the variance, use $\langle z_n^2 \rangle = 1$, while $\langle z_n \rangle = 0$, and verify that the above definition is compatible with these moments. Integrals over disjoint time intervals are independent because the noise is Gaussian and white.

$$\langle z_n^2 \rangle = \frac{1}{2\Delta k_B T} \int_{t_n}^{t_{n+1}} dt \int_{t_n}^{t_{n+1}} dt' \langle \eta(t)\eta(t') \rangle = \tag{2.124}$$

$$= \frac{2k_B T}{2\Delta k_B T} \int_{t_n}^{t_{n+1}} dt \int_{t_n}^{t_{n+1}} dt' \delta(t - t') = \frac{1}{\Delta} \int_{t_n}^{t_{n+1}} dt = 1 \tag{2.125}$$

Thus, the full *discretized Langevin* equation becomes:

$$\boxed{x_{n+1} = x_n + \Delta f(x_n) + \sqrt{2\Delta k_B T} z_n} \tag{2.126}$$

Notice that now, the only parameter that controls the sampling is the time-step Δ. This can be seen as a hyperparameter for the proposal. However, as we are going to demonstrate, the continuous-time Langevin process has the Boltzmann distribution as its stationary density, when this density is normalizable and appropriate boundary conditions are imposed. Its Euler discretization generally has a step-size bias, which vanishes as $\Delta \to 0$ under suitable stability and regularity assumptions.

Finite-step bias: a harmonic example. For $V(x) = \frac{1}{2}\omega^2 x^2$, the force is $f(x) = -\omega^2 x$ and the discretized equation becomes

$$x_{n+1} = (1 - \omega^2\Delta)x_n + \sqrt{2\Delta k_B T}\, z_n. \tag{2.127}$$

Its mean tends to zero from an arbitrary initial mean in the stable range $|1 - \omega^2\Delta| < 1$, or $0 < \omega^2\Delta < 2$. Independence of z_n from x_n gives the variance recursion

$$\mathrm{var}(x_{n+1}) = (1 - \omega^2\Delta)^2 \mathrm{var}(x_n) + 2\Delta k_B T. \tag{2.128}$$

At stationarity, denote this variance by σ_Δ^2. Then

$$\sigma_\Delta^2[1 - (1 - \omega^2\Delta)^2] = 2\Delta k_B T, \tag{2.129}$$

$$\sigma_\Delta^2 = \frac{2\Delta k_B T}{2\omega^2\Delta - \omega^4\Delta^2} = \frac{k_B T}{\omega^2(1 - \omega^2\Delta/2)}. \tag{2.130}$$

The exact Boltzmann variance is $k_B T/\omega^2$. Thus the finite-step chain converges to a Gaussian with the wrong width, even after arbitrarily many iterations. Decreasing Δ controls this bias; adding the acceptance step below removes it from the stationary distribution. For forces growing faster than linearly, finite-step stability can also be problematic, so the existence of a continuous-time equilibrium does not by itself guarantee a stable Euler sampler.

Metropolis-Langevin. Let us still consider finite Δ and use the Langevin step as a trial proposal for a Metropolis algorithm. The unadjusted proposal generally does not satisfy detailed balance with respect to the desired Boltzmann density. In d dimensions, write

$$\mathbf{x}' = \mathbf{x} + \Delta\, \mathbf{f}(\mathbf{x}) + \sqrt{2\Delta k_B T}\, \boldsymbol{\eta}, \tag{2.131}$$

where $\mathbf{f}(\mathbf{x}) = -\nabla U(\mathbf{x})$ is the force vector and $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{1})$. Here U denotes the potential energy in several dimensions. The trial density is

$$T(\mathbf{x}'|\mathbf{x}) = \frac{1}{(4\pi\Delta k_B T)^{d/2}} \exp\left[-\frac{\|\mathbf{x}' - \mathbf{x} - \Delta\mathbf{f}(\mathbf{x})\|^2}{4\Delta k_B T}\right]. \tag{2.132}$$

The drift has a plus sign in front of $\Delta\mathbf{f}$, since the force itself is minus the gradient.

To impose detailed balance with $\pi(\mathbf{x}) \propto e^{-U(\mathbf{x})/(k_B T)}$, introduce the Metropolis–Hastings acceptance probability

$$\alpha(\mathbf{x} \to \mathbf{x}') = \min\left\{1, \frac{\pi(\mathbf{x}')T(\mathbf{x}|\mathbf{x}')}{\pi(\mathbf{x})T(\mathbf{x}'|\mathbf{x})}\right\}. \tag{2.133}$$

Substituting the expressions for π and T, we obtain

$$\begin{aligned}\alpha(\mathbf{x} \to \mathbf{x}') = \min\Bigg\{1, \exp\Bigg[&- \frac{U(\mathbf{x}') - U(\mathbf{x})}{k_B T} \\ &+ \frac{1}{4\Delta k_B T}\Big(\|\mathbf{x}' - \mathbf{x} - \Delta\mathbf{f}(\mathbf{x})\|^2 \\ &\qquad - \|\mathbf{x} - \mathbf{x}' - \Delta\mathbf{f}(\mathbf{x}')\|^2\Big)\Bigg]\Bigg\}.\end{aligned} \tag{2.134}$$

In particular, the reverse trial density is in the numerator. Expanding both norms explicitly,

$$\|\mathbf{x}' - \mathbf{x} - \Delta\mathbf{f}(\mathbf{x})\|^2 = \|\mathbf{x}' - \mathbf{x}\|^2 - 2\Delta(\mathbf{x}' - \mathbf{x}) \cdot \mathbf{f}(\mathbf{x}) + \Delta^2\|\mathbf{f}(\mathbf{x})\|^2, \tag{2.135}$$

$$\|\mathbf{x} - \mathbf{x}' - \Delta\mathbf{f}(\mathbf{x}')\|^2 = \|\mathbf{x}' - \mathbf{x}\|^2 + 2\Delta(\mathbf{x}' - \mathbf{x}) \cdot \mathbf{f}(\mathbf{x}') + \Delta^2\|\mathbf{f}(\mathbf{x}')\|^2. \tag{2.136}$$

Subtracting and simplifying gives

$$\begin{aligned}\alpha(\mathbf{x} \to \mathbf{x}') = \min\Bigg\{1, \exp\Bigg[&- \frac{U(\mathbf{x}') - U(\mathbf{x})}{k_B T} \\ &- \frac{(\mathbf{x}' - \mathbf{x}) \cdot [\mathbf{f}(\mathbf{x}) + \mathbf{f}(\mathbf{x}')]}{2k_B T} \\ &+ \frac{\Delta}{4k_B T}\big(\|\mathbf{f}(\mathbf{x})\|^2 - \|\mathbf{f}(\mathbf{x}')\|^2\big)\Bigg]\Bigg\}.\end{aligned} \tag{2.137}$$

As in the original Metropolis algorithm, a rejected proposal leaves the state unchanged and this repeated state contributes to measurements. The full transition kernel includes these rejections. Its stationary density is the desired π at finite Δ; convergence and efficiency still require suitable ergodicity and proposal choices.

For the one-dimensional harmonic potential, substitution of $f(x) = -\omega^2 x$ gives the useful check

$$\alpha(x \to x') = \min\left\{1, \exp\left[-\frac{\omega^4\Delta}{4k_B T}(x'^2 - x^2)\right]\right\}. \tag{2.138}$$

The potential-energy term and the term linear in the force cancel, leaving the finite-step correction shown here.

Exercise 2.7.1: Metropolis vs Langevin sampling

Consider an harmonic classical potential $V = \frac{1}{2}\omega^2 x^2$, at a finite temperature T.

1. Use the Metropolis sampling to evaluate the expectation values of $\langle V\rangle$ and $\langle x\rangle$.
2. Use the Langevin dynamics and check that it converges to the exact values in the limit $\Delta \to 0$.
3. Add the Metropolis–Langevin correction and compare at finite Δ. The exact values are $\langle x\rangle = 0$, $\langle x^2\rangle = k_B T/\omega^2$, and $\langle V\rangle = k_B T/2$. Compare the unadjusted result with the finite-step variance derived above, and use direct Gaussian draws as an independent-sampling benchmark. Distinguish statistical error from discretization bias and compare the methods at equal computational cost. For Langevin dynamics also distinguish time in updates from physical sampling time $t = n\Delta$.

2.7.3 From the Langevin Dynamics to Fokker-Planck equation

The unadjusted discretized Langevin equation induces a stochastic map from x_n to x_{n+1}, which defines a transition kernel $K(x'|x)$ describing the conditional probability of reaching configuration x' starting from x after one time step. Since $z \sim \mathcal{N}(0,1)$,

$$K(x'|x) = \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2}\delta(x' - x - \Delta f(x) - \sqrt{2\Delta k_B T}\, z) \tag{2.139}$$

This kernel represents the probability density of making a transition from x to x' in one time step, by considering the probability of extracting a random displacement z. We can perform (see Appendix B) the integral finding

$$K(x'|x) = \frac{1}{\sqrt{4\pi\Delta k_B T}} \exp\left[-\frac{(x' - x - \Delta f(x))^2}{4\Delta k_B T}\right] \tag{2.140}$$

In Appendix B we further prove that this lead to the Fokker-Planck equation that is a differential equation for the evolution of the probability $\rho(x,t)$. This is useful to prove convergence to the Boltzmann distribution and the draw interesting comparison between quantum mechanics and classical statistical mechanics. The computation is not straightforward and several texts use different methods. Here we choose the simplest way, without use of Fourier transform, or using stochastic calculus. The final result reads,

$$\boxed{\frac{\partial \rho(x,t)}{\partial t} = -\frac{\partial}{\partial x}\left[f(x)\rho(x,t)\right] + k_B T \frac{\partial^2 \rho(x,t)}{\partial x^2}} \tag{2.141}$$

The stationary solution $\rho(x) = \pi(x)$ is easily verified by direct substitution[8]. This verifies stationarity for the continuous-time equation, not exact Boltzmann sampling by the finite-step Euler scheme.

The same statement can be expressed through the probability current:

$$\partial_t \rho(x,t) = -\partial_x J(x,t), \qquad J(x,t) = f(x)\rho(x,t) - k_B T\, \partial_x \rho(x,t). \tag{2.142}$$

For the Boltzmann density, $J = 0$. Decay at infinity or reflecting boundary conditions ensure no loss of probability. On an unbounded domain one also needs $Z < \infty$; for example, free diffusion on the entire real line has no normalizable equilibrium density. Convergence requires additional conditions on the dynamics, and a positive spectral gap gives exponential convergence as discussed next.

2.7.4 From the Fokker-Planck to the Schrödinger Equations

The Fokker-Planck equation can be formally recasted into the Schrödinger equation. We now write $P(x,t) \equiv \rho(x,t)$ and $P_0(x) = \pi(x)$, preserving the notation of the following transformation. This connection is obtained by plugging this ansatz in Eq. 2.141:

$$P(x,t) = \psi_0(x)\, \Phi(x,t) \tag{2.143}$$

where $\psi_0(x) = \sqrt{P_0(x)} = \frac{1}{\sqrt{Z}} e^{-V(x)/2k_B T}$, which, being $\int dx\, \psi_0^2(x) = 1$, is a normalized state of the auxiliary Schrödinger problem.
Indeed, inserting this into the Fokker-Planck equation we obtain an equation for $\Phi(x,t)$ which is an imaginary-time Schrödinger Eq. with an effective Hamiltonian $\mathcal{H}$:

$$\frac{\partial}{\partial t}\Phi(x,t) = -\mathcal{H}\Phi(x,t) = -\left[-k_B T \frac{\partial^2}{\partial x^2} + \mathcal{V}(x)\right]\Phi(x,t) \tag{2.144}$$

[8] Set $\partial_t \rho = 0$ and $\pi(x) = Z^{-1} e^{-V(x)/(k_B T)}$. Then $\partial_x \pi = -V'\pi/(k_B T)$ and $f\pi = -V'\pi = k_B T\, \partial_x \pi$. Thus $-\partial_x(f\pi) = -k_B T\, \partial_x^2 \pi$, which cancels the diffusion term.

where

$$\mathcal{V}(x) = \frac{1}{4k_BT}\left(\frac{\partial V(x)}{\partial x}\right)^2 - \frac{1}{2}\frac{\partial^2 V(x)}{\partial x^2}. \tag{2.145}$$

The cancellation of first-derivative terms can be seen explicitly. Since

$$\frac{\psi_0'}{\psi_0} = -\frac{V'}{2k_BT}, \qquad \frac{\psi_0''}{\psi_0} = \frac{(V')^2}{4(k_BT)^2} - \frac{V''}{2k_BT}, \tag{2.146}$$

substitution of $P = \psi_0\Phi$ into $\partial_t P = \partial_x(V'P) + k_BT\,\partial_x^2 P$ gives

$$\psi_0\partial_t\Phi = V''\psi_0\Phi + V'\psi_0'\Phi + V'\psi_0\Phi' + k_BT\big(\psi_0''\Phi + 2\psi_0'\Phi' + \psi_0\Phi''\big), \tag{2.147}$$

$$\partial_t\Phi = k_BT\Phi'' + \left(V' + 2k_BT\frac{\psi_0'}{\psi_0}\right)\Phi' + \left(V'' + V'\frac{\psi_0'}{\psi_0} + k_BT\frac{\psi_0''}{\psi_0}\right)\Phi \tag{2.148}$$

$$= k_BT\Phi'' + \left[\frac{1}{2}V'' - \frac{(V')^2}{4k_BT}\right]\Phi. \tag{2.149}$$

The coefficient of Φ' vanishes, leaving precisely the auxiliary Hamiltonian above.

The extrema of the effective potential need not coincide with those of $V(x)$. Indeed,

$$\mathcal{V}'(x) = \frac{V'(x)V''(x)}{2k_BT} - \frac{1}{2}V'''(x), \tag{2.150}$$

so $V'(x) = 0$ does not generally imply $\mathcal{V}'(x) = 0$. The ground state of this effective Hamiltonian can be found exactly and is simply $\psi_0(x)$. Moreover, the corresponding ground state energy is $E_0 = 0$ (notice that the effective hamiltonian spectrum is not defined up to a constant)[9].

The positivity of this auxiliary Hamiltonian can be checked directly. Define

$$Q = \sqrt{k_BT}\,\partial_x + \frac{V'(x)}{2\sqrt{k_BT}}, \qquad Q^\dagger = -\sqrt{k_BT}\,\partial_x + \frac{V'(x)}{2\sqrt{k_BT}}. \tag{2.151}$$

With decay at infinity, or boundary conditions for which the integration-by-parts boundary term vanishes,

$$Q^\dagger Q = -k_BT\,\partial_x^2 + \frac{V'(x)^2}{4k_BT} - \frac{1}{2}V''(x) = \mathcal{H}, \tag{2.152}$$

$$\langle\Phi|\mathcal{H}|\Phi\rangle = \langle Q\Phi|Q\Phi\rangle \geq 0, \tag{2.153}$$

$$Q\psi_0 = \left[\sqrt{k_BT}\left(-\frac{V'}{2k_BT}\right) + \frac{V'}{2\sqrt{k_BT}}\right]\psi_0 = 0. \tag{2.154}$$

On a finite interval, the operator domains must respect the transformed boundary conditions: a reflecting endpoint has $Q\Phi = 0$, which is equivalent to zero probability current. Hence $E_0 = 0$ and all eigenvalues are nonnegative. This operator generates relaxation in imaginary time; its eigenvalues have units of inverse time in the mobility convention used here. It is the

[9] If you use $V(x) = \omega^2x^2/2$ then you will notice that the second term in $\mathcal{V}(x)$, the 'hessian' term, will provide a negative shift that will cancel out the zero-point-motion energy that a 'normal' quantum harmonic oscillator would have.

continuous-space counterpart of the symmetric transformation of P in Section 2.2.3; these are auxiliary relaxation eigenvalues, not the physical quantum energy levels of the original potential.

Therefore, the full solution of the Schrödinger equation, and thus of the corresponding Fokker-Planck equation, has a formal spectral representation, once the eigenfunctions and eigenvalues are known. For a discrete spectrum with a complete real orthonormal eigenbasis, expand the initial condition as:

$$\frac{P(x,t=0)}{\psi_0(x)} = \Phi(x,t=0) = \sum_n a_n \psi_n(x), \tag{2.155}$$

with coefficients

$$a_n = \int dx\, \psi_n(x) \frac{P(x,t=0)}{\psi_0(x)}, \tag{2.156}$$

implying $a_0 = 1$ from the normalization condition on $P(x,t=0)$.
The time evolution of the probability distribution is then given by:

$$P(x,t) = \sum_n a_n \psi_n(x) \psi_0(x) e^{-E_n t}, \tag{2.157}$$

For an initial condition admitting this expansion and a unique stationary state separated from the rest of the spectrum by a positive gap, the nonstationary modes decay exponentially and $P(x,t)$ tends to $\psi_0^2(x)$. If there is a continuous part of the spectrum, the corresponding spectral integrals must be included; a positive gap is not automatic.

The slowest exponential relaxation time τ is given by the inverse of the gap of this continuous-time operator:

$$\tau = \frac{1}{E_1 - E_0} = \frac{1}{E_1}. \tag{2.158}$$

An initial distribution or observable that has no overlap with the first excited mode can relax faster. The integrated autocorrelation time of a particular observable need not equal $1/E_1$.

This justifies continuous-time Langevin sampling after equilibration, under the stated assumptions. In an actual discretized simulation one must additionally control the step-size bias or use the Metropolis correction; running longer alone cannot remove that bias. This pathway can be further explored to establish more connection between classical statistical mechanics and quantum formalism, but we will not go further here.

More explicitly, for a real observable $A(x)$ in the stationary continuous process, define

$$b_n = \int dx\, \psi_n(x) \psi_0(x) \big[A(x) - \langle A \rangle_\pi\big]. \tag{2.159}$$

Here $A(x)$ denotes the measured observable, so $f(x)$ continues to denote the force. The spectral expansion gives

$$C_A(t) = \langle A(x(0)) A(x(t)) \rangle_\pi - \langle A \rangle_\pi^2 = \sum_{n \geq 1} b_n^2 e^{-E_n t}. \tag{2.160}$$

For nonzero variance and a positive gap, integration yields

$$\tau_{\mathrm{int},A}^{\mathrm{cont}} = \int_0^\infty \frac{C_A(t)}{C_A(0)}\,dt = \sum_{n\geq 1} \frac{b_n^2}{C_A(0)}\frac{1}{E_n} \leq \frac{1}{E_1}. \tag{2.161}$$

Equality holds when the observable couples only to the slowest nonstationary eigenspace. This is the continuous-time counterpart of the observable-dependent integrated time derived for a discrete Markov chain.

Harmonic example: gap and observable dependence. For the same potential $V(x) = \omega^2 x^2/2$, the effective Hamiltonian is

$$\mathcal{H} = -k_B T\,\partial_x^2 + \frac{\omega^4 x^2}{4k_B T} - \frac{\omega^2}{2}. \tag{2.162}$$

Its spectrum is $E_n = n\omega^2$, so the slowest relaxation time is $1/\omega^2$. The stationary Langevin process has

$$C_x(t) = \langle x(0)x(t)\rangle = \frac{k_B T}{\omega^2} e^{-\omega^2 t}. \tag{2.163}$$

Since the joint distribution at two times is Gaussian, its fourth moment gives

$$\langle x(0)^2 x(t)^2\rangle = \langle x^2\rangle^2 + 2\langle x(0)x(t)\rangle^2, \tag{2.164}$$

$$C_{x^2}(t) = \langle x(0)^2 x(t)^2\rangle - \langle x^2\rangle^2 = 2\left(\frac{k_B T}{\omega^2}\right)^2 e^{-2\omega^2 t}. \tag{2.165}$$

Thus x sees the first excited mode, whereas the centered observable $x^2 - \langle x^2\rangle$ sees the second. Their continuous-time integrated autocorrelation times are

$$\tau_{\mathrm{int},x}^{\mathrm{cont}} = \int_0^\infty \frac{C_x(t)}{C_x(0)}\,dt = \frac{1}{\omega^2}, \qquad \tau_{\mathrm{int},x^2}^{\mathrm{cont}} = \frac{1}{2\omega^2}. \tag{2.166}$$

For a single exponential mode with rate $E > 0$, measurements of the continuous process spaced by a time Δ instead have

$$\tau_{\mathrm{int}}^{\mathrm{steps}} = \frac{1}{2} + \sum_{n=1}^{\infty} e^{-En\Delta} = \frac{1 + e^{-E\Delta}}{2(1 - e^{-E\Delta})}, \qquad \Delta\tau_{\mathrm{int}}^{\mathrm{steps}} \longrightarrow \frac{1}{E}. \tag{2.167}$$

2.8 Exponential Divergence of Autocorrelation Time in Disordered Systems

In many disordered or frustrated statistical systems, Markov Chain Monte Carlo simulations suffer from extremely long autocorrelation times (cfn. Sect 2.2.3), especially at low temperatures. When the free-energy landscape contains many deep local minima separated by barriers, the system becomes trapped in metastable states. For local dynamics dominated by activated barrier crossing, a common estimate is $\tau \sim e^{\Delta F/(k_B T)}$. If the relevant free-energy barrier grows as L^θ, this gives the conditional scaling:

$$\tau \sim \exp(\Delta F/k_B T) \sim \exp\Big(cL^\theta\Big),$$

where L is a size parameter, c depends on temperature and the model, and $\theta > 0$ is a barrier exponent. This is a physical scaling estimate, not a universal law for all disordered models or sampling algorithms.

A well-studied example is the two-dimensional $\pm J$ Ising spin glass, with Hamiltonian

$$E(\{s_i\}) = -\sum_{\langle i,j \rangle} J_{ij} s_i s_j, \quad s_i = \pm 1, \quad J_{ij} \in \{-1, +1\} \text{ randomly assigned.} \tag{2.168}$$

At low temperatures, the system has a highly degenerate and rugged energy landscape, leading to a proliferation of metastable states. The transition between these configurations becomes extremely slow with increasing system size, and local MCMC methods like Metropolis sampling are quickly rendered ineffective. The Houdayer cluster move was introduced in 2001 and is particularly effective for certain two-dimensional spin glasses. Its usefulness depends on the geometry and percolation of the clusters; variants have also been applied in higher dimensions.[10]

Another archetypal case is the Sherrington-Kirkpatrick (SK) model, a fully connected spin glass described by

$$E(\{s_i\}) = -\sum_{i<j} J_{ij} s_i s_j, \quad J_{ij} \sim \mathcal{N}(0, 1/L). \tag{2.169}$$

This model also exhibits a glassy phase with exponentially many low-energy states. Below the spin-glass transition temperature, local dynamics can become extremely slow. Numerical studies of equilibrium relaxation in this model find a typical logarithmic time scale growing roughly as $L^{1/3}$ in the studied low-temperature regime, with strong disorder dependence; $\tau \sim e^{cL}$ is not a general result. Here L is the number of spins.[11]

Similar behavior is observed in random constraint satisfaction problems, such as random k-SAT and graph coloring near the satisfiability threshold. In these models, the solution space fractures into clusters separated by large Hamming distances. Local algorithms can struggle to navigate between such clusters, and exponential equilibration times can occur in appropriate regimes. The conclusion depends on the ensemble, parameters and algorithm.

These difficulties are not limited to discrete spin systems. In continuous-space models, such as those describing classical particles in disordered media or structural glasses, similar challenges arise. For example, in a Lennard-Jones mixture used to model glass-forming liquids, particles experience an amorphous potential landscape with deep wells and barriers.

These challenges have motivated the development of enhanced sampling techniques such as umbrella sampling, parallel tempering (replica exchange), metadynamics, and path sampling methods. While such techniques can mitigate the severity of critical slowing down or glassy dynamics, they do not provide a general guarantee of efficient equilibration in deeply frustrated or strongly first-order systems.

[10] J. Houdayer, *A Cluster Monte Carlo Algorithm for 2-Dimensional Spin Glasses*, `https://arxiv.org/abs/cond-mat/0101116`; Z. Zhu, A. J. Ochoa and H. G. Katzgraber, *Efficient Cluster Algorithm for Spin Glasses in Any Space Dimension*, `https://arxiv.org/abs/1501.05630`.

[11] A. Billoire, *Distribution of time scales in the Sherrington–Kirkpatrick model*, `https://arxiv.org/abs/1003.6086`.

Exercise 2.8.1: Mixing times of Metropolis for the SK model (*)

Consider the SK model of E 2.169. The couplings have standard deviation $1/\sqrt{L}$. This normalization keeps the variance of the random local field finite and gives an extensive equilibrium energy scale in the *all-to-all* model. Set $k_B = 1$. The thermodynamic critical temperature for this coupling normalization is $T_c = 1$; study temperatures both above and below it, using several disorder realizations.

- Construct the full P_{ij} matrix on the 2^L configurations for a local spin-flip proposal: $P_{ij} = T_{ij}A_{ij}$ for $i \neq j$, with the diagonal fixed by row normalization. Here i, j label configurations, not lattice sites.
- Diagonalize it and find the gap δ.
- Perform a scaling analysis for system sizes between about 4 to 8.
- Investigate how the gap varies with L and with the disorder realization. These small sizes do not establish an asymptotic exponential law; do not assume one in the fit.
- Keep the system's size fixed, e.g. $L = 6$ and now vary the temperature. How does the plot looks like?
- (Bonus) Repeat and compared with a different choice of the transition matrix T_{ij}, the uniform distribution, namely, choosing a full configuration j uniformly, irrespective of i. Compare its error per unit computational cost with local updates, particularly at high temperatures. At infinite temperature this proposal produces independent uniform configurations.

Chapter 3

Variational Monte Carlo

In this chapter we show how to compute the energy of correlated trial states, such as those presented in Sect. 1.4, and optimize their parameters. We then connect the same wavefunction representations to learning from measurement data.

3.1 Local Energy and Zero Variance Property

Variational Monte Carlo (VMC) is a stochastic method for computing properties of a trial[1] wavefunction $\psi(x)$, where x denotes a configuration in Hilbert space. The configuration can represent a spin state on a lattice or a set of continuous particle coordinates. Optimizing the trial state makes VMC a method for approximating ground-state properties.

Throughout this chapter, ψ may be complex and need not be normalized. The Hamiltonian $\hat{H}$ is Hermitian. We use real coordinates $\boldsymbol{\theta}$ for the variational parameters: if a network weight is complex, its real and imaginary parts are two independent real coordinates. Particular exercises may use real trial states as special cases.

The central quantity is the variational energy,

$$E_{\mathrm{var}} = \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle}, \tag{3.1}$$

which is real and provides an upper bound to the ground-state energy. Let $\mathcal{N} = \langle\psi|\psi\rangle$. Inserting a resolution of the identity gives, explicitly for a complex wavefunction,

$$E_{\mathrm{var}} = \frac{1}{\mathcal{N}} \sum_x \langle\psi|x\rangle\langle x|\hat{H}|\psi\rangle \tag{3.2}$$

$$= \frac{1}{\mathcal{N}} \sum_x \psi^*(x)(\hat{H}\psi)(x) \tag{3.3}$$

$$= \sum_x \frac{|\psi(x)|^2}{\mathcal{N}} \frac{(\hat{H}\psi)(x)}{\psi(x)} \equiv \sum_x P(x) E_L(x). \tag{3.4}$$

Here we define

$$P(x) = \frac{|\psi(x)|^2}{\mathcal{N}}, \qquad E_L(x) = \frac{\langle x|\hat{H}|\psi\rangle}{\psi(x)}. \tag{3.5}$$

[1] Previously also denoted as ψ_T.

The local energy is defined where $\psi(x) \neq 0$. At configurations of zero probability it need not be assigned a value. Sums are replaced by integrals in continuous space.

Although $E_L(x)$ can be complex for an individual configuration, Hermiticity implies

$$\langle E_L \rangle_P = E_{\text{var}} \in \mathbb{R}, \qquad \langle \operatorname{Im} E_L \rangle_P = 0. \tag{3.6}$$

Thus we estimate the real energy by

$$\boxed{\widehat{E}_{\text{var}} = \frac{1}{N} \sum_{n=1}^{N} \operatorname{Re} E_L(x_n), \qquad x_n \sim P.} \tag{3.7}$$

The sampled imaginary part should approach zero within statistical uncertainty. As in Chapter 2, samples from a Markov chain are correlated, and error bars must account for their autocorrelation. The variance and autocorrelation relevant to the error bar on this estimator are those of $\operatorname{Re} E_L$. Metropolis sampling only requires ratios of $|\psi|^2$, so the normalization $\mathcal{N}$ is not needed.

A particularly important property is the *zero variance principle*. If $\hat{H}\psi = E\psi$, then, on the sampled support,

$$E_L(x) = \frac{\langle x|\hat{H}|\psi\rangle}{\langle x|\psi\rangle} = \frac{E\langle x|\psi\rangle}{\langle x|\psi\rangle} = E. \tag{3.8}$$

The local energy is then independent of the configuration and its variance vanishes.

For a complex random variable, the nonnegative variance uses an absolute square:

$$\operatorname{Var}_P(E_L) = \left\langle |E_L - E_{\text{var}}|^2 \right\rangle_P = \langle |E_L|^2 \rangle_P - E_{\text{var}}^2. \tag{3.9}$$

Assume first that $\psi(x)$ is nonzero throughout the Hilbert-space sector being sampled. Then

$$\langle |E_L|^2 \rangle_P = \sum_x \frac{|\psi(x)|^2}{\mathcal{N}} \frac{|\langle x|\hat{H}|\psi\rangle|^2}{|\psi(x)|^2} \tag{3.10}$$

$$= \frac{1}{\mathcal{N}} \sum_x \langle \psi|\hat{H}^\dagger|x\rangle\langle x|\hat{H}|\psi\rangle \tag{3.11}$$

$$= \frac{\langle \psi|\hat{H}^\dagger\hat{H}|\psi\rangle}{\mathcal{N}} = \frac{\langle \psi|\hat{H}^2|\psi\rangle}{\mathcal{N}}. \tag{3.12}$$

Consequently,

$$\operatorname{Var}_P(E_L) = \frac{\langle \psi|\hat{H}^2|\psi\rangle}{\mathcal{N}} - \left(\frac{\langle \psi|\hat{H}|\psi\rangle}{\mathcal{N}} \right)^2 \tag{3.13}$$

$$= \frac{\langle \psi|(\hat{H} - E_{\text{var}})^2|\psi\rangle}{\mathcal{N}} = \frac{\|(\hat{H} - E_{\text{var}})|\psi\rangle\|^2}{\mathcal{N}}. \tag{3.14}$$

This is the quantum variance of the Hamiltonian. Notice that $\langle E_L^2 \rangle$ cannot replace $\langle |E_L|^2 \rangle$ for a general complex wavefunction. Also,

$$\operatorname{Var}_P(E_L) = \operatorname{Var}_P(\operatorname{Re} E_L) + \operatorname{Var}_P(\operatorname{Im} E_L), \tag{3.15}$$

so this quantum variance is not generally the variance entering the error bar of the real-energy estimator.

Under these assumptions, zero variance is equivalent to being an eigenstate, which need not be the ground state.

3.1.1 Lattice Models

In lattice systems, such as quantum spin models or lattice fermion models, the Hilbert space is discrete. A configuration is denoted by $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_L)$, representing the spin states at each site of the system; fermionic configurations instead specify occupations of the spin orbitals.
The Hamiltonian $\hat{H}$ in these models usually contains terms that act locally or on small clusters of neighboring spins. In this setting, the local energy can be evaluated as a sum over configurations $\boldsymbol{\sigma}'$ that are connected to the current configuration $\boldsymbol{\sigma}$ by non-zero matrix elements of the Hamiltonian:

$$E_L(\boldsymbol{\sigma}) = \sum_{\boldsymbol{\sigma}'} \frac{\langle \boldsymbol{\sigma} | \hat{H} | \boldsymbol{\sigma}' \rangle \psi(\boldsymbol{\sigma}')}{\psi(\boldsymbol{\sigma})}. \tag{3.16}$$

Because most Hamiltonians of interest are local, the number of connected configurations grows polynomially, often linearly, with system size. Evaluation is efficient provided that the needed wavefunction ratios are also efficient to compute. Notice that this way, we can make use of the definition of the Hamiltonian in terms of its matrix elements.
More generally, the same strategy applies to the computation of the expectation value of any operator $\hat{A}$. Defining the corresponding *local estimator* as

$$A_L(\boldsymbol{\sigma}) = \sum_{\boldsymbol{\sigma}'} \frac{\langle \boldsymbol{\sigma} | \hat{A} | \boldsymbol{\sigma}' \rangle \psi(\boldsymbol{\sigma}')}{\psi(\boldsymbol{\sigma})}, \tag{3.17}$$

one can estimate $\langle \hat{A} \rangle$ as the average of $A_L(\boldsymbol{\sigma})$ over the sampled configurations. No complex conjugation is inserted in the wavefunction ratio. For Hermitian A, the expectation is real, so the real part of the sampled estimator can be used. This generalization is crucial for measuring observables other than the energy.

Exercise 3.1.1: Hopping particle on a lattice

Consider a spinless particle that can hop on a lattice made of L sites. The hopping amplitude is parametrized by $t > 0$, and there is also a linear potential. This system is a lattice discretized version of a particle. The Hamiltonian is

$$H = -t \sum_{i=1}^{L-1} (c_{i+1}^\dagger c_i + h.c.) + V \sum_{i=1}^{L} (i-1) c_i^\dagger c_i. \tag{3.18}$$

The particle cannot move outside the range $[1, L]$. Consider a trial function of the form

$$\psi_T(i) \propto e^{-gi}, \tag{3.19}$$

and system parameters $V/|t| = 10$, $L = 30$.

1. Write the local energy.
2. Write a VMC code that calculates the expectation value of the energy at fixed parameter g.
3. Find the optimal value of g with simple grid search.

Exercise 3.1.2: Heisenberg model in 1D (*)

Consider the one-dimensional Heisenberg model with an even number L of sites:

$$H = J \sum_{i=1}^{L} \vec{\sigma}_i \cdot \vec{\sigma}_{i+1} \tag{3.20}$$

with $J > 0$ and periodic boundary conditions (PBC) $\vec{\sigma}_{i+L} = \vec{\sigma}_i$. A variational wavefunction for the ground state of H can be defined in the basis of configurations $\{s_i^z, \cdots, s_L^z\}$, where each site i has a definite value of $s_i^z = \pm 1$, and the total spin projection on the z-axis vanishes: $\sum_i s_i^z = 0$. In this basis, the wavefunction can be written as:

$$\psi(x) = \psi(s_1^z, \ldots, s_L^z) = \mathrm{Sign}(x) \times \exp\left(\alpha \sum_{i \neq j} v_{ij}\, s_i^z s_j^z\right) \tag{3.21}$$

where: α is a variational parameter, $v_{ij} = \ln\left(d_{ij}^2\right)$ is the spin Jastrow factor, which depends on the chord distance d_{ij} between two sites, defined as:

$$d_{ij} = 2 \sin\left(\frac{\pi |i-j|}{L}\right),$$

$\mathrm{Sign}(s_i^z, \cdots, s_L^z) = (-1)^P$, where $P = \sum_{i=1}^{L/2} \left(s_{2i}^z + 1\right)/2$ is the so-called Marshall sign, determined by the number of up spins on even sites.

1. Write down the matrix element of the Hamiltonian and the local energy.
2. Write a code that performs a VMC for a system of $L = 100$.
3. Find the best variational parameter α.
4. The exact energy per site in the infinite-chain limit is $e_0 = J(1 - 4\log 2) \simeq -4J \times 0.44314718$, since the Hamiltonian uses Pauli matrices. Perform a finite-size convergence study of E_{var}/L. Separate the finite-size dependence from the remaining variational error; the optimized ansatz need not attain the exact energy.

A fermionic example: the Hubbard model. For lattice fermions, a configuration x specifies occupations $n_{i\sigma} \in \{0, 1\}$, rather than one spin per site. Consider

$$H = -t \sum_{\langle ij \rangle, \sigma} \left(c_{i\sigma}^\dagger c_{j\sigma} + c_{j\sigma}^\dagger c_{i\sigma}\right) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \qquad t > 0. \tag{3.22}$$

Each undirected bond is counted once. Work at fixed $N_\uparrow, N_\downarrow$ and choose a fixed order of spin orbitals in the occupation basis. One convenient order lists all up-spin orbitals and then all down-spin orbitals. Fermionic matrix elements must use this same order throughout.

A simple Slater–Jastrow, or Gutzwiller, ansatz is

$$\psi_g(x) = e^{-gD(x)} \mathcal{D}_\uparrow(x) \mathcal{D}_\downarrow(x), \qquad D(x) = \sum_i n_{i\uparrow} n_{i\downarrow}, \tag{3.23}$$

where g is real and the determinants may be complex. Positive g suppresses double occupancy. The diagonal part of the local energy is $UD(x)$. The off-diagonal part involves every allowed single-particle hop:

$$E_L(x) = UD(x) - t \sum_{x' \text{ connected to } x} \eta(x,x') \frac{\psi_g(x')}{\psi_g(x)}. \tag{3.24}$$

The sum is over hopping processes, and $\eta = \pm 1$ is the fermionic sign. For a hop between occupied and empty spin orbitals with indices p and q, it is $(-1)^{N_{\text{between}}}$, where N_{between} counts occupied spin orbitals strictly between them in the chosen ordering. Moreover,

$$\frac{\psi_g(x')}{\psi_g(x)} = e^{-g[D(x')-D(x)]} \frac{\mathcal{D}_\uparrow(x')}{\mathcal{D}_\uparrow(x)} \frac{\mathcal{D}_\downarrow(x')}{\mathcal{D}_\downarrow(x)}, \qquad O_g(x) = \partial_g \log \psi_g(x) = -D(x). \tag{3.25}$$

Only the determinant of the hopping spin species changes. A symmetric particle-number-conserving proposal can select a bond and a spin uniformly and reject invalid hops; choosing uniformly only among currently allowed hops generally requires a Hastings proposal ratio.

The half-filled Hubbard dimer. For two sites joined by one bond, with $N_\uparrow = N_\downarrow = 1$, use the bonding orbital for both determinants. In the singlet subspace define

$$|S\rangle = \frac{c_{1\uparrow}^\dagger c_{2\downarrow}^\dagger + c_{2\uparrow}^\dagger c_{1\downarrow}^\dagger}{\sqrt{2}} |0\rangle, \tag{3.26}$$

$$|D_+\rangle = \frac{c_{1\uparrow}^\dagger c_{1\downarrow}^\dagger + c_{2\uparrow}^\dagger c_{2\downarrow}^\dagger}{\sqrt{2}} |0\rangle. \tag{3.27}$$

The plus sign in $|S\rangle$ uses the displayed creation-operator ordering; in site-ordered spin notation this is the usual spin singlet. The Hamiltonian and normalized trial state are

$$H\big|_{\{S,D_+\}} = \begin{pmatrix} 0 & -2t \\ -2t & U \end{pmatrix}, \qquad |\psi_g\rangle = \frac{|S\rangle + e^{-g}|D_+\rangle}{\sqrt{1+e^{-2g}}}. \tag{3.28}$$

It follows directly that

$$E(g) = \frac{Ue^{-2g} - 4te^{-g}}{1+e^{-2g}}. \tag{3.29}$$

For $U \geq 0$, minimization gives

$$e^{-g_*} = \frac{\sqrt{U^2+16t^2} - U}{4t}, \qquad E(g_*) = \frac{U - \sqrt{U^2+16t^2}}{2} = E_0. \tag{3.30}$$

This variational family contains the exact dimer ground state. The example therefore connects double occupancy, energy optimization, and the zero variance property.

Exercise 3.1.3: VMC for the Hubbard model (**)

1. Implement the dimer at fixed $N_\uparrow = N_\downarrow = 1$, including fermionic signs. Compute the local energies for singly occupied and doubly occupied configurations.
2. Sample $|\psi_g|^2$, compare the measured energy with $E(g)$, and verify that the local-energy variance vanishes at g_*. Try $U/t = 0, 1, 4, 10$.
3. After studying the gradient formulas below, optimize g using $O_g = -D$ and compare with g_*.
4. Optional: extend the calculation to a small chain and compare with exact diagonalization in the same particle-number sector. The one-parameter ansatz is no longer generally exact. State the boundary conditions and check any determinant zeros and sampling restrictions explicitly.

3.1.2 Continuous Systems

In systems with particles in continuous space, the Hilbert space is infinite-dimensional, and configurations are denoted by $\boldsymbol{x} = (\vec{x}_1, \ldots, \vec{x}_L)$, where $\vec{x}_i \in \mathbb{R}^d$ is the position of particle i.

A typical Hamiltonian takes the form:

$$\hat{H} = -\sum_{i=1}^{L} \frac{\hbar^2}{2m} \nabla_i^2 + V(\boldsymbol{x}), \tag{3.31}$$

where $V(\boldsymbol{x})$ is the potential energy, which may include both external and interaction terms.

The corresponding local energy reads:

$$E_L(\boldsymbol{x}) = -\sum_{i=1}^{L} \frac{\hbar^2}{2m} \frac{\nabla_i^2 \psi(\boldsymbol{x})}{\psi(\boldsymbol{x})} + V(\boldsymbol{x}). \tag{3.32}$$

This expression involves second derivatives of the wavefunction with respect to spatial coordinates, which can be cumbersome and error-prone to compute analytically. In modern implementations, these derivatives are often evaluated using *automatic differentiation*, which provides a flexible and accurate way to obtain gradients and Laplacians, especially when the wavefunction is expressed as a computational graph (e.g., using neural networks).

The same expression applies to complex ψ. Away from nodes, introduce a local function $\ell(\boldsymbol{x}) = \log \psi(\boldsymbol{x})$. The useful identity

$$\frac{\nabla_i^2 \psi}{\psi} = \nabla_i^2 \ell + (\nabla_i \ell) \cdot (\nabla_i \ell) \tag{3.33}$$

contains a dot product *without complex conjugation*, not $|\nabla_i \ell|^2$. Thus

$$E_L = -\sum_i \frac{\hbar^2}{2m} \left[\nabla_i^2 \ell + (\nabla_i \ell) \cdot (\nabla_i \ell) \right] + V. \tag{3.34}$$

For a real potential, writing $\psi = e^{A+i\Phi}$ locally, with $A = \log|\psi|$ and Φ real, makes the amplitude and phase contributions explicit:

$$\mathrm{Re}\, E_L = -\sum_i \frac{\hbar^2}{2m}\left[\nabla_i^2 A + |\nabla_i A|^2 - |\nabla_i \Phi|^2\right] + V, \tag{3.35}$$

$$\mathrm{Im}\, E_L = -\sum_i \frac{\hbar^2}{2m}\left[\nabla_i^2 \Phi + 2\nabla_i A \cdot \nabla_i \Phi\right]. \tag{3.36}$$

Exercise 3.1.4: Local energy of a harmonic oscillator

Consider the following Hamiltonian

$$H = -\frac{d^2}{dx^2} + x^2, \tag{3.37}$$

which has ground state energy of $E_0 = 1$, and ground state wf, $\psi_0(x) = \frac{1}{\pi^{1/4}} e^{-x^2/2}$. Suppose that we don't know this but guess simply the functional form of normalized the ground-state,

$$\psi_T(x) = \left(\frac{\theta}{\pi}\right)^{1/4} e^{-\theta x^2/2}, \qquad \theta > 0. \tag{3.38}$$

1. Write analytically the local energy $E_L(x;\theta)$.
2. Show that the local energy is constant when the ψ_T becomes the exact ground state.
3. (bonus) Write a VMC Metropolis code and numerically check that the variance of the energy decreases as $\theta \to 1$.

3.1.3 Electronic systems: Jastrow and Cusp Conditions

Let us consider a possibly complex trial wavefunction $\psi(\boldsymbol{x})$, where a configuration is defined as $\boldsymbol{x} = (\boldsymbol{r}_1, \ldots, \boldsymbol{r}_L)$, with $\boldsymbol{r}_i \in \mathbb{R}^3$ representing the spatial coordinate of particle i. In this subsection we use atomic units and three-dimensional Coulomb interactions, with fixed nuclei. To distinguish spatial and spin degrees of freedom, we explicitly write the configuration of a single particle as $x = (\boldsymbol{r}, \sigma)$. A widely used class of variational wavefunctions takes the form of a product between a symmetric correlation factor, known as the Jastrow factor, and a symmetrization or antisymmetrization component. For fermionic systems, the latter is typically a Slater determinant ensuring antisymmetry under particle exchange:

$$\psi(\boldsymbol{x}) = \mathcal{D}(\boldsymbol{x}) \times \exp\left(-\sum_{i<j} u(r_{ij})\right), \tag{3.39}$$

where $\mathcal{D}(\boldsymbol{x})$ denotes a Slater determinant and $u(r_{ij})$ is a two-body correlation function, often chosen to include variational parameters. A popular choice for $u(r)$ is the Padé form, which captures short-range correlations effectively. This structure is also suitable for bosonic systems, such as liquid ^{4}He, by replacing the Slater determinant with a fully symmetric function (e.g., a constant or a permanent).

In fermionic systems with equal numbers of spin-up and spin-down electrons, the Slater determinant is often decomposed into a product of two determinants, one for each spin species. That is, the Slater-Jastrow wavefunction is written as

$$\psi(\boldsymbol{x}) = \det\left[\phi_k^{\uparrow}(\boldsymbol{r}_i)\right] \cdot \det\left[\phi_k^{\downarrow}(\boldsymbol{r}_j)\right] \cdot e^{-U(\boldsymbol{x})}, \tag{3.40}$$

where the orbitals ϕ_k^{σ} may be complex and are spin-dependent, with $1 \leq i \leq L/2$ and $L/2+1 \leq j \leq L$. This product is the fixed-spin component of the antisymmetric state constructed from spin orbitals; the electrons remain identical fermions. Unequal populations simply give determinants of sizes $N_{\uparrow}$ and $N_{\downarrow}$. $U(\boldsymbol{r})$ is the total Jastrow exponent.

The *local energy* is constant for an exact eigenfunction, with the converse subject to the support and regularity assumptions given above. In practice, however, the trial wavefunction is approximate, and E_L fluctuates, especially near coalescence points where particles approach each other or a nucleus. Known exact properties of the wavefunction can be imposed here. The *Kato cusp conditions* specify the leading behavior as an electron approaches a nucleus or another electron. They apply to complex wavefunctions as well. The **electron-nucleus** cusp is

$$\left.\frac{\partial\overline{\psi}}{\partial r_{iI}}\right|_{r_{iI}=0} = -Z_I\overline{\psi}(0), \tag{3.41}$$

where $r_{iI} = |\boldsymbol{r}_i - \boldsymbol{R}_I|$, Z_I is the nuclear charge, and $\overline{\psi}$ is the spherical average about that nucleus, with the other particles held fixed away from further coalescences. The entire many-electron wavefunction need not be radial. The derivation is in Appendix C.

The electron–electron cusp depends on the leading relative-angular-momentum channel. For opposite spins, a nonzero spatial component at coalescence is allowed and the usual condition concerns its s-wave part. Opposite spin labels do not by themselves guarantee that this component is nonzero. For parallel spins, antisymmetry forces a node at coalescence; generically the leading component is linear in the relative coordinate. The derivations are in Appendix C.

An operational way to derive the cusp conditions is by avoiding divergence in the local energy. We assume a functional for the trial function

$$\psi(\boldsymbol{r}) = e^{-u(r)}f(\boldsymbol{r}), \tag{3.42}$$

where $u(r)$ is a Jastrow correlation factor depending only on the interelectronic distance $r = |\boldsymbol{r}_i - \boldsymbol{r}_j|$, and f is a smooth, possibly complex factor containing the remaining spatial dependence and the required exchange symmetry.

- For two electrons with **antiparallel spins** and a finite nonzero leading s-wave component of f, the cusp condition is

$$\left.\frac{du}{dr_{ij}}\right|_{r_{ij}\to 0} = -\frac{1}{2}. \tag{3.43}$$

- For two electrons with **parallel spins**, the Pauli exclusion principle enforces a node at coalescence, i.e., $f \to 0$ as $r_{ij} \to 0$. For a nonzero leading linear component, the cusp

condition becomes

$$\left.\frac{du}{dr_{ij}}\right|_{r_{ij}\to 0} = -\frac{1}{4}. \tag{3.44}$$

Exercise 3.1.5: VMC for the Helium atom (*/)**

Consider the Hamiltonian for the Helium atom

$$\hat{H} = -\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 - \frac{2}{r_1} - \frac{2}{r_2} + \frac{1}{r_{12}}. \tag{3.45}$$

We can define the (spatial) trial state $\psi_T(\boldsymbol{r}_1, \boldsymbol{r}_2)$, parametrized with $a, c > 0$ and $b \geq 0$ (to avoid a pole at positive r_{12}),

$$\psi_T(\boldsymbol{r}_1, \boldsymbol{r}_2) = \exp\left[-a \cdot (r_1 + r_2) + \frac{r_{12}}{c\,(1 + b r_{12})}\right]. \tag{3.46}$$

This symmetric spatial state is combined with the antisymmetric spin singlet, so the full electronic state has the correct fermionic antisymmetry. In a fixed-spin representation, the one-body factor is a product of two 1×1 determinants with orbitals $\phi(r) = e^{-ar}$, as in Eq. 3.40.

1. Use the cusp conditions to fix a and c, leaving only b as real variational parameter.
2. (**)Write a VMC code and find the variational energy as a function of b, and compare with the benchmark $E_0 \simeq -2.903724377$ Ha for this nonrelativistic, infinite-nuclear-mass Hamiltonian.[a]

   Hint: you can use some symbolic calculus software to calculate E_L: it is a bit cumbersome.

[a] C. Schwartz, *Further computations of the He atom ground state*, arXiv:math-ph/0605018 (2006).

3.2 Optimization of the Wavefunction

The quality of the variational energy depends critically on the choice of the trial wavefunction $\psi(x; \{\theta_k\})$, which is typically parametrized by a set of variational parameters $\{\theta_k\}$. The goal is to find the set of parameters that minimizes E_{var}, ideally reaching the ground-state energy.

In traditional approaches, the number of parameters was small, but in modern variational ansätze, such as neural quantum states (NQS), the number of parameters can reach thousands or even millions. This makes optimization both challenging and computationally demanding.

One difficulty arises when trying to evaluate differences between two independently estimated stochastic quantities. When the difference becomes small, statistical noise may dominate, obscuring the optimization direction. More sophisticated techniques are therefore required.

3.2.1 Correlated Sampling

To mitigate noise in comparing closely related wavefunctions, one can use *correlated sampling*. This method allows estimating energy differences more accurately by reusing the same sample of configurations.

Let $\psi_0(x)$ be a reference wavefunction, and consider a new wavefunction $\psi_1(x)$. If we sample configurations x according to $|\psi_0(x)|^2$, we can reweight the estimates using:

$$E_{\text{var}}[\psi_1] \approx \frac{\sum_{x\sim|\psi_0|^2} w(x)\,\text{Re}\,E_L^{(1)}(x)}{\sum_{x\sim|\psi_0|^2} w(x)}, \quad w(x) = \left|\frac{\psi_1(x)}{\psi_0(x)}\right|^2. \tag{3.47}$$

Here $E_L^{(1)} = (H\psi_1)/\psi_1$ must be evaluated with the new wavefunction. The weights alone do not change the reference local energy into the new one. This is the self-normalized reweighting method of Chapter 2. It requires that the reference probability cover the support of the target, and is useful only when the two distributions have sufficient overlap. The ratio estimator is generally biased at finite sample size, but consistent under the usual integrability and sampling assumptions. Using common samples can reduce the uncertainty of energy differences because their fluctuations are correlated. For instance $\psi_0 = \psi(\theta)$, while $\psi_1 = \psi(\theta + \delta\theta)$.

3.2.2 Gradients

An efficient alternative to comparing nearby energy values is to compute derivatives with respect to the variational parameters. We allow a complex wavefunction and take θ_k to be real coordinates, including separate real and imaginary parts of any complex weights.

Introduce the logarithmic derivative

$$O_k(x) = \frac{1}{\psi(x)}\frac{\partial\psi(x)}{\partial\theta_k} = \frac{\partial \log\psi(x)}{\partial\theta_k}. \tag{3.48}$$

The first expression is the definition and avoids any ambiguity about branches of the complex logarithm. Define

$$F_k = \langle O_k^* E_L\rangle_P - \langle O_k^*\rangle_P\langle E_L\rangle_P. \tag{3.49}$$

Then the gradient and the force are, respectively,

$$\boxed{\partial_k E_{\text{var}} = 2\,\text{Re}\,F_k, \qquad f_k = -\partial_k E_{\text{var}} = -2\,\text{Re}\,F_k.} \tag{3.50}$$

Equivalently,

$$f_k = -2\,\text{Re}\,\langle (O_k - \langle O_k\rangle_P)^*(E_L - \langle E_L\rangle_P)\rangle_P\,. \tag{3.51}$$

For samples $x_n \sim P$, store both $E_L(x_n)$ and $O_k(x_n)$, including their complex parts. With bars denoting sample means,

$$\widehat{f_k} = -2\,\text{Re}\left[\frac{1}{M}\sum_{n=1}^{M}\big(O_k(x_n) - \overline{O}_k\big)^*\big(E_L(x_n) - \overline{E}_L\big)\right]. \tag{3.52}$$

This is a consistent Monte Carlo estimator; a finite sample does not make it an exact equality. In particular, taking $\text{Re}\,E_L$ before forming this covariance would generally discard part of the gradient.

Proof for a complex wavefunction. Writing $\mathcal{N} = \langle\psi|\psi\rangle$, differentiation of the numerator and denominator gives

$$\partial_k E_{\mathrm{var}} = \frac{\langle\partial_k\psi|H|\psi\rangle + \langle\psi|H|\partial_k\psi\rangle}{\mathcal{N}} - E_{\mathrm{var}}\frac{\langle\partial_k\psi|\psi\rangle + \langle\psi|\partial_k\psi\rangle}{\mathcal{N}} \tag{3.53}$$

$$= 2\,\mathrm{Re}\left[\frac{\langle\partial_k\psi|H|\psi\rangle}{\mathcal{N}} - E_{\mathrm{var}}\frac{\langle\partial_k\psi|\psi\rangle}{\mathcal{N}}\right]. \tag{3.54}$$

Here Hermiticity of H makes the two terms in each numerator complex conjugates. We have not assumed that the operator product $O_k H$ is Hermitian; it generally is not.

Using $\partial_k\psi(x) = O_k(x)\psi(x)$, we obtain

$$\frac{\langle\partial_k\psi|H|\psi\rangle}{\mathcal{N}} = \frac{1}{\mathcal{N}}\sum_x \big[\partial_k\psi(x)\big]^*(H\psi)(x) \tag{3.55}$$

$$= \sum_x \frac{|\psi(x)|^2}{\mathcal{N}} O_k^*(x)\frac{(H\psi)(x)}{\psi(x)} = \langle O_k^* E_L\rangle_P, \tag{3.56}$$

$$\frac{\langle\partial_k\psi|\psi\rangle}{\mathcal{N}} = \sum_x P(x)O_k^*(x) = \langle O_k^*\rangle_P. \tag{3.57}$$

Substitution yields Eq. (3.50). The covariance form follows by expanding

$$\langle(O_k - \langle O_k\rangle)^*(E_L - \langle E_L\rangle)\rangle = \langle O_k^* E_L\rangle - \langle O_k^*\rangle\langle E_L\rangle - \langle O_k^*\rangle\langle E_L\rangle + \langle O_k^*\rangle\langle E_L\rangle \tag{3.58}$$

$$= \langle O_k^* E_L\rangle - \langle O_k^*\rangle\langle E_L\rangle. \tag{3.59}$$

Once the force has been estimated, a gradient-descent step is

$$\theta_k' = \theta_k + \delta\theta_k = \theta_k + \Delta f_k, \qquad \Delta > 0. \tag{3.60}$$

The normalization derivative is essential because ψ is not assumed normalized.

3.2.3 Natural Gradients and Stochastic Reconfiguration

The basic gradient-descent method can be improved by using information about the geometry of the variational manifold. This leads to *stochastic reconfiguration* (SR), closely related to quantum natural-gradient descent.[2]

For the real coordinates θ_i introduced above, define the quantum geometric tensor (QGT)

$$S_{ij} = \langle(O_i - \langle O_i\rangle)^*(O_j - \langle O_j\rangle)\rangle = \langle O_i^* O_j\rangle - \langle O_i^*\rangle\langle O_j\rangle. \tag{3.61}$$

It is Hermitian and positive semidefinite. Its real part is the metric for real parameter displacements. Hence the update is

$$\boldsymbol{\theta}' = \boldsymbol{\theta} + \Delta(\mathrm{Re}\,\boldsymbol{S})^{-1}\boldsymbol{f}, \tag{3.62}$$

[2] S Sorella, *Generalized Lanczos algorithm for variational quantum Monte Carlo*, Phys. Rev. B 64, 024512 (2001), https://doi.org/10.1103/PhysRevB.64.024512

when the inverse exists. In practice, solve the linear system rather than explicitly forming an inverse:

$$\boxed{\sum_j \operatorname{Re} S_{ij}\, \delta\theta_j = \Delta f_i.} \tag{3.63}$$

Taking the real part is essential for a general complex wavefunction parametrized by real coordinates. For a real wavefunction, S is already real. The pure-state quantum Fisher information matrix, with its usual normalization, is $4 \operatorname{Re} S$.

Iterative linear solvers can apply the sampled covariance matrix to a vector without storing all its entries. This reduces memory requirements; the overall cost still depends on the number of samples, parameters, and solver iterations.

The matrix may be singular because of redundant parameters, a pure normalization or global-phase direction, or insufficient sample rank, as well as being ill-conditioned because of statistical noise. A commonly used diagonal rescaling is

$$S_{ii} \longrightarrow S_{ii}(1+\epsilon). \tag{3.64}$$

It helps when the diagonal entries are nonzero, but does not remove a null direction whose diagonal entry vanishes. A more general regularized solve uses

$$G_{\text{reg}} = \operatorname{Re} S + \epsilon \operatorname{diag}(\operatorname{Re} S) + \eta I, \qquad \epsilon \geq 0, \quad \eta > 0, \tag{3.65}$$

The shift and step size are practical parameters to be adjusted.

Justification as a metric in curved space. Let $|\psi_{\boldsymbol{\theta}}\rangle$ denote a possibly unnormalized complex state. A small Euclidean change in its parameters need not produce a small physical change in the state. The Fubini–Study distance between the corresponding states is

$$d^2_{\text{FS}}(\psi_{\boldsymbol{\theta}}, \psi_{\boldsymbol{\theta}'}) = \arccos^2\left(\sqrt{\frac{\langle\psi_{\boldsymbol{\theta}}|\psi_{\boldsymbol{\theta}'}\rangle\langle\psi_{\boldsymbol{\theta}'}|\psi_{\boldsymbol{\theta}}\rangle}{\langle\psi_{\boldsymbol{\theta}}|\psi_{\boldsymbol{\theta}}\rangle\langle\psi_{\boldsymbol{\theta}'}|\psi_{\boldsymbol{\theta}'}\rangle}}\right). \tag{3.66}$$

Let $\theta_i' = \theta_i + d\theta_i$, with real $d\theta_i$. Keeping second-order terms explicitly,

$$|\psi_{\boldsymbol{\theta}'}\rangle = |\psi\rangle + \sum_i d\theta_i |\partial_i \psi\rangle + \frac{1}{2}\sum_{ij} d\theta_i d\theta_j |\partial_i \partial_j \psi\rangle + \mathcal{O}(\|d\boldsymbol{\theta}\|^3). \tag{3.67}$$

Here $\psi = \psi_{\boldsymbol{\theta}}$. Recall

$$O_i(x) = \frac{\partial_i \psi(x)}{\psi(x)}, \qquad Q_{ij}(x) = \frac{\partial_i \partial_j \psi(x)}{\psi(x)}. \tag{3.68}$$

For the first inner product we obtain

$$\langle\psi|\psi_{\boldsymbol{\theta}'}\rangle = \mathcal{N} + \sum_i d\theta_i \langle\psi|\partial_i\psi\rangle + \frac{1}{2}\sum_{ij} d\theta_i d\theta_j \langle\psi|\partial_i\partial_j\psi\rangle + \mathcal{O}(\|d\boldsymbol{\theta}\|^3) \tag{3.69}$$

$$= \mathcal{N}\left[1 + \sum_i d\theta_i \langle O_i\rangle + \frac{1}{2}\sum_{ij} d\theta_i d\theta_j \langle Q_{ij}\rangle\right] + \mathcal{O}(\|d\boldsymbol{\theta}\|^3). \tag{3.70}$$

We divided by $\mathcal{N}$ to express the terms as averages over $P = |\psi|^2/\mathcal{N}$. Similarly, the norm of the perturbed state is

$$\langle \psi_{\boldsymbol{\theta}'} | \psi_{\boldsymbol{\theta}'} \rangle = \mathcal{N}\Big[1 + \sum_i d\theta_i \langle O_i^* \rangle + \sum_j d\theta_j \langle O_j \rangle + \sum_{ij} d\theta_i d\theta_j \langle O_i^* O_j \rangle + \operatorname{Re} \sum_{ij} d\theta_i d\theta_j \langle Q_{ij} \rangle \Big] + \mathcal{O}(\|d\boldsymbol{\theta}\|^3). \tag{3.71}$$

To display the cancellation of the second derivatives, abbreviate

$$a = \sum_i d\theta_i \langle O_i \rangle, \qquad b = \sum_{ij} d\theta_i d\theta_j \langle Q_{ij} \rangle, \qquad c = \sum_{ij} d\theta_i d\theta_j \langle O_i^* O_j \rangle. \tag{3.72}$$

Here c is real, since it equals $\langle |\sum_i d\theta_i O_i|^2 \rangle$. The normalized squared overlap is

$$R = \frac{|\langle \psi | \psi_{\boldsymbol{\theta}'} \rangle|^2}{\mathcal{N} \langle \psi_{\boldsymbol{\theta}'} | \psi_{\boldsymbol{\theta}'} \rangle} \tag{3.73}$$

$$= \frac{1 + a + a^* + |a|^2 + \operatorname{Re} b}{1 + a + a^* + c + \operatorname{Re} b} + \mathcal{O}(\|d\boldsymbol{\theta}\|^3) \tag{3.74}$$

$$= 1 + |a|^2 - c + \mathcal{O}(\|d\boldsymbol{\theta}\|^3) \tag{3.75}$$

$$= 1 - \sum_{ij} d\theta_i d\theta_j \left[\langle O_i^* O_j \rangle - \langle O_i^* \rangle \langle O_j \rangle \right] + \mathcal{O}(\|d\boldsymbol{\theta}\|^3). \tag{3.76}$$

The terms containing Q_{ij} have canceled. Since $\arccos^2(\sqrt{1-x}) = x + \mathcal{O}(x^2)$ for small $x \geq 0$, the result is

$$d_{\mathrm{FS}}^2 = \sum_{ij} d\theta_i \, \operatorname{Re} S_{ij} \, d\theta_j + \mathcal{O}(\|d\boldsymbol{\theta}\|^3). \tag{3.77}$$

The imaginary part of a Hermitian S is antisymmetric and drops out for real displacements. In comparison, the Euclidean metric is

$$d_{\mathrm{Euclidean}}^2 = \sum_i (d\theta_i)^2 = \sum_{ij} \delta_{ij} \, d\theta_i d\theta_j. \tag{3.78}$$

Digression: covariant steepest descent. Why should the metric precondition the gradient?[3] Write $G = \operatorname{Re} S$ and consider a real loss $L(\theta)$, with $g_\theta = \nabla_\theta L$. The natural-gradient direction is

$$\Delta\theta = -\eta G_\theta^{-1} g_\theta. \tag{3.79}$$

Let $\theta = \theta(\phi)$ be a smooth invertible change of real coordinates, with Jacobian

$$J_{ai} = \frac{\partial \theta_a}{\partial \phi_i}. \tag{3.80}$$

Infinitesimal displacements obey $\Delta\theta = J \Delta\phi$. The chain rule and the metric transformation give

$$g_\phi = J^\top g_\theta, \qquad G_\phi = J^\top G_\theta J. \tag{3.81}$$

[3] G. Mazzola, A. Zen, S. Sorella, *Finite-temperature electronic simulations without the Born-Oppenheimer constraint*, J. Chem. Phys. **137**, 134112 (2012), https://doi.org/10.1063/1.4755992.

Hence

$$\Delta\phi = -\eta G_\phi^{-1} g_\phi \tag{3.82}$$
$$= -\eta (J^\top G_\theta J)^{-1} J^\top g_\theta \tag{3.83}$$
$$= -\eta J^{-1} G_\theta^{-1} (J^\top)^{-1} J^\top g_\theta \tag{3.84}$$
$$= -\eta J^{-1} G_\theta^{-1} g_\theta. \tag{3.85}$$

Mapping this direction back,

$$\Delta\theta = J\Delta\phi = -\eta J J^{-1} G_\theta^{-1} g_\theta = -\eta G_\theta^{-1} g_\theta. \tag{3.86}$$

Thus the continuous flow, or infinitesimal direction, is independent of the chosen parametrization.

Justification as linearized imaginary-time propagation. For a short imaginary-time step,

$$e^{-\delta\tau H}|\psi\rangle = |\psi\rangle - \delta\tau H|\psi\rangle + \mathcal{O}(\delta\tau^2). \tag{3.87}$$

SR chooses the parameter change whose tangent-state displacement best approximates this evolution. Since normalization and global phase do not change the physical state, allow an additional complex coefficient δc_0:

$$(1+\delta c_0)|\psi\rangle + \sum_j d\theta_j |\partial_j \psi\rangle \simeq |\psi\rangle - \delta\tau H|\psi\rangle. \tag{3.88}$$

The real increments $d\theta_j$ and complex δc_0 minimize the norm of the residual

$$|R\rangle = \delta c_0 |\psi\rangle + \sum_j d\theta_j |\partial_j\psi\rangle + \delta\tau H|\psi\rangle. \tag{3.89}$$

Variation of the real and imaginary parts of δc_0 gives $\langle\psi|R\rangle = 0$. Thus

$$\delta c_0 \mathcal{N} + \sum_j d\theta_j \langle\psi|\partial_j\psi\rangle = -\delta\tau\langle\psi|H|\psi\rangle, \tag{3.90}$$
$$\delta c_0 = -\delta\tau E_{\text{var}} - \sum_j d\theta_j \langle O_j\rangle. \tag{3.91}$$

Variation of each real $d\theta_i$ gives

$$\mathrm{Re}\langle\partial_i\psi|R\rangle = 0. \tag{3.92}$$

Dividing by $\mathcal{N}$ and substituting δc_0,

$$0 = \mathrm{Re}\left[\left(-\delta\tau E_{\text{var}} - \sum_j d\theta_j\langle O_j\rangle\right)\langle O_i^*\rangle + \sum_j d\theta_j\langle O_i^* O_j\rangle + \delta\tau\langle O_i^* E_L\rangle\right] \tag{3.93}$$
$$= \sum_j \mathrm{Re}\left[\langle O_i^* O_j\rangle - \langle O_i^*\rangle\langle O_j\rangle\right] d\theta_j + \delta\tau\,\mathrm{Re}\left[\langle O_i^* E_L\rangle - \langle O_i^*\rangle E_{\text{var}}\right]. \tag{3.94}$$

Therefore

$$\sum_j \mathrm{Re}\, S_{ij}\, d\theta_j = -\delta\tau\, \mathrm{Re}\, F_i = \frac{\delta\tau}{2} f_i. \tag{3.95}$$

This is Eq. (3.63) with $\Delta = \delta\tau/2$. The factor of two follows from the definition of the force, and can be absorbed into the optimization step size.

Connection with the Fisher information in machine learning. Let $P_\theta = |\psi_\theta|^2/\mathcal{N}_\theta$. For real parameter coordinates,

$$\partial_i \log P_\theta(x) = 2\,\mathrm{Re}\left[O_i(x) - \langle O_i\rangle\right]. \tag{3.96}$$

The classical Fisher information of computational-basis measurements is

$$I_{ij} = \langle \partial_i \log P_\theta\, \partial_j \log P_\theta\rangle = 4\,\mathrm{Cov}_P(\mathrm{Re}\, O_i, \mathrm{Re}\, O_j). \tag{3.97}$$

By contrast,

$$\mathrm{Re}\, S_{ij} = \mathrm{Cov}_P(\mathrm{Re}\, O_i, \mathrm{Re}\, O_j) + \mathrm{Cov}_P(\mathrm{Im}\, O_i, \mathrm{Im}\, O_j). \tag{3.98}$$

For a positive wavefunction, or a parameter-independent configuration phase, the second covariance vanishes and $I = 4S$. Thus SR and classical natural-gradient geometry coincide up to a constant factor in that case. For a general complex wavefunction, computational-basis probabilities do not contain the missing phase information, whereas the quantum metric does. This distinction will reappear in the tomography section.

Exercise 3.2.1: VMC optimization with gradients (*)

Consider the TFIM Hamiltonian in 1D with PBC:

$$H = -\sum_i^L \sigma_i^z \sigma_{i+1}^z - \sum_i^L \sigma_i^x, \tag{3.99}$$

and define the ansatz

$$\psi_T(s_1, \cdots, s_L) = \exp\left(\theta_1 \sum_i s_i s_{i+1} + \theta_2 \sum_i s_i s_{i+2}\right), \tag{3.100}$$

where $s_i \in \{-1, +1\}$ represents the spin z-value at site i, and θ_1, θ_2 are variational parameters encoding nearest and next-nearest neighbor interactions, respectively.

1. Write a VMC code with gradient optimization to find the best variational energy, use $L = 20$. Reasonable values for parameters are $\Delta = 0.1 - 0.01$, and order of 1000s MC sweeps (include also thermalization sweeps). You need to identify and evaluate O_1 and O_2, and accumulate the necessary averages.

2. Plot the energy vs optimization steps and compare with the exact diagonalization value, -25.491.

3. Implement also the natural gradient descent, computing also the covariance matrix (2×2 in this case). Use a reasonable value of $\epsilon = 0.001$. Around 100 steps is a useful initial budget, but convergence depends on sampling noise, regularization, initialization, and step size; monitor it rather than assume it. For this real ansatz, use the regularized real covariance matrix in `delta = np.linalg.solve(S, f)`, where `delta` is a two-dimensional array that contains the parameter's displacement $\delta\theta/\Delta$.

3.3 Restricted Boltzmann Machines

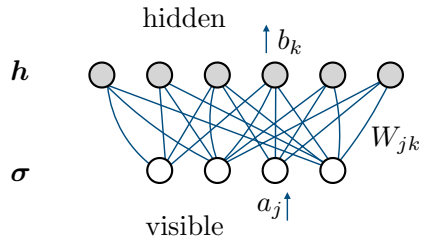


Figure 3.1: RBM network with visible and hidden neurons. Notice the absence of intra-layer connections.

In the last decade we have seen the convergence of fields that were previously seen distant: quantum Monte Carlo and Machine Learning (ML). A pioneering example is the neural-network wavefunction introduced by Carleo and Troyer.[4] A useful architecture is the Restricted Boltzmann Machine (RBM). The RBM is defined by a set of *visible* units $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_N)$, representing physical degrees of freedom (e.g., spin-1/2 variables with $\sigma_j \in \{-1, +1\}$), and a set of *hidden* units $\mathbf{h} = (h_1, \ldots, h_M)$, also binary-valued (see Fig. 3.1).

The joint energy of a configuration of visible and hidden units is given by:

$$E(\boldsymbol{\sigma}, \mathbf{h}) = -\sum_{j=1}^{N} a_j \sigma_j - \sum_{k=1}^{M} b_k h_k - \sum_{j=1}^{N}\sum_{k=1}^{M} W_{jk}\sigma_j h_k, \tag{3.101}$$

where a_j, b_k, and W_{jk} are variational parameters: visible biases, hidden biases, and connection weights. They may be complex. With complex parameters, E is a formal network function and e^{-E} is not a classical joint probability; it defines the wavefunction ansatz below.

The RBM wavefunction is obtained by marginalizing over the hidden degrees of freedom:

$$\Psi(\boldsymbol{\sigma}) = \sum_{\mathbf{h}} e^{-E(\boldsymbol{\sigma}, \mathbf{h})}. \tag{3.102}$$

Due to the bipartite structure of the RBM, the sum over hidden variables can be performed exactly:

$$\Psi(\boldsymbol{\sigma}) = \exp\left(\sum_{j=1}^{N} a_j \sigma_j\right) \prod_{k=1}^{M} \sum_{h_k = \pm 1} \exp\left(h_k \left[b_k + \sum_{j=1}^{N} W_{jk}\sigma_j\right]\right) \tag{3.103}$$

$$= \exp\left(\sum_{j=1}^{N} a_j \sigma_j\right) \prod_{k=1}^{M} 2\cosh\left(b_k + \sum_{j=1}^{N} W_{jk}\sigma_j\right). \tag{3.104}$$

[4] G. Carleo and M. Troyer, *Solving the quantum many-body problem with artificial neural networks*, Science **355**, 602–606 (2017), doi:10.1126/science.aag2302.

Thus, the final visible-only form of the variational wavefunction becomes:

$$\Psi(\boldsymbol{\sigma}) = \exp\left(\sum_j a_j \sigma_j\right) \prod_k 2\cosh\left(b_k + \sum_j W_{jk}\sigma_j\right). \tag{3.105}$$

This RBM ansatz can be interpreted as a compact representation of a correlated quantum state. The product of cosh terms introduces multi-spin correlations of arbitrary range and order. In particular, the hidden units mediate effective interactions between visible spins in a compact, efficiently parameterized form.

In VMC, we require the logarithmic derivatives of the wavefunction with respect to each variational parameter:

$$\mathcal{O}_\theta(\boldsymbol{\sigma}) = \frac{1}{\Psi(\boldsymbol{\sigma})}\frac{\partial \Psi(\boldsymbol{\sigma})}{\partial \theta} = \frac{\partial \log \Psi(\boldsymbol{\sigma})}{\partial \theta}. \tag{3.106}$$

From the log-wavefunction:

$$\log \Psi(\boldsymbol{\sigma}) = \sum_j a_j \sigma_j + \sum_k \log\left[2\cosh\left(b_k + \sum_j W_{jk}\sigma_j\right)\right], \tag{3.107}$$

we derive the following:

- Visible bias a_j:

$$\mathcal{O}_{a_j}(\boldsymbol{\sigma}) = \sigma_j \tag{3.108}$$

- Hidden bias b_k:

$$\mathcal{O}_{b_k}(\boldsymbol{\sigma}) = \tanh\left(b_k + \sum_j W_{jk}\sigma_j\right) \tag{3.109}$$

- Weight W_{jk}:

$$\mathcal{O}_{W_{jk}}(\boldsymbol{\sigma}) = \sigma_j \cdot \tanh\left(b_k + \sum_i W_{ik}\sigma_i\right) \tag{3.110}$$

For complex parameters these are holomorphic derivatives. For any weight $z = u + iv$, use $O_u = O_z$ and $O_v = iO_z$ to construct the real-coordinate force and QGT derived above. Thus both the phase and amplitude can be optimized. The covariance of the logarithmic derivatives is S, and the pure-state quantum Fisher matrix is $4\,\mathrm{Re}\,S$. Near a zero of the amplitude, logarithmic derivatives and ratios need care; they are only evaluated on the sampled support.

With real parameters this ansatz is strictly positive. Complex parameters allow sign and phase structure without changing the displayed architecture. Neural-network ansätze provide flexible variational families, but their accuracy and optimization difficulty depend on the problem and representation.

Efficient Sampling: Use of Lookup Tables

Computing the full wavefunction ratio from scratch at each Metropolis iteration can be expensive for large systems. To accelerate this, we store a *lookup table* of the effective arguments of the cosh, which we denote by:

$$\theta_k(\boldsymbol{\sigma}) = b_k + \sum_j W_{jk}\sigma_j. \tag{3.111}$$

Notice that the usage of look-up tables is recommended also for the Jastrow function of the previous exercises.

These quantities appear repeatedly in both the wavefunction and its derivatives:

$$\Psi(\boldsymbol{\sigma}) \propto e^{\sum_j a_j\sigma_j} \prod_k \cosh\left(\theta_k(\boldsymbol{\sigma})\right), \quad \mathcal{O}_{b_k} = \tanh\left(\theta_k\right), \quad \mathcal{O}_{W_{jk}} = \sigma_j \tanh\left(\theta_k\right). \tag{3.112}$$

When a single spin σ_s is flipped, from σ_s to $-\sigma_s$, the configuration changes from $\boldsymbol{\sigma}$ to $\boldsymbol{\sigma}^{\text{flip}\,s}$, and each θ_k updates as:

$$\theta_k^{'} = \theta_k - 2W_{sk}\sigma_s. \tag{3.113}$$

This rule is derived by observing that the spin flip changes $\sigma_s \to -\sigma_s$, which affects only the terms in the sum over j where $j = s$. Hence, instead of recomputing all θ_k from scratch, we can efficiently update them using the cached values.
The corresponding amplitude ratio, including the visible bias, is

$$\frac{\Psi(\boldsymbol{\sigma}^{\text{flip}\,s})}{\Psi(\boldsymbol{\sigma})} = e^{-2a_s\sigma_s} \prod_k \frac{\cosh(\theta_k - 2W_{sk}\sigma_s)}{\cosh\theta_k}. \tag{3.114}$$

For a symmetric spin-flip trial, Metropolis accepts with $\min(1, |\Psi'/\Psi|^2)$, including for complex amplitudes. Cached fields reduce this ratio calculation from $O(NM)$ to $O(M)$. Constructing all NM weight derivatives still requires $O(NM)$ output, although caching avoids recomputing their shared fields.

In implementation, one typically:

- Initializes θ_k at the start of sampling.
- Proposes a spin flip, computes the updated θ_k' using Eq. (3.113), and calculates the acceptance probability.
- If the move is accepted, overwrite the current θ_k values with the updated ones.

Exercise 3.3.1: VMC with a RBM (**)

Consider the same TFIM of Exercise 3.2.1 but the RBM of Eq. 3.105 as ansatz. For the positive TFIM ground state, real parameters suffice in this exercise. Choose $M = \alpha N$, start with $\alpha = 1$ and $L = 8$ to not have too many parameters W's.

1. Write the VMC code with the RBM ansatz. Start with random and small parameters distributed with a Gaussian distribution or uniformly in the range [-0.01,0.01].
2. Find the best variational energy and compare with the Jastrow case (using the same L, obviously). Try to increase α.

3.4 Quantum State Tomography

In VMC we choose a parametrized wavefunction and learn its parameters by minimizing the energy of a known Hamiltonian. Another use of QMC beyond finding grond states can be to reconstruct a state from measurements. This is *quantum state tomography*, and find application in the development of real quantum simulators and quantum computers (basically, one would like to know which quantum state has been prepared in hardware). A parametrized wavefunction can be trained by fitting measurement probabilities instead of minimizing the energy. Neural-network quantum state tomography provides a direct connection between quantum Monte Carlo and generative learning.[5]

3.4.1 Simple case: reconstructing a positive-valued state

For a general normalized complex pure state,

$$\Psi_*(\boldsymbol{\sigma}) = \sqrt{p_*(\boldsymbol{\sigma})}e^{i\Phi_*(\boldsymbol{\sigma})}, \tag{3.115}$$

measurements in the σ^z basis sample $p_* = |\Psi_*|^2$ and contain no information about the configuration-dependent phase Φ_*. A snapshot is an entire configuration $(\sigma_1, \ldots, \sigma_L)$, not only the mean spin at each site. Even perfect knowledge of p_* does not determine a general complex state or mixed-state density matrix.

A useful special case is a state known in advance to be pure and positive in this basis. For the finite transverse-field Ising model

$$H = -J\sum_{i=1}^{L}\sigma_i^z\sigma_{i+1}^z - h\sum_{i=1}^{L}\sigma_i^x, \qquad h > 0, \tag{3.116}$$

with periodic boundary conditions, the off-diagonal matrix elements are $-h$. Single-spin flips connect all computational-basis configurations. The ground state is unique and can be chosen strictly positive. Consequently,

$$\Psi_*(\boldsymbol{\sigma}) = \sqrt{p_*(\boldsymbol{\sigma})} \tag{3.117}$$

determines that ground state up to an overall phase. The assumption is about the pure ground state; it is not justified for arbitrary finite-temperature snapshots. Also, the positive-ground-state argument does not give uniqueness at $h = 0$.

We use this special case for the example below. The general VMC formulas of this chapter still allow complex states; the single-basis tomography exercise explicitly restricts the target family.

3.4.2 KL divergence and maximum likelihood

Let $\mathcal{D} = \{\boldsymbol{\sigma}^{(n)}\}_{n=1}^{N_\mathcal{D}}$ be independent measurement snapshots from p_*. Parametrize a normalized model probability q_θ and, for the positive-state reconstruction, set

$$\psi_\theta(\boldsymbol{\sigma}) = \sqrt{q_\theta(\boldsymbol{\sigma})}. \tag{3.118}$$

[5] G. Torlai, G. Mazzola, J. Carrasquilla, M. Troyer, R. Melko and G. Carleo, *Neural-network quantum state tomography*, Nature Physics **14**, 447–450 (2018), doi:10.1038/s41567-018-0048-5; arXiv:1703.05334.

The objective is the forward Kullback–Leibler divergence,

$$\boxed{D_{\mathrm{KL}}(p_*\|q_\theta) = \sum_{\boldsymbol{\sigma}} p_*(\boldsymbol{\sigma}) \log \frac{p_*(\boldsymbol{\sigma})}{q_\theta(\boldsymbol{\sigma})}.} \tag{3.119}$$

It is nonnegative and vanishes precisely when the two distributions agree. The term $\sum p_* \log p_*$ does not depend on θ. Replacing the remaining expectation by a data average gives the negative log-likelihood

$$\boxed{\mathcal{L}_{\mathcal{D}}(\theta) = -\frac{1}{N_{\mathcal{D}}} \sum_{n=1}^{N_{\mathcal{D}}} \log q_\theta(\boldsymbol{\sigma}^{(n)}), \qquad \theta_* \in \arg\min_\theta \mathcal{L}_{\mathcal{D}}(\theta).} \tag{3.120}$$

This is exactly equivalent to minimizing $D_{\mathrm{KL}}(\widehat{p}_{\mathcal{D}}\|q_\theta)$, where $\widehat{p}_{\mathcal{D}}$ is the empirical distribution. Finite data, limited expressivity, and imperfect optimization can all limit the reconstruction. A separate validation set helps distinguish fitting the training sample from learning its underlying distribution.

For a general complex state measured in several bases b, the same principle applies to

$$q_\theta^{[b]}(y) = \frac{|\langle y; b|\psi_\theta\rangle|^2}{\mathcal{N}_\theta}, \qquad \mathcal{L}(\theta) = -\sum_b \frac{\omega_b}{N_b} \sum_{n=1}^{N_b} \log q_\theta^{[b]}(y_n), \tag{3.121}$$

with specified positive weights ω_b and sufficiently informative measurement bases. We will not implement this phase-learning problem here. Its likelihood gradient is stated in Appendix D, so that the relation to the complex-state formalism remains explicit.

3.4.3 An RBM probability model and its gradients

We use the same spin convention as in the preceding RBM section: visible spins $\sigma_j \in \{-1, +1\}$ and hidden spins $h_k \in \{-1, +1\}$. The visible biases are a_j, the hidden biases are b_k, and W_{jk} couples visible spin j to hidden spin k. Here all probability-model parameters are real, and θ denotes their collection. Define

$$q_\theta(\boldsymbol{\sigma}, \boldsymbol{h}) = \frac{1}{Z_\theta} \exp\left(\sum_j a_j\sigma_j + \sum_k b_k h_k + \sum_{jk} W_{jk}\sigma_j h_k\right), \tag{3.122}$$

$$\theta_k(\boldsymbol{\sigma}) = b_k + \sum_j W_{jk}\sigma_j, \tag{3.123}$$

$$A_\theta(\boldsymbol{\sigma}) = \sum_j a_j\sigma_j + \sum_k \log\left[2\cosh\theta_k(\boldsymbol{\sigma})\right]. \tag{3.124}$$

The quantities $\theta_k(\boldsymbol{\sigma})$ are the configuration-dependent hidden fields. Summing over the hidden layer gives

$$q_\theta(\boldsymbol{\sigma}) = \frac{e^{A_\theta(\boldsymbol{\sigma})}}{Z_\theta}, \qquad Z_\theta = \sum_{\boldsymbol{\sigma}} e^{A_\theta(\boldsymbol{\sigma})}, \qquad \psi_\theta(\boldsymbol{\sigma}) = \frac{e^{A_\theta(\boldsymbol{\sigma})/2}}{\sqrt{Z_\theta}}. \tag{3.125}$$

The unnormalized amplitude is $e^{A/2}$. In the previous VMC section, the RBM marginal itself was used as an amplitude; its Born probability is the square modulus of that amplitude. These are two different parametrization conventions. In particular, for the present unnormalized amplitude, $O_\mu = \frac{1}{2}\partial_\mu A$, not $\partial_\mu A$, where ∂_μ denotes differentiation with respect to a model parameter.

Write $\langle\cdot\rangle_{\mathcal{D}}$ for an average over data. The loss and its exact gradient are

$$\mathcal{L}_{\mathcal{D}}(\theta) = \log Z_\theta - \langle A_\theta\rangle_{\mathcal{D}}. \tag{3.126}$$

$$\boxed{\partial_\mu \mathcal{L}_{\mathcal{D}} = \langle \partial_\mu A_\theta\rangle_{q_\theta} - \langle \partial_\mu A_\theta\rangle_{\mathcal{D}}.} \tag{3.127}$$

The data average is often called the positive phase and the model average the negative phase when discussing ascent of the log-likelihood. Here we minimize its negative, so the gradient is *model minus data*.

Using $d\log(2\cosh x)/dx = \tanh x$, we obtain

$$\partial_{a_j} A = \sigma_j, \qquad \partial_{b_k} A = \tanh\theta_k, \qquad \partial_{W_{jk}} A = \sigma_j \tanh\theta_k. \tag{3.128}$$

Therefore all gradients needed to write the code are

$$\partial_{a_j}\mathcal{L} = \langle\sigma_j\rangle_q - \langle\sigma_j\rangle_{\mathcal{D}}, \tag{3.129}$$

$$\partial_{b_k}\mathcal{L} = \langle\tanh\theta_k\rangle_q - \langle\tanh\theta_k\rangle_{\mathcal{D}}, \tag{3.130}$$

$$\partial_{W_{jk}}\mathcal{L} = \langle\sigma_j\tanh\theta_k\rangle_q - \langle\sigma_j\tanh\theta_k\rangle_{\mathcal{D}}. \tag{3.131}$$

The fields in each average are evaluated at that average's visible configuration. The derivation, including the differentiation of Z_θ, is in Appendix D.

3.4.4 Block Gibbs sampling and contrastive divergence

The bipartite structure makes the conditional distributions factorize within each layer:

$$q(h_k = +1 \mid \boldsymbol{\sigma}) = \frac{e^{\theta_k(\boldsymbol{\sigma})}}{2\cosh\theta_k(\boldsymbol{\sigma})} = \frac{1+\tanh\theta_k(\boldsymbol{\sigma})}{2}, \tag{3.132}$$

$$q(\sigma_j = +1 \mid \boldsymbol{h}) = \frac{1+\tanh(a_j + \sum_k W_{jk}h_k)}{2}. \tag{3.133}$$

In each case the probability of -1 is the complementary probability. At fixed parameters, alternating draws of the entire hidden layer and the entire visible layer defines a block Gibbs chain with stationary visible distribution q_θ. Independent units can be drawn in parallel within each conditional layer. Equilibrium model averages require sufficient equilibration and sampling; conditional factorization does not guarantee rapid mixing of the full chain.

Contrastive divergence with m steps, abbreviated CD-m, replaces this equilibrium calculation by short chains initialized at the data. For each visible configuration in a minibatch of size B, perform

$$\boldsymbol{\sigma}^{(0)} = \boldsymbol{\sigma}_{\text{data}}, \qquad \boldsymbol{h}^{(\ell)} \sim q(\boldsymbol{h} \mid \boldsymbol{\sigma}^{(\ell)}), \qquad \boldsymbol{\sigma}^{(\ell+1)} \sim q(\boldsymbol{\sigma} \mid \boldsymbol{h}^{(\ell)}), \quad \ell = 0, \dots, m-1. \tag{3.134}$$

Use the final visible states for the model contribution. Writing

$$\overline{h}_k^{(n,\ell)} = \mathbb{E}[h_k \mid \boldsymbol{\sigma}^{(n,\ell)}] = \tanh\theta_k(\boldsymbol{\sigma}^{(n,\ell)}), \tag{3.135}$$

the estimated loss gradients are

$$\widehat{g}_{a_j}^{\mathrm{CD}} = \frac{1}{B}\sum_{n=1}^{B}\left[\sigma_j^{(n,m)} - \sigma_j^{(n,0)}\right], \tag{3.136}$$

$$\widehat{g}_{b_k}^{\mathrm{CD}} = \frac{1}{B}\sum_{n=1}^{B}\left[\overline{h}_k^{(n,m)} - \overline{h}_k^{(n,0)}\right], \tag{3.137}$$

$$\widehat{g}_{W_{jk}}^{\mathrm{CD}} = \frac{1}{B}\sum_{n=1}^{B}\left[\sigma_j^{(n,m)}\overline{h}_k^{(n,m)} - \sigma_j^{(n,0)}\overline{h}_k^{(n,0)}\right]. \tag{3.138}$$

Update each parameter with

$$\theta \longleftarrow \theta - \eta\widehat{g}^{\mathrm{CD}}, \qquad \eta > 0. \tag{3.139}$$

Hidden conditional means can be used in these averages instead of sampled hidden spins. Notice that $\mathbb{E}[h_k \mid \boldsymbol{\sigma}] = \tanh\theta_k$ is a mean spin, whereas $q(h_k = +1 \mid \boldsymbol{\sigma}) = (1 + \tanh\theta_k)/2$ is a probability. Actual hidden spins are still sampled to advance the Gibbs chain. All states within a CD update use the same parameter values, and the parameter update occurs after the minibatch statistics have been accumulated.

CD-m is generally a *biased approximation* to the exact likelihood gradient because the final states need not follow q_θ. For fixed parameters, the bias disappears as $m \to \infty$ if the chain mixes. The usual finite-m update is also not the full derivative of the contrastive-divergence functional; Appendix D makes this distinction explicit.

3.4.5 Reconstructing observables

Once trained, the positive wavefunction can be sampled using Gibbs sampling or Metropolis. Ratios do not require Z_θ:

$$\frac{\psi_\theta(\boldsymbol{\sigma}')}{\psi_\theta(\boldsymbol{\sigma})} = \exp\left[\frac{A_\theta(\boldsymbol{\sigma}') - A_\theta(\boldsymbol{\sigma})}{2}\right]. \tag{3.140}$$

For diagonal observables, such as $\sigma_i^z\sigma_j^z$, the estimator is simply $\sigma_i\sigma_j$. Let $\boldsymbol{\sigma}^{\mathrm{flip}\,i}$ denote the configuration with σ_i replaced by $-\sigma_i$. Then

$$\langle\sigma_i^x\rangle_{\psi_\theta} = \left\langle\frac{\psi_\theta(\boldsymbol{\sigma}^{\mathrm{flip}\,i})}{\psi_\theta(\boldsymbol{\sigma})}\right\rangle_{q_\theta} = \left\langle\sqrt{\frac{q_\theta(\boldsymbol{\sigma}^{\mathrm{flip}\,i})}{q_\theta(\boldsymbol{\sigma})}}\right\rangle_{q_\theta}, \tag{3.141}$$

$$E_L(\boldsymbol{\sigma}) = -J\sum_i \sigma_i\sigma_{i+1} - h\sum_i \exp\left[\frac{A_\theta(\boldsymbol{\sigma}^{\mathrm{flip}\,i}) - A_\theta(\boldsymbol{\sigma})}{2}\right]. \tag{3.142}$$

The reconstruction of off-diagonal observables from single-basis snapshots relies on the known positivity and purity of the target. For an unrestricted complex state, these estimators

would additionally require the learned phase differences. The Hamiltonian is used here to generate synthetic data and evaluate the reconstruction; it does not enter the likelihood-training gradient.

Exercise 3.4.1: Write a tomography code for the TFIM

Use a chain of $L = 8$ sites with periodic boundary conditions, $J = h = 1$, and a real ± 1 RBM with $M = L$ hidden units.

1. Diagonalize the TFIM and generate, for example, 5000 independent training snapshots and a separate validation set from its normalized ground-state probabilities. Use $\sigma_i = \pm 1$ for every snapshot and keep the site ordering consistent in the Hamiltonian, snapshots, and observables.

2. Implement $A_\theta(\boldsymbol{\sigma})$, the conditional probabilities in Eq. (3.133), and alternating block Gibbs sampling. Initialize parameters with small random values. Use $\tanh x$ for the conditional means and evaluate $\log(2\cosh x)$ stably as $|x| + \log\bigl(1 + e^{-2|x|}\bigr)$.

3. Implement minibatch CD-m training using Eq. (3.138), starting with $B = 100$, $m = 5$, and a tunable learning rate. Return and apply the minibatch-averaged gradient with the displayed descent sign.

4. For this small system, enumerate all 2^L visible states to compute Z_θ and the exact model averages. Check the exact likelihood gradient against finite differences, and compare it with averages of CD estimates for $m = 1, 5, 20$. Finite differences test the exact gradient, not the biased CD estimate.

5. Plot training and validation negative log-likelihood versus epoch. Exact normalization is feasible at this size; for large RBMs it is generally not, so exact validation likelihood is not automatically available. Select a checkpoint using validation data rather than the unknown target-state overlap.

6. Compare diagonal correlations, transverse magnetization, and energy with the exact results. At this small size also compute the squared fidelity

$$\mathcal{F} = |\langle \Psi_* | \psi_\theta \rangle|^2 = \left[\sum_{\boldsymbol{\sigma}} \sqrt{p_*(\boldsymbol{\sigma}) q_\theta(\boldsymbol{\sigma})} \right]^2 . \tag{3.143}$$

   The unsquared expression is the overlap amplitude, not this squared fidelity. Use the exact target only for benchmarking and synthetic-data generation.

7. Generate fresh snapshots from the trained model, equilibrating independently of the short CD chains, and estimate observables with autocorrelation-aware error bars. Repeat training with different datasets or seeds to investigate reconstruction uncertainty, which is distinct from the Monte Carlo error at fixed trained parameters.

Chapter 4

Diffusion and Green's Function Monte Carlo

In this Chapter we describe the simplest idea to go beyond the variational method. We refer to the continuous-space version, named Diffusion Monte Carlo. The lattice version of the method is usually dubbed Green's function Monte Carlo. We use units in which $\hbar = m = 1$. The positive-walker algorithms below assume a real Hamiltonian with a local potential and real wavefunctions; a general complex state requires a separate treatment of its phase.

4.1 Imaginary-time projection

Consider a quantum system with Hamiltonian $\hat{H}$, acting on a Hilbert space of states $\phi_i(\boldsymbol{r})$, which are eigenfunctions of the Hamiltonian:

$$\hat{H}\phi_i(\boldsymbol{r}) = E_i\phi_i(\boldsymbol{r}), \tag{4.1}$$

For simplicity, assume a discrete spectrum with a nondegenerate ground state, $E_0 < E_1 \leq E_2 \leq \ldots$. The vector $\boldsymbol{r} \in \mathbb{R}^{3N}$ denotes the positions of all N particles in continuous space.

Given an initial wavefunction $\psi_{\text{init}}(\boldsymbol{r})$ that admits the decomposition

$$\psi_{\text{init}}(\boldsymbol{r}) = \sum_i c_i\phi_i(\boldsymbol{r}), \tag{4.2}$$

we define an imaginary-time evolution:

$$\boxed{-\frac{\partial}{\partial\tau}\Psi(\boldsymbol{r},\tau) = \left(\hat{H} - E_T\right)\Psi(\boldsymbol{r},\tau)} \tag{4.3}$$

with formal solution

$$\Psi(\boldsymbol{r},\tau) = e^{-\tau(\hat{H}-E_T)}\psi_{\text{init}}(\boldsymbol{r}) = \sum_i c_i e^{-\tau(E_i-E_T)}\phi_i(\boldsymbol{r}). \tag{4.4}$$

As $\tau \to \infty$, excited-state components decay exponentially faster than the ground state, yielding

$$e^{\tau(E_0-E_T)}\Psi(\boldsymbol{r},\tau) \xrightarrow[\tau\to\infty]{} c_0\,\phi_0(\boldsymbol{r}). \tag{4.5}$$

The parameter E_T is a *reference energy*. In this formal solution it is constant; choosing $E_T = E_0$ removes the asymptotic exponential growth or decay of the norm. In a simulation it can instead be adjusted to control the walker population. This factor does not affect the spatial form of the projected ground state $\phi_0(\boldsymbol{r})$, only its overall normalization.

The imaginary-time propagation thus filters out the ground state from any initial state ψ_{init} that has a non-zero overlap with ϕ_0, exponentially suppressing all contributions from higher-energy eigenstates.

This is the continuous-time equivalent of the power method of Sect. 1.5. The aim is to avoid storing the full wavefunction by representing it stochastically; this does not guarantee efficient sampling for every Hamiltonian.

Let us consider a generic wavefunction $\Psi(\boldsymbol{r}, \tau)$. Its imaginary-time evolution (over an imaginary-time interval Δ_τ, not necessarily small) can be written in integral form using the Green's function $G(\boldsymbol{r}, \boldsymbol{r}', \Delta_\tau)$:

$$\Psi(\boldsymbol{r}, \tau + \Delta_\tau) = \int d\boldsymbol{r}'\, G(\boldsymbol{r}, \boldsymbol{r}', \Delta_\tau)\, \Psi(\boldsymbol{r}', \tau). \tag{4.6}$$

The Green's function is defined as the matrix element (in coordinate basis) of the imaginary-time evolution operator,

$$G(\boldsymbol{r}, \boldsymbol{r}', \Delta_\tau) = \langle \boldsymbol{r} | e^{-\Delta_\tau (\hat{H} - E_T)} | \boldsymbol{r}' \rangle. \tag{4.7}$$

We now write the Hamiltonian as the sum of a kinetic and a potential term:

$$\hat{H} = -\frac{1}{2} \sum_{i=1}^{N} \nabla_i^2 + V(\boldsymbol{r}), \tag{4.8}$$

where $V(\boldsymbol{r})$ is a general potential energy acting on the full configuration $\boldsymbol{r} \in \mathbb{R}^{3N}$. For this purpose we do not need to specify the different terms.

We now analyze its form in two limiting cases.

- *Purely kinetic case:* neglecting the potential, the Hamiltonian becomes $\hat{H} = -\frac{1}{2} \sum_i \nabla_i^2$. Define the normalized free-particle propagator without the energy offset:

$$G_{\text{kin}}(\boldsymbol{r}, \boldsymbol{r}', \Delta_\tau) = \left(\frac{1}{2\pi \Delta_\tau} \right)^{\frac{3N}{2}} \exp\left(-\frac{|\boldsymbol{r} - \boldsymbol{r}'|^2}{2\Delta_\tau} \right). \tag{4.9}$$

  This corresponds to a diffusion kernel in configuration space. If $E_T \neq 0$, the full purely kinetic propagator in Eq. (4.7) is $e^{\Delta_\tau E_T} G_{\text{kin}}$. In the splitting below we include this offset in the potential factor instead. This can be derived directly inserting plane waves into Eq. (4.7) with $E_T = 0$ or by using the results we derived in Sect. 2.7.3 with the following substitutions $K \to G_{\text{kin}}$; $f = 0$; $\Delta \to \Delta_\tau$, $k_B T \to 1/2$, namely recognize that the imaginary time Schrödinger equation is formally the heat-equation, or a Fokker-Planck without drift.

- *Purely potential case:* when the kinetic term is neglected and $\hat{H} = V(\boldsymbol{r})$, the Green's function becomes diagonal in real space:

$$G_{\text{pot}}(\boldsymbol{r}, \boldsymbol{r}', \Delta_\tau) = \delta(\boldsymbol{r} - \boldsymbol{r}') \exp\left[-\Delta_\tau \left(V(\boldsymbol{r}) - E_T\right)\right]. \tag{4.10}$$

  This leads to a local exponential rescaling of the wavefunction amplitude. For regions where $V(\boldsymbol{r}) > E_T$, the wavefunction decays, while it grows where $V(\boldsymbol{r}) < E_T$.

These two limiting forms describe, respectively, *diffusion* and *amplitude modulation*, and will be used to construct an approximate propagator valid for small Δ_τ.

4.2 Trotter Formulas

We now introduce the Lie–Trotter product formula, which allows us to approximate the exponential of a sum of non-commuting operators. The exponential of a sum of operators $\hat{A} + \hat{B}$ can be expressed using the *Lie product formula*:

$$e^{-\tau(\hat{A}+\hat{B})} = \lim_{n\to\infty} \left(e^{-\frac{\tau}{n}\hat{A}} e^{-\frac{\tau}{n}\hat{B}}\right)^n. \tag{4.11}$$

This motivates the first-order *Lie–Trotter approximation*[1], which holds for small time steps $\Delta_\tau = \tau/n$:

$$e^{-\Delta_\tau(\hat{A}+\hat{B})} = e^{-\Delta_\tau \hat{A}} e^{-\Delta_\tau \hat{B}} + \mathcal{O}(\Delta_\tau^2). \tag{4.12}$$

In the context of imaginary-time evolution, we identify:

$$\hat{A} = -\frac{1}{2}\sum_{i=1}^{N} \nabla_i^2, \quad \hat{B} = V(\boldsymbol{r}) - E_T,$$

so that $\hat{H} - E_T = \hat{A} + \hat{B}$.

A more accurate second-order formula, known as the symmetric Trotter–Suzuki decomposition, is given by:

$$e^{-\Delta_\tau(\hat{A}+\hat{B})} = e^{-\frac{1}{2}\Delta_\tau \hat{B}} e^{-\Delta_\tau \hat{A}} e^{-\frac{1}{2}\Delta_\tau \hat{B}} + \mathcal{O}(\Delta_\tau^3). \tag{4.13}$$

This formula requires the same number of steps, plus one additional for $\hat{B}$, but achieves a higher accuracy in Δ_τ.[2]

[1] For bounded operators the product formula and the displayed local error orders follow from ordinary operator expansions. For unbounded kinetic and potential operators, convergence requires suitable assumptions on the operator sum or quadratic forms and the associated semigroups. The stated error orders additionally require sufficient regularity and control of the relevant commutators on the states being propagated; they are not automatic operator-norm bounds for singular potentials.

[2] the $B/2$ terms at the boundaries merge into a full B step, except for the initial and final half-step.

Using the kinetic propagator and performing the integrals over the delta functions in the potential kernels, we obtain the approximate full Green's function:

$$G(\boldsymbol{r}, \boldsymbol{r}', \Delta_\tau) \approx e^{-\frac{\Delta_\tau}{2}[V(\boldsymbol{r})-E_T]} G_{\text{kin}}(\boldsymbol{r}, \boldsymbol{r}', \Delta_\tau) e^{-\frac{\Delta_\tau}{2}[V(\boldsymbol{r}')-E_T]}, \tag{4.14}$$

more explicitly,

$$G(\boldsymbol{r}, \boldsymbol{r}', \Delta_\tau) \approx \left(\frac{1}{2\pi\Delta_\tau}\right)^{\frac{3N}{2}} \exp\left[-\frac{(\boldsymbol{r}-\boldsymbol{r}')^2}{2\Delta_\tau}\right] \exp\left[-\frac{\Delta_\tau}{2}\left(V(\boldsymbol{r})+V(\boldsymbol{r}')-2E_T\right)\right]. \tag{4.15}$$

4.3 Implementing the Projection Stochastically: Simple Algorithm

After splitting the total imaginary time τ into n intervals of size Δ_τ, the time-evolved wavefunction is expressed as an n-fold convolution of Green's functions:

$$\begin{aligned}\Psi(\boldsymbol{r}_n, \tau) = \int d\boldsymbol{r}_{n-1} \cdots \int d\boldsymbol{r}_1 \int d\boldsymbol{r}_0 \; G(\boldsymbol{r}_n, \boldsymbol{r}_{n-1}, \Delta_\tau) \cdots \\ \cdots G(\boldsymbol{r}_2, \boldsymbol{r}_1, \Delta_\tau) G(\boldsymbol{r}_1, \boldsymbol{r}_0, \Delta_\tau) \psi_{\text{init}}(\boldsymbol{r}_0),\end{aligned} \tag{4.16}$$

Here $\psi_{\text{init}}(\boldsymbol{r}_0)$ is the initial state. This convolution is exact for the exact G. Replacing each factor by the symmetric short-time kernel gives an accumulated error of order Δ_τ^2 at fixed total time τ, under the regularity assumptions stated above. The whole idea behind DMC is to perform this multi-dimensional integral stochastically.

4.3.1 Single Walker DMC

We introduce the concept of a *walker*: which is a pair of a multi-particle configuration $\boldsymbol{r}_k$ and a weight ω_k that evolves in imaginary time: $(\boldsymbol{r}_k, \omega_k)$. The index k denotes the imaginary-time slice. For this positive-walker construction, assume $\psi_{\text{init}} \geq 0$ and $C = \int d\boldsymbol{r}\, \psi_{\text{init}}(\boldsymbol{r})$ is finite and nonzero. At the beginning sample $\boldsymbol{r}_0$ from $\psi_{\text{init}}(\boldsymbol{r}_0)/C$, and set $\omega_0 = 1$. This samples the initial amplitude, not its square. We can separately consider the action of the two Green functions, to a configuration:

- *Diffusion step*: The walker is displaced according to the kinetic part of the Green's function. This corresponds to a random Gaussian displacement:

  $$\boldsymbol{r}_{k+1} = \boldsymbol{r}_k + \boldsymbol{\eta}, \quad \text{with } \eta_i \sim \mathcal{N}(0, \Delta_\tau) \;\text{ where }\; 1 \leq i \leq 3N.$$

  We have already seen in Sect. 2.7.3 that the kinetic Green function applied is equivalent to a diffusive dynamics.

- *Branching step*: The weight ω_k of the walker is updated based on the potential energy at the current and previous positions. According to the Green's function,

  $$\omega_{k+1} = \omega_k \times \exp\left[-\frac{\Delta_\tau}{2}\left(V(\boldsymbol{r}_{k+1}) + V(\boldsymbol{r}_k) - 2E_T\right)\right].$$

Thus, each walker propagates by random displacement and accumulates a weight depending on the local potential.

Formally, the propagated wavefunction can be reconstructed from the weighted distribution at the final time $k = n$:

$$\Psi(\boldsymbol{r}, \tau = n\Delta_\tau) \approx C \, \langle \omega_n \delta(\boldsymbol{r}_n - \boldsymbol{r}) \rangle \, . \tag{4.17}$$

The average is over independent trajectories of duration n steps, and the approximation is the short-time error of the chosen kernel. The factor C is unnecessary if only the normalized shape is required. This follows from Eq. (4.16). At long projection time, the normalized shape tends to the ground state.

However, it is extremely inefficient to calculate properties from the wavefunction alone, so one is usually interested in the average value of the energy. We are going to provide a method to calculate the energy even for the single-walker method later, but before let's discuss a (small) improvement of the naive algorithm.

4.3.2 Multi-Walker DMC

The main problem of the above method is that the weight at step n is a product of n factors and can develop very large relative fluctuations. A few trajectories can then dominate the weighted averages, even if the mean total weight is controlled.

Now imaginary-time projection is obtained by evolving an ensemble of walkers $\{\boldsymbol{r}_k^{(i)}, \omega_k^{(i)}\}_{i=1}^{N_w}$, where N_w is the number of walkers at time slice k.

Instead of keeping explicit weights for each walker, the *many-walker formulation* simulates the effect of the potential via stochastic replication or deletion of walkers (Fig 4.1). This process is often called **branching**, and replaces the continuous weight update with a discrete population change:

- After the diffusion step $\boldsymbol{r}_{k+1}^{(i)} = \boldsymbol{r}_k^{(i)} + \boldsymbol{\eta}^{(i)}$, each walker's weight factor is computed:

$$w^{(i)} = \exp\left[-\frac{\Delta_\tau}{2}\left(V(\boldsymbol{r}_{k+1}^{(i)}) + V(\boldsymbol{r}_k^{(i)}) - 2E_T\right)\right].$$

- The number of copies $M^{(i)}$ of walker i to retain at step $k+1$ is determined by stochastic rounding:

$$M^{(i)} = \left\lfloor w^{(i)} + z^{(i)} \right\rfloor, \quad z^{(i)} \sim \mathcal{U}[0,1).$$

  Walkers with $M^{(i)} = 0$ are removed; those with $M^{(i)} > 1$ are duplicated accordingly.[3]

However, the number of walkers can grow without bound or the population can die out. Walkers descended from the same parent are correlated. To stabilize the population size N_w, the reference energy E_T is dynamically adjusted during the evolution, ensuring that the

[3]This choice reflects that a walker can be spawned or killed probabilistically. If the weight is vanishing the survival probability is also vanishing, however, walkers with $0 < w < 1$ survive with probability w. In general, $\mathbb{E}[M^{(i)} \mid w^{(i)}] = w^{(i)}$.

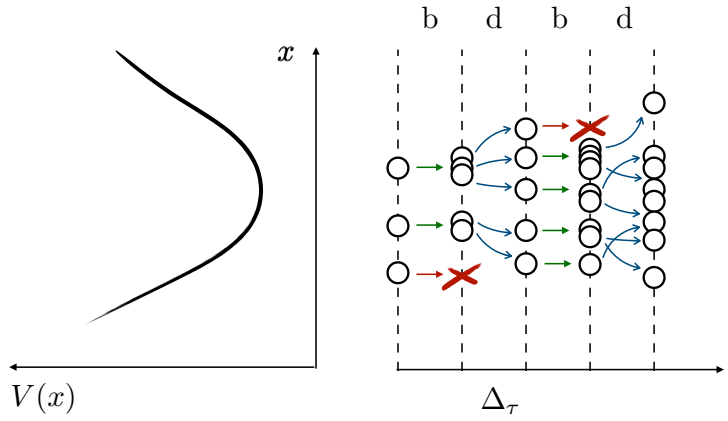


Figure 4.1: Some steps of the multi-walker DMC algorithm, with branching (b) and diffusion (d) steps.

average walker number remains approximately constant. The long-time population growth can be used to estimate the energy, but finite-population feedback introduces a population-control bias. The reference energy is not itself an unbiased energy measurement at each step. This direct algorithm is often inefficient, especially for singular potentials.

To summarize, the problems of the DMC algorithms as presented so far are

- $V(\boldsymbol{r}) - E_T$ may be unbounded in some region, causing wild fluctuation of the weights, and therefore the walker population.
- Estimating the energy from population growth can be noisy and sensitive to population control.
- So far, the formalism only works for nonnegative wavefunctions.

4.4 Sign-Problem in Fermionic DMC and Fixed Node Approximation

4.4.1 1D Example

The diffusion Monte Carlo method evolves a wavefunction $\Psi(\boldsymbol{r}, \tau)$ in imaginary time toward the ground state $\phi_0(\boldsymbol{r})$ of the Hamiltonian. However, for fermionic systems, the full fermionic state is antisymmetric. For the real spatial wavefunctions considered here, exchange of identical same-spin particles generally requires sign changes in configuration space. This introduces the notorious *sign problem*.

To illustrate this, consider a one-dimensional relative coordinate x with a parity-symmetric Hamiltonian. The lowest odd state $\phi_0^{(\mathrm{F})}(x)$, of energy $E_0^{(\mathrm{F})}$, models the antisymmetric spatial state of two identical spin-polarized fermions and has positive and negative regions (see Fig. 4.2). The unconstrained, even, nodeless ground state of the same Hamiltonian, $\phi_0^{(\mathrm{B})}(x)$,

has $E_0^{(\mathrm{B})} < E_0^{(\mathrm{F})}$. This is a one-coordinate illustration; the configuration space of N particles moving on a line is N-dimensional.

We can define the positive and negative parts of a wavefunction, $\psi(x)$ as:

$$\psi_+(x) = \frac{1}{2}\left(|\psi(x)| + \psi(x)\right), \qquad \psi_-(x) = \frac{1}{2}\left(|\psi(x)| - \psi(x)\right). \tag{4.18}$$

Clearly, $\psi(x) = \psi_+(x) - \psi_-(x)$, and these initial components are non-overlapping and non-negative (i.e., $\psi_+(x) \geq 0$, $\psi_-(x) \geq 0$).

Due to the linearity of the Schrödinger equation, one might hope to propagate ψ_+ and ψ_- independently and then subtract the solutions:

$$\psi(x,\tau) = \psi_+(x,\tau) - \psi_-(x,\tau), \tag{4.19}$$

each obeying its own imaginary-time evolution under the same Hamiltonian. After propagation their supports generally overlap; they are no longer the disjoint positive and negative parts of the evolved difference.

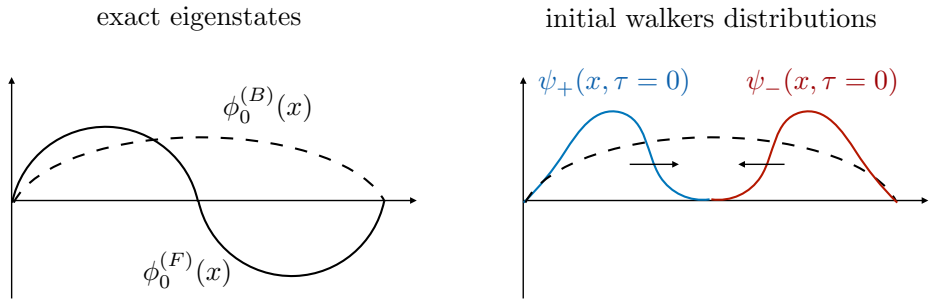


Figure 4.2: The shape of the wavefunctions discussed in the text.

For an odd initial ψ, the sum $\psi_+ + \psi_-$ is even and the difference $\psi_+ - \psi_-$ is odd. Their spectral expansions generally contain many eigenstates, not just the lowest even and odd states. Using Sect. 4.1, their leading long-time terms are

$$\psi_+(x,\tau) + \psi_-(x,\tau) \sim 2c^B e^{-\tau(E_0^{(\mathrm{B})} - E_T)} \phi_0^{(\mathrm{B})}(x), \tag{4.20}$$

$$\psi_+(x,\tau) - \psi_-(x,\tau) \sim 2c^F e^{-\tau(E_0^{(\mathrm{F})} - E_T)} \phi_0^{(\mathrm{F})}(x), \tag{4.21}$$

provided the corresponding overlaps are nonzero. Consequently, away from zeros of the leading terms, the relative signal obeys

$$\frac{|\psi_+(x,\tau) - \psi_-(x,\tau)|}{|\psi_+(x,\tau) + \psi_-(x,\tau)|} \propto e^{-\tau(E_0^{(\mathrm{F})} - E_0^{(\mathrm{B})})}. \tag{4.22}$$

The two large nonnegative contributions must therefore cancel with exponentially increasing relative precision. In many-body systems the Bose–Fermi energy difference is often extensive, so this difficulty typically also grows with system size; extensivity is not a universal statement about every Hamiltonian.

4.4.2 Fixing the Nodal Structure

Algorithmically what happens is that the "positive" walkers, initialized in the ψ_+ region freely diffuse toward the negative region (of the exact ground state) and the opposite.

In principle, knowing the nodal surface would let us constrain the positive and negative walkers to their assigned regions. In the one-coordinate odd-state example the node is $x = 0$, fixed by symmetry; in a general many-body problem the nodes are not known.

For a real wavefunction in dN coordinates, a regular part of its nodal set is generically $(dN-1)$-dimensional. Coincidence of a specified pair of identical same-spin particles, $\boldsymbol{r}_i = \boldsymbol{r}_j$, imposes d constraints and defines a $(dN - d)$-dimensional set. This is a pair-coincidence condition, not simultaneous coincidence of all N particles. Antisymmetry forces the nodal set to contain these pair-coincidence sets.

In one spatial dimension, the ground state of identical spin-polarized fermions with a regular local Hamiltonian on a line has no additional nodes inside each connected particle-ordering sector. Its nodes are therefore fixed by the pair coincidences. This removes an unknown-node approximation in that setting; equality of dimensions alone is not a proof, and it does not mean that every one-dimensional fermion problem has an analytic solution.

The fixed-node approximation imposes a chosen nodal surface, usually that of a real antisymmetric trial wavefunction such as a Slater determinant. It solves the imaginary-time problem with zero (Dirichlet) boundary values on that surface. In the direct diffusion algorithm, trajectories reaching the boundary are absorbed. Checking only the endpoints of a finite diffusion step can miss a crossing, producing an additional time-step error.

The fixed-node energy is variational for this real local Hamiltonian:

$$E_{\mathrm{FN}} \geq E_0^{(\mathrm{F})}. \tag{4.23}$$

This statement concerns the converged fixed-node problem; finite time steps, finite walker populations, and statistical uncertainty can obscure the bound in numerical estimates. Exact ground-state nodes recover the exact fermionic ground-state energy.

4.5 Importance Sampling DMC

The algorithm above misses the important zero-variance property that VMC has. We aim to have a more stable algorithm, and we need a different object to sample from. With a fixed real guiding function ψ_T, we define the importance-sampled function:

$$f(\boldsymbol{r}, \tau) = \psi_T(\boldsymbol{r})\psi(\boldsymbol{r}, \tau). \tag{4.24}$$

For a positive state, or within a fixed nodal pocket where ψ_T and ψ have the same sign, f is nonnegative. Its normalized version $f/\int f$ can be sampled as a probability density; f itself need not integrate to one. Again the trial function $\psi_T(\boldsymbol{r})$ is a fixed state, that could be the final state of a VMC simulation, and does not depend on time. It is possible to see that the imaginary time equation

$$-\frac{\partial\psi(\boldsymbol{r}, \tau)}{\partial\tau} = (\hat{H} - E_T)\psi(\boldsymbol{r}, \tau) = -\frac{1}{2}\nabla^2\psi(\boldsymbol{r}, \tau) + [V(\boldsymbol{r}) - E_T]\psi(\boldsymbol{r}, \tau), \tag{4.25}$$

Here $\nabla^2 = \sum_i \nabla_i^2$ acts on all $3N$ coordinates. Multiplying by ψ_T and substituting $\psi = f/\psi_T$ gives an equation for f (the detailed calculation is in Appendix E). The equation reads

$$\boxed{-\frac{\partial}{\partial\tau} f(\boldsymbol{r},\tau) = -\frac{1}{2}\nabla^2 f(\boldsymbol{r},\tau) + \nabla\cdot(\mathbf{v} f) + [E_L(\boldsymbol{r}) - E_T] f(\boldsymbol{r},\tau)}, \tag{4.26}$$

where we define the so-called *quantum-velocity*,

$$\mathbf{v}(\boldsymbol{r}) = \frac{\nabla\psi_T(\boldsymbol{r})}{\psi_T(\boldsymbol{r})} = \nabla \log|\psi_T(\boldsymbol{r})|, \qquad E_L(\boldsymbol{r}) = \frac{\hat{H}\psi_T(\boldsymbol{r})}{\psi_T(\boldsymbol{r})}, \tag{4.27}$$

where ∇ is the full $3N$-dimensional gradient. These ratios are defined away from the nodes.

Without the branching term, Eq. (4.26) is a Fokker–Planck equation with drift, as in Sect. 2.7.3. With the convention that $\boldsymbol{r}'$ is the starting point and $\boldsymbol{r}$ the endpoint, its short-time Euler drift–diffusion kernel is

$$G_{\mathrm{dd}}(\boldsymbol{r},\boldsymbol{r}',\Delta_\tau) \approx (2\pi\Delta_\tau)^{-3N/2} \exp\left[-\frac{|\boldsymbol{r}-\boldsymbol{r}'-\Delta_\tau\mathbf{v}(\boldsymbol{r}')|^2}{2\Delta_\tau}\right]. \tag{4.28}$$

This Gaussian is exact for constant drift and is a short-time approximation otherwise. In time-slice notation the corresponding update is

$$\boldsymbol{r}_{k+1} = \boldsymbol{r}_k + \Delta_\tau\mathbf{v}(\boldsymbol{r}_k) + \boldsymbol{\eta}, \qquad \eta_i \sim \mathcal{N}(0,\Delta_\tau), \tag{4.29}$$

$$\boldsymbol{r}_{k+1} = \boldsymbol{r}_k + \Delta_\tau\mathbf{v}(\boldsymbol{r}_k) + \sqrt{\Delta_\tau}\,\boldsymbol{z}, \qquad z_i \sim \mathcal{N}(0,1), \qquad 1 \le i \le 3N. \tag{4.30}$$

The components are independent. The Euler approximation of the drift does not in general inherit the second-order global accuracy of the earlier symmetric kinetic–potential splitting.

The *branching* (b) Green's function becomes instead

$$G_{\mathrm{b}}(\boldsymbol{r},\boldsymbol{r}',\Delta_\tau) = \delta(\boldsymbol{r}-\boldsymbol{r}')\exp\left[-\Delta_\tau\left(E_L(\boldsymbol{r}) - E_T\right)\right], \tag{4.31}$$

with the same energy shift E_T as in the evolution equation. We observe that

- The branching term features $E_L(\boldsymbol{r})$ rather than $V(\boldsymbol{r})$. For a good trial state, particularly one satisfying the cusp conditions, its fluctuations are usually much smaller. It need not be bounded near imperfect nodes or uncancelled singularities. If ψ_T is an exact eigenstate, E_L is constant and the energy estimator has zero variance; choosing $E_T = E_L$ also removes branching fluctuations.

- The drift pushes walkers toward regions of larger $|\psi_T|$ and, near a regular node, away from the nodal surface. Nevertheless, a finite Gaussian step can cross a node. A fixed-node calculation must explicitly enforce its nodal boundary condition; in an importance-sampled implementation, proposals ending in a different-sign region are rejected. Endpoint checks do not detect every possible crossing at finite time step. [4]

[4] A Metropolis–Hastings correction using the asymmetric drift proposal can make the unweighted position chain sample $|\psi_T|^2$ at finite step size; see Eq. (2.133). This correction alone neither forbids a sign-changing proposal nor eliminates all DMC time-step errors. The nodal restriction is separate, and the branching approximation still has a time-step bias.

4.5.1 Computing the energy

Another advantage is that the energy is an average of the trial local energy, as in VMC, but with a different sampling distribution. For this derivation take $\psi(0) = \psi_T$, assume a nonzero ground-state overlap, and use self-adjointness and the commutation of $\hat{H}$ with its exponential:

$$E_0 = \frac{\langle\phi_0|\hat{H}|\phi_0\rangle}{\langle\phi_0|\phi_0\rangle} \tag{4.32}$$

$$= \lim_{\tau\to\infty} \frac{\langle\psi_T|e^{-\tau\hat{H}/2}\hat{H}e^{-\tau\hat{H}/2}|\psi_T\rangle}{\langle\psi_T|e^{-\tau\hat{H}/2}e^{-\tau\hat{H}/2}|\psi_T\rangle} \tag{4.33}$$

$$= \lim_{\tau\to\infty} \frac{\langle\psi_T|e^{-\tau\hat{H}}\hat{H}|\psi_T\rangle}{\langle\psi_T|e^{-\tau\hat{H}}|\psi_T\rangle} = \frac{\langle\phi_0|\hat{H}|\psi_T\rangle}{\langle\phi_0|\psi_T\rangle} \tag{4.34}$$

$$= \lim_{\tau\to\infty} \frac{\int d\boldsymbol{r}\, f(\boldsymbol{r},\tau)E_L(\boldsymbol{r})}{\int d\boldsymbol{r}\, f(\boldsymbol{r},\tau)} \approx \frac{1}{M_{\mathrm{smp}}}\sum_{m=1}^{M_{\mathrm{smp}}} E_L(\boldsymbol{r}_m). \tag{4.35}$$

The offset E_T cancels from each ratio. The last average assumes that the configurations have equal weights and sample the normalized mixed distribution $f/\int f$. For explicitly weighted walkers it must instead be $\sum_m \omega_m E_L(\boldsymbol{r}_m)/\sum_m \omega_m$.

For fixed-node DMC, replace ϕ_0, E_0 by $\phi_{\mathrm{FN}}, E_{\mathrm{FN}}$ and the propagator by that of the Dirichlet fixed-node Hamiltonian. The trial state obeys the same nodal boundary condition; its local energy inside a pocket still uses the original local differential operator. The identities refer to converged projection, vanishing time-step error, and controlled population bias.

The mixed estimator is exact for the energy and, more generally, for observables for which the projected state is an eigenstate, including operators commuting with $\hat{H}$ when the ground state is nondegenerate. It is generally biased for other observables. For a Hermitian $\hat{A}$, a useful extrapolation is

$$\frac{\langle\phi_0|\hat{A}|\phi_0\rangle}{\langle\phi_0|\phi_0\rangle} \approx 2\,\mathrm{Re}\,\frac{\langle\phi_0|\hat{A}|\psi_T\rangle}{\langle\phi_0|\psi_T\rangle} - \frac{\langle\psi_T|\hat{A}|\psi_T\rangle}{\langle\psi_T|\psi_T\rangle}. \tag{4.36}$$

For normalized states with aligned overall phase, the remaining error is of order $\|\psi_T - \phi_0\|^2$, assuming the matrix elements admit this expansion. For real states and real observables the mixed matrix element is real, so the real-part symbol can be omitted. This extrapolation is not an exact or variational estimator for a general observable; in fixed-node DMC it refers to ϕ_{FN}.

4.5.2 Single-Walker Implementation

We can now return to the single-walker implementation. The position follows the drift–diffusion process without branching, while weights are retained for estimating projected averages.

Instead of independent trajectories of ℓ steps, generate one long trajectory of length $\mathcal{N} \gg \ell$ and use overlapping windows. The unweighted continuous-time drift process has stationary density proportional to $|\psi_T|^2$, not the mixed distribution f. A Metropolis-corrected position

chain has that same stationary density at finite step size; an uncorrected Euler chain generally has a finite-step error. Resetting the weight to one at the start of each window initiates a new finite projection from the stationary position distribution. The endpoint weighting produces the corresponding projected distribution.

For a simple implementation, use the first-order branching factor at the *departure* point of each step. During the trajectory store

- the local energies $E_L(\boldsymbol{r}_i) \equiv e_i$;
- the one-step factors
$$b(\boldsymbol{r}_i) = \exp[-\Delta_\tau (E_L(\boldsymbol{r}_i) - E_T)] \equiv b_i. \tag{4.37}$$

The energies are stored for $0 \le i \le \mathcal{N}$ and departure factors for $0 \le i < \mathcal{N}$. Hold E_T fixed for this post-processing construction. The departure-point factor is a first-order choice; a symmetric branching weight would instead depend on both endpoints of each link.

After discarding an initial equilibration interval of i_0 steps, estimate

$$\widehat{E}_{\ell,\Delta_\tau} = \frac{\displaystyle\sum_{i=i_0+\ell}^{\mathcal{N}} \mathcal{G}_i^\ell e_i}{\displaystyle\sum_{i=i_0+\ell}^{\mathcal{N}} \mathcal{G}_i^\ell}, \qquad \mathcal{G}_i^\ell = \prod_{j=1}^{\ell} b_{i-j}. \tag{4.38}$$

The weights cover the ℓ departures ending at point i. All windows must start after equilibration. A fixed E_T contributes the same factor to every window of length ℓ and therefore cancels from this ratio.

For an ergodic chain and well-defined weighted expectations, the formal exact limit is

$$E_0 = \lim_{\Delta_\tau \to 0} \lim_{\ell \to \infty} \lim_{\mathcal{N} \to \infty} \widehat{E}_{\ell,\Delta_\tau}, \tag{4.39}$$

with E_0 replaced by E_{FN} for fixed nodes. In practice one checks convergence in the projection time $\ell\Delta_\tau$ while keeping many well-sampled windows, and separately checks the time-step bias. Taking $\ell \to \infty$ at fixed trajectory length is not meaningful and usually leads to domination by a few weights.

The error bars must account for both the trajectory correlations and the overlap of adjacent windows. Block the pairs $(\mathcal{G}_i^\ell e_i, \mathcal{G}_i^\ell)$ and estimate the uncertainty of their ratio, for example with a block jackknife. Binning e_i alone does not give the uncertainty of this weighted estimator. Blocks must be long enough to resolve the correlations, which can extend over the projection window and beyond.

Exercise 4.5.1: Single-walker DMC for the Helium atom (/***)**

Consider the Hamiltonian and trial wavefunction of Exercise 3.1.5.

1. Write a single-walker importance sampling code, implementing the drift-diffusion for the walker position dynamics, with e_i, and b_i as output.
2. Postprocess the weighted energy and its uncertainty. Check convergence as $\ell\Delta_\tau$ increases and as Δ_τ decreases. A very large ℓ can make a few windows dominate the result.
3. (Bonus) Implement the Metropolis–Hastings correction for the asymmetric drift proposal. It corrects position sampling, but a DMC time-step extrapolation is still required.
4. Compare with the best VMC energy and the exact value. Does the converged DMC energy of this nodeless helium ground state depend on the parameters of a positive trial function? Distinguish this question from dependence of the efficiency and finite-projection or finite-step errors on those parameters.

4.6 Green's Function Monte Carlo

The counterpart of DMC for lattice models is called *Green's function MC* (GFMC). We are not going to reintroduce the whole idea of stochastically sampling the imaginary-time evolution, but simply adapt the concept to the discrete world. The algorithm is the stochastic implementation of the power method of Sect. 1.5. We are going to directly introduce the importance sampling variant. We work in a finite discrete basis and assume H is real symmetric and that the chosen basis permits nonnegative propagation weights. The real guiding function is taken strictly positive on the sampled connected sector. Given a configuration $|x\rangle$, we denote its connected elements by $|x'\rangle$. These are the configurations for which the matrix element

$$H_{x,x'} = \langle x'|H|x\rangle \tag{4.40}$$

is nonzero. Their number depends on the Hamiltonian; it is not generally exactly L. The diagonal configuration x must also be included when constructing the propagator.

The power-method projection is based on the unnormalized kernel

$$G_{x,x'} = \langle x'|\Lambda\mathbb{I} - H|x\rangle = \Lambda\delta_{x,x'} - H_{x,x'}. \tag{4.41}$$

The matrix elements must be *entrywise nonnegative* for positive sampling weights; this is different from positive definiteness. The kernel is not yet a transition matrix because its rows need not sum to one. For the positive guide assumed here, nonpositive off-diagonal $H_{x,x'}$ are required. Increasing Λ changes only the diagonal and cannot repair an off-diagonal sign problem.

For a finite connected sector with these signs, choosing $\Lambda > \max_x H_{x,x}$ gives positive diagonal entries and an irreducible, aperiodic nonnegative kernel. Its dominant eigenvector is the ground state of H. Powers of this kernel project onto that state; they are not an exact exponential propagator at a specified finite imaginary-time step.

We then introduce the *importance sampling* GFMC:

$$\bar{G}_{x,x'} = \frac{\psi_T(x')}{\psi_T(x)} G_{x,x'} \tag{4.42}$$

The transition probability matrix used to update a walker is given by

$$p_{x,x'} = \bar{G}_{x,x'}/b_x \tag{4.43}$$

$$b_x = \sum_{x'} \bar{G}_{x,x'} = \Lambda - E_L(x) > 0 \tag{4.44}$$

Here

$$E_L(x) = \frac{\langle x|H|\psi_T\rangle}{\psi_T(x)} = \sum_{x'} H_{x,x'} \frac{\psi_T(x')}{\psi_T(x)}, \tag{4.45}$$

where the second equality uses the real symmetry of H with the index convention above. The normalization b_x is the one-step weight at the departure configuration x. The entries $p_{x,x'} \geq 0$ sum to one, so they define a transition probability. Notice that we can numerically compute $\bar{G}_{x,x'}$ because we can evaluate to numerical precision the ratio $\psi_T(x')/\psi_T(x)$.

In practice, at configuration x, we need to select one of the connected configuration x' with probability $p_{x,x'}$. Operatively, this is simply done by computing and stacking all the $p_{x,x'}$'s (which sum gives 1) and extract a random number z between 0 and 1 and see in which interval it falls. For instance, suppose that $a, b, \cdots$ are the connected elements to x. We ideally divide the segment [0,1) as it follows

$$[0, p_{x,x}) \bigcup [p_{x,x}, p_{x,x} + p_{x,a}) \bigcup [p_{x,x} + p_{x,a}, p_{x,x} + p_{x,a} + p_{x,b}) \cdots . \tag{4.46}$$

The single-walker implementation uses the weighted-window recipe of Sect. 4.5.2, with $e_i = E_L(x_i)$ and $b_i = b_{x_i}$ recorded *before* sampling x_{i+1}. The unweighted chain does not by itself sample the mixed ground-state distribution. Indeed, for the real symmetric kernel above, detailed balance gives its stationary distribution proportional to $\psi_T(x)^2 b_x$. Weighting each window by the product of its departure factors reconstructs powers of $\bar{G}$ and approaches the mixed distribution for long projection windows. Here ℓ counts power iterations; there is no continuum time-step error from a Trotter approximation, although finite projection and statistical errors remain.

Exercise 4.6.1: GFMC for the TFIM (**)

Consider the TFIM of Exercise 3.2.1.

1. Write a single-walker GFMC code that implements the walk in imaginary time. Use $L = 20$ to begin with.

2. Postprocess the e_x and b_x value to obtain the ground state energy.

Chapter 5

Path Integral Monte Carlo

In this Chapter we are going to present QMC algorithms based on the path-integral formalism of quantum mechanics. These are used to simulate finite-temperature systems but also ground states. As usual, we cover both continuous-variable and lattice models. We use $\hbar = k_B = 1$ throughout. The mass m is kept explicit. Unless otherwise stated, positive-weight sampling assumes a real scalar potential in continuous space, or a sign-free Hamiltonian in the chosen lattice basis. The trace identities remain valid more generally, but negative or complex weights require sign or phase reweighting.

5.1 Finite Temperature Formalism

The equilibrium properties of a quantum system at temperature $T = 1/\beta$ are encoded in the unnormalized density operator:

$$\hat{\rho} = e^{-\beta \hat{H}}, \tag{5.1}$$

where $\hat{H}$ is the Hamiltonian operator. The normalized thermal density operator is $\hat{\rho}/Z$. The partition function is given by

$$Z = \mathrm{Tr}(e^{-\beta \hat{H}}). \tag{5.2}$$

Let $\{\phi_i\}$ denote the complete orthonormal set of eigenstates of $\hat{H}$, such that

$$\hat{H}|\phi_i\rangle = E_i|\phi_i\rangle. \tag{5.3}$$

In this basis, the density matrix can be written as

$$\hat{\rho} = \sum_i e^{-\beta E_i} |\phi_i\rangle\langle\phi_i|, \tag{5.4}$$

and the partition function becomes

$$Z = \sum_i e^{-\beta E_i}. \tag{5.5}$$

If we instead use the coordinate basis $\{|x\rangle\}$, where x denotes a configuration of the system, the matrix elements of the density operator are given by

$$\rho(x, x') = \langle x|e^{-\beta\hat{H}}|x'\rangle. \tag{5.6}$$

In the case of discrete coordinates, the trace becomes a sum:

$$Z = \sum_x \rho(x, x), \tag{5.7}$$

while for continuous degrees of freedom, it becomes an integral:

$$Z = \int dx\, \rho(x, x). \tag{5.8}$$

The expectation value of an operator $\hat{O}$ at finite temperature is given by

$$\langle\hat{O}\rangle = \frac{1}{Z}\mathrm{Tr}\left(\hat{O}e^{-\beta\hat{H}}\right), \tag{5.9}$$

which in the basis of the eigenstates becomes

$$\langle\hat{O}\rangle = \frac{1}{Z}\sum_i \langle\phi_i|\hat{O}|\phi_i\rangle e^{-\beta E_i} \tag{5.10}$$

whereas, in the coordinate basis, this becomes

$$\langle\hat{O}\rangle = \frac{1}{Z}\int dx\, \langle x|\hat{O}e^{-\beta\hat{H}}|x\rangle, \tag{5.11}$$

or, when $\hat{O}$ is diagonal in the coordinate basis,

$$\langle\hat{O}\rangle = \frac{1}{Z}\int dx\, O(x)\, \rho(x, x). \tag{5.12}$$

In general, the Hamiltonian may consist of several non-commuting parts,

$$\hat{H} = \hat{H}_1 + \hat{H}_2 + \hat{H}_Q, \tag{5.13}$$

where $\hat{H}_1$, $\hat{H}_2$, and $\hat{H}_Q$ do not commute with each other. In the following, we will explicitly develop the formalism needed to handle such cases directly in the basis chosen.

5.2 Path Integrals in Continuous Variable Models

We start from the continuous variable case since we have most of the ingredients already prepared in Chapter 4. Let's start from the density matrix,

$$\rho(\boldsymbol{r}, \boldsymbol{r'}; \beta) = \langle\boldsymbol{r}|e^{-\beta\hat{H}}|\boldsymbol{r'}\rangle = \sum_i \phi_i(\boldsymbol{r})\phi_i^*(\boldsymbol{r'})e^{-\beta E_i}, \tag{5.14}$$

where we explicitly indicate the parameter β, and we used Eq 5.4 for the last step. However, given that we do not know[1] in principle the eigenstates of the Hamiltonian, we need to try to calculate the matrix element directly in the coordinate basis. Gladly we already encountered this object in the previous Chapter, see Eq. 4.7, namely the imaginary-time Green's function. If $G(\beta) = e^{-\beta(\hat{H}-E_T)}$, the present kernel is $\rho(\beta) = e^{-\beta E_T} G(\beta)$. Thus the two coincide when $E_T = 0$, which we assume in this comparison.

We observe that the β parameters can be understood as an *imaginary time* finite interval, therefore we split it in P smaller intervals[2] of size $\Delta_\tau = \beta/P = 1/(PT)$. The matrix element between two configurations $\boldsymbol{r} \equiv \boldsymbol{r}_0$ and $\boldsymbol{r'} \equiv \boldsymbol{r}_P$ can be written in the same we wrote Eq. 4.16,

$$\rho(\boldsymbol{r}_0, \boldsymbol{r}_P; \beta) = \int d\boldsymbol{r}_1 \int d\boldsymbol{r}_2 \ \cdots \int d\boldsymbol{r}_{P-1} \ \rho(\boldsymbol{r}_0, \boldsymbol{r}_1, \Delta_\tau)\rho(\boldsymbol{r}_1, \boldsymbol{r}_2, \Delta_\tau) \cdots \rho(\boldsymbol{r}_{P-1}, \boldsymbol{r}_P, \Delta_\tau), \tag{5.15}$$

with the difference that, this time we do not integrate over $\boldsymbol{r}_0$ since it is a fixed input. Performing the Trace simply means to identify $\boldsymbol{r}_0 = \boldsymbol{r}_P$ and integrate over this missing coordinate, hence

$$Z_\beta = \int d\boldsymbol{r}_0 \int d\boldsymbol{r}_1 \int d\boldsymbol{r}_2 \ \cdots \int d\boldsymbol{r}_{P-1} \ \rho(\boldsymbol{r}_0, \boldsymbol{r}_1, \Delta_\tau)\rho(\boldsymbol{r}_1, \boldsymbol{r}_2, \Delta_\tau) \cdots \rho(\boldsymbol{r}_{P-1}, \boldsymbol{r}_0, \Delta_\tau). \tag{5.16}$$

So far, this is still exact. The reason why we introduce this format it because we hope to compute the *short-time* density matrix $\rho(\boldsymbol{r}_k, \boldsymbol{r}_{k+1}, \Delta_\tau)$, with $0 \leq k \leq P-1$, using the Trotter-Lie approximation, thus separating the two non-commuting parts of $\hat{H}$.

For N distinguishable particles of equal mass, with $\hat{H} = -\nabla^2/(2m) + V(\boldsymbol{r})$, the primitive short-time approximation gives $Z_P \to Z$ as $P \to \infty$ at fixed β. Inserting the results obtained in Chapter 4, we obtain

$$Z_P = \left(\frac{m}{2\pi\Delta_\tau}\right)^{\frac{3NP}{2}} \int \prod_{j=0}^{P-1} d\boldsymbol{r}_j \ \exp\left[-\sum_{j=0}^{P-1}\left(\frac{m}{2\Delta_\tau}(\boldsymbol{r}_{j+1} - \boldsymbol{r}_j)^2 + \Delta_\tau \, V(\boldsymbol{r}_j)\right)\right], \tag{5.17}$$

where we assumed that the position $\boldsymbol{r}$, collects all the $3N$ system's coordinates. In arbitrary dimensions one simply replaces $3 \to d$. Adding $P-1$ replicas converts the quantum trace into an integral of products of short-time kernels. This local structure makes Monte Carlo sampling possible even when the full density matrix cannot be constructed.

5.2.1 Ring-polymer isomorphism

In the path integral formulation of quantum statistical mechanics, the thermal properties of a particle are obtained by summing over all possible paths $\boldsymbol{r}(\tau)$ the particle can take in imaginary time $\tau \in [0, \beta]$, where $\beta = 1/T$. A *path* refers to the projection of trajectory of the

[1] This representation is useful to get exact results from exact diagonalization of small systems.

[2] we use the symbol Δ_τ to maintain consistency with the previous Chapter. The β of this Chapter is the τ of DMC.

particle in configuration space, parameterized by imaginary time. The quantum partition function is expressed as a path integral over these trajectories, weighted by the Euclidean action:

$$Z = \int \mathcal{D}[\boldsymbol{r}(\tau)]\, e^{-S[\boldsymbol{r}(\tau)]}, \tag{5.18}$$

where the action is defined as

$$S[\boldsymbol{r}(\tau)] = \int_0^\beta \left[\frac{m}{2} \left(\frac{d\boldsymbol{r}}{d\tau} \right)^2 + V(\boldsymbol{r}(\tau)) \right] d\tau, \tag{5.19}$$

with periodic boundary conditions. This is a formal continuum notation: typical paths have Brownian fluctuations and are not differentiable. The kinetic action and the measure are defined through the discretized propagators.

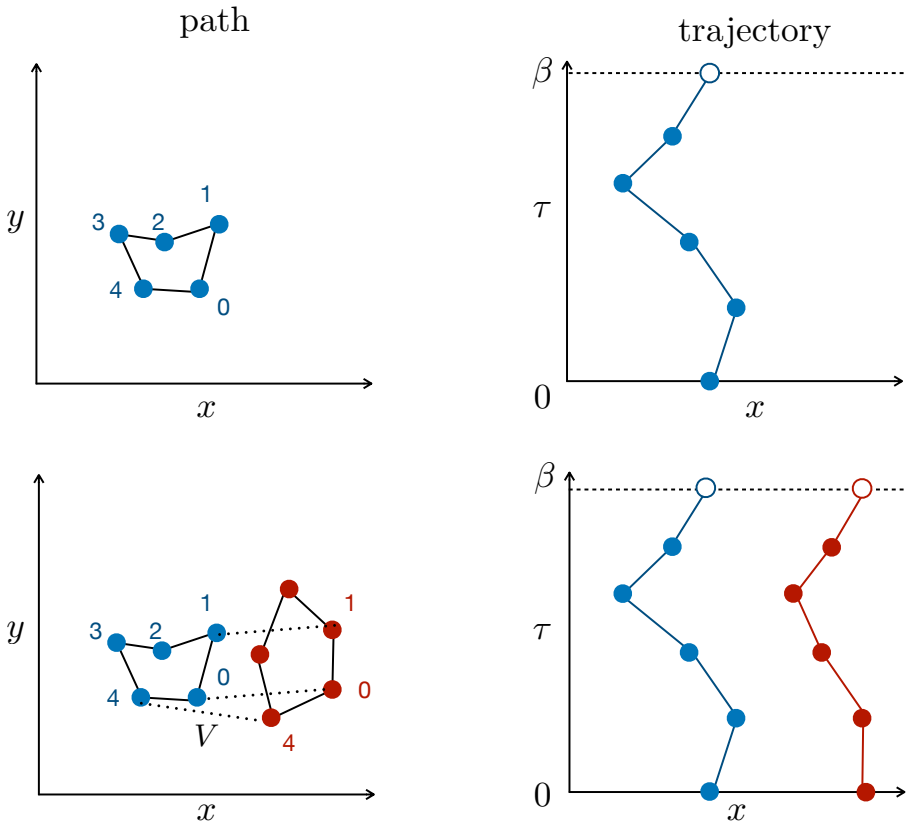


Figure 5.1: Path and trajectories in imaginary time for a single and two particles systems. The time is discretized using 5 steps for visualization sake. The inter-particle potential V acts only at *equal-times*. Replicas at different imaginary times do not interact directly through the instantaneous potential. The schematic illustrates the connectivity, not a physical real-time trajectory. The trajectory plot is also called *world-line* representation. Notice the periodic-boundary conditions in imaginary-time.

To evaluate the path integral numerically, we discretized imaginary time into P slices of width $\Delta_\tau = \beta/P$, and approximate the path $\boldsymbol{r}(\tau)$ by a set of P coordinates $\{\boldsymbol{r}_0, \boldsymbol{r}_1, \ldots, \boldsymbol{r}_{P-1}\}$. These coordinates are interpreted as the positions of P replicas (or "beads") of the particle at successive imaginary time points. The discrete action given by

$$S_P[\{\boldsymbol{r}_j\}] = \sum_{j=0}^{P-1} \left[\frac{m}{2\Delta_\tau} (\boldsymbol{r}_{j+1} - \boldsymbol{r}_j)^2 + \Delta_\tau \, V(\boldsymbol{r}_j) \right], \tag{5.20}$$

and periodic boundary conditions $\boldsymbol{r}_P = \boldsymbol{r}_0$ to reflect the trace in the partition function.

The discretized path integral has a compelling classical analogy: the discretized weight maps exactly to a classical ring of P replicas. For N physical particles this means N polymers, each with P beads; $\boldsymbol{r}_j$ collects all particles at slice j. The remaining difference from the quantum system is the finite-P approximation. Each bead $\boldsymbol{r}_j$ represents the configuration of the quantum particle at the imaginary time slice $\tau_j = j\Delta_\tau$, and the harmonic terms $(\boldsymbol{r}_{j+1} - \boldsymbol{r}_j)^2$ arise from the kinetic energy operator. The potential energy $V(\boldsymbol{r}_j)$ is evaluated independently at each bead, as if each bead "feels" the external potential separately. This structure is known as the *ring-polymer*.

It's important to distinguish between *imaginary time* τ and the *simulation time* used during the Monte Carlo process. Imaginary time serves as a label indexing the beads in the ring-polymer, each bead corresponds to a different point in the discretized path in imaginary time. In contrast, simulation time refers to the sequence of configurations generated during the Monte Carlo sampling process. In PIMC, we use suitable Monte Carlo moves to explore the configuration space of paths. The goal of this sampling is to generate an ensemble of ring-polymers whose statistical distribution reproduces the quantum thermal density matrix.

From the classical isomorphism, we gain an intuitive picture of quantum fluctuations. At fixed P, increasing β reduces the kinetic coefficient $m/(2\Delta_\tau)$, so the beads are less tightly constrained to coincide. At fixed Δ_τ, instead, increasing β adds beads. In the high-temperature classical regime the internal polymer extent becomes small. In a confining potential its low-temperature extent approaches a finite quantum scale rather than growing without bound. The internal width and the fluctuations of the centroid are distinct. Defining

$$\boldsymbol{r}_c = \frac{1}{P} \sum_{j=0}^{P-1} \boldsymbol{r}_j, \tag{5.21}$$

one has the exact decomposition

$$\left\langle \frac{1}{P} \sum_j |\boldsymbol{r}_j - \langle \boldsymbol{r}_c \rangle|^2 \right\rangle = \left\langle \frac{1}{P} \sum_j |\boldsymbol{r}_j - \boldsymbol{r}_c|^2 \right\rangle + \left\langle |\boldsymbol{r}_c - \langle \boldsymbol{r}_c \rangle|^2 \right\rangle. \tag{5.22}$$

Thus the internal spread measures quantum delocalization, while the centroid also contributes to the total positional fluctuations.

5.2.2 A Simple Energy Estimator

Physical observables are computed as ensemble averages over the space of discretized paths. For the total energy E, one commonly used approach is the *thermodynamic estimator*, which is derived by taking the derivative of the partition function with respect to the inverse temperature β. Specifically, the total energy is given by:

$$E = -\frac{\partial}{\partial \beta} \ln Z. \tag{5.23}$$

Differentiating Z_P at fixed P (so $\partial_\beta \Delta_\tau = 1/P$) yields the thermodynamic estimator for N particles in d dimensions:

$$E_{P,\text{thermo}} = \left\langle \frac{dN}{2\Delta_\tau} - \frac{m}{2P} \sum_{j=0}^{P-1} \frac{(\boldsymbol{r}_{j+1} - \boldsymbol{r}_j)^2}{\Delta_\tau^2} + \frac{1}{P} \sum_{j=0}^{P-1} V(\boldsymbol{r}_j) \right\rangle, \tag{5.24}$$

where the angle brackets denote an ensemble average over ring-polymer configurations sampled during the Monte Carlo simulation.

The estimator consists of two contributions: a kinetic term (involving the squared distances between adjacent beads) and a potential term (averaged over all beads). This estimator is not very efficient at the interesting limit of small Δ_τ, as the first term clearly diverges as $1/\Delta_\tau$ and this needs to be cancelled by the also diverging second term. A useful alternative is the *centroid-virial estimator*. For the distinguishable-particle paths considered here, in Cartesian space with a sufficiently regular confining potential, it reads

$$E_{P,\text{cv}} = \left\langle \frac{dN}{2\beta} + \frac{1}{P} \sum_{j=0}^{P-1} \left[V(\boldsymbol{r}_j) + \frac{1}{2} (\boldsymbol{r}_j - \boldsymbol{r}_c) \cdot \nabla V(\boldsymbol{r}_j) \right] \right\rangle. \tag{5.25}$$

Both estimators give $-\partial_\beta \ln Z_P$, but the second avoids cancelling two terms of order P. Its statistical performance depends on the system; the important practical advantage is that its variance does not suffer the same growth with the number of beads in the usual smooth-potential applications. Appendix F.1 derives the identity. Boundary terms, winding paths in a periodic box, and exchange cycles require appropriate modifications to that derivation.

5.2.3 Monte Carlo Sampling and Path Updates

Efficient sampling of the path space is essential for the convergence the simulations. The goal is to construct a Markov chain whose stationary distribution is proportional to the discretized path integral weight e^{-S_P}. Various Monte Carlo move types are employed to explore the configuration space of ring polymers, ranging from simple local updates to more global proposals.

The most basic type of Monte Carlo move is the **local** bead displacement. In this move, a single bead $\boldsymbol{r}_j$ is randomly displaced to a new position $\boldsymbol{r}'_j = \boldsymbol{r}_j + \delta$, where δ is drawn from a uniform or Gaussian distribution centered at zero. The Metropolis acceptance probability for this move is given by

$$A = \min\left(1, \exp\left[-\Delta S_j\right]\right), \tag{5.26}$$

where $\Delta S_j = S_P[\ldots, \boldsymbol{r}'_j, \ldots] - S_P[\ldots, \boldsymbol{r}_j, \ldots]$ is the change in the discretized action resulting from the displacement of bead j. Only the terms in the action involving $\boldsymbol{r}_{j-1}, \boldsymbol{r}_j, \boldsymbol{r}_{j+1}$ need to be recomputed.

Given that the system is purely classical we can use all range of **molecular dynamics** methods, with a thermostat. For instance we can use a simple overdamped Langevin dynamics as described in Sect 2.7.2, which for bead j, takes the form

$$\boldsymbol{r}'_j = \boldsymbol{r}_j - \gamma \nabla_{\boldsymbol{r}_j} S_P + \sqrt{2\gamma}\,\eta, \tag{5.27}$$

where γ is an algorithmic step-size parameter[3], $\nabla_{\boldsymbol{r}_j} S_P$ is the local gradient of the action, and η is a Gaussian noise term with zero mean and unit variance.

This Euler discretization has a finite-step sampling bias. To sample e^{-S_P} exactly at a nonzero algorithmic step, regard it as a Gaussian trial density $T(X'|X)$ and accept with

$$A(X \to X') = \min\left\{1, e^{-S_P(X')+S_P(X)} \frac{T(X|X')}{T(X'|X)}\right\}. \tag{5.28}$$

Alternatively, extrapolate the algorithmic step to zero. This sampling error is independent of the physical imaginary-time discretization Δ_τ.

To improve the efficiency of sampling long-wavelength fluctuations in the path, global updates such as the **bisection algorithm** are used. This method leverages the exact sampling of the free-particle (kinetic) path integral and updates multiple beads simultaneously.

In the bisection algorithm, a path segment between two fixed endpoints $\boldsymbol{r}_j$ and $\boldsymbol{r}_{j+\ell}$ is updated by recursively sampling intermediate points using the exact free-particle propagator. The total imaginary time between the endpoints is $\Delta = \ell \Delta_\tau$.

The midpoint $\boldsymbol{r}_{j+\ell/2}$, assuming ℓ is even, is sampled from the conditional probability distribution:

$$\boldsymbol{r}_{j+\ell/2} \sim \mathcal{N}\left(\frac{\boldsymbol{r}_j + \boldsymbol{r}_{j+\ell}}{2}, \frac{\Delta}{4m}\right), \tag{5.29}$$

where the variance in each Cartesian component $\sigma^2 = \Delta/(4m) = \ell\Delta_\tau/(4m)$ takes into account that the imaginary time step that separates the mid-point from the extrema is $\ell\Delta_\tau/2$, and is bigger than Δ_τ.
After proposing this midpoint, repeat recursively on the two subsegments until all interior beads have been replaced. Choosing $\ell = 2^q$ makes all subdivisions equal; more general segment lengths require the corresponding unequal-time Brownian-bridge conditionals. In absence of a potential, this new path configuration would be accepted with probability 1. Once the full segment is proposed using the kinetic propagator, the move is accepted or rejected based on the change in potential energy along the segment:

$$A = \min\left(1, \exp\left[-\Delta S_V\right]\right), \tag{5.30}$$

where $\Delta S_V = \sum_{\text{interior } j} \Delta_\tau [V(\boldsymbol{r}_j^{\text{new}}) - V(\boldsymbol{r}_j^{\text{old}})]$. The free bridge is not a symmetric proposal: its reverse/forward ratio cancels the change of kinetic weight exactly. This leaves only the potential factor, provided the segment-selection rule is symmetric.

[3] we use this symbol to avoid confusion with the imaginary time step Δ_τ

5.2.4 Path Integral Ground State

The Path Integral Ground State (PIGS) method, also known as Variational Path Integral (VPI), projects a ground state from a trial wavefunction ψ_T. In contrast to finite-temperature PIMC, its paths are open and their endpoint weights are determined by ψ_T.

Assume a nondegenerate ground state and $\langle\psi_0|\psi_T\rangle \neq 0$. The normalized projection is

$$|\psi_\tau\rangle = \frac{e^{-\tau\hat{H}}|\psi_T\rangle}{\sqrt{\langle\psi_T|e^{-2\tau\hat{H}}|\psi_T\rangle}} \longrightarrow e^{i\varphi}|\psi_0\rangle \qquad (\tau\to\infty), \tag{5.31}$$

where $e^{i\varphi}$ is an irrelevant phase. For a degenerate ground space the projection selects the component of ψ_T in that space. To compute expectation values, the projection acts on both sides of the observable:

$$\langle\hat{O}\rangle_\tau = \frac{\langle\psi_T|e^{-\tau\hat{H}}\hat{O}e^{-\tau\hat{H}}|\psi_T\rangle}{\langle\psi_T|e^{-2\tau\hat{H}}|\psi_T\rangle}. \tag{5.32}$$

At finite τ this is a projected trial-state expectation; the ground-state value requires $\tau\to\infty$.

The denominator uses the same bulk propagator as a thermal path integral of length 2τ, but it is a boundary-to-boundary amplitude, not a trace. Put $2\tau = P\Delta_\tau$, with P even. Inserting intermediate configurations gives

$$\langle\psi_T|e^{-2\tau\hat{H}}|\psi_T\rangle = \int\prod_{j=0}^{P} d\boldsymbol{r}_j\ \psi_T^*(\boldsymbol{r}_0)\prod_{j=0}^{P-1}\rho(\boldsymbol{r}_j,\boldsymbol{r}_{j+1};\Delta_\tau)\psi_T(\boldsymbol{r}_P). \tag{5.33}$$

Both endpoints are integrated over and sampled, rather than held at fixed coordinates. This identity permits complex trial states, but ordinary positive-weight PIGS requires nonnegative bulk kernels and endpoint weights. In the following sampling formulas we therefore take ψ_T real and positive on the sampled domain. Fermions require a separate sign treatment or a nodal approximation.

Using a symmetric primitive propagator, define the internal action

$$\begin{aligned} S_{\text{open}} = &\sum_{j=0}^{P-1}\frac{m|\boldsymbol{r}_{j+1}-\boldsymbol{r}_j|^2}{2\Delta_\tau} \\ &+\Delta_\tau\left[\frac{1}{2}V(\boldsymbol{r}_0)+\sum_{j=1}^{P-1}V(\boldsymbol{r}_j)+\frac{1}{2}V(\boldsymbol{r}_P)\right]. \end{aligned} \tag{5.34}$$

The endpoint potentials have half weight. Including the trial state gives the full sampling action

$$S_{\text{PIGS}} = S_{\text{open}} - \ln\psi_T(\boldsymbol{r}_0) - \ln\psi_T(\boldsymbol{r}_P). \tag{5.35}$$

For an observable diagonal in the coordinate basis, measuring $O(\boldsymbol{r}_{P/2})$ implements the symmetric estimator. Off-diagonal observables require a kernel insertion or another appropriate estimator; evaluating a coordinate function is not sufficient.

Monte Carlo sampling uses the same local and bridge ideas as PIMC. A general trial move $T(X'|X)$ has acceptance

$$A = \min\left\{1, \frac{\psi_T(\boldsymbol{r}_0')\psi_T(\boldsymbol{r}_P')}{\psi_T(\boldsymbol{r}_0)\psi_T(\boldsymbol{r}_P)}e^{-S_{\text{open}}(X')+S_{\text{open}}(X)}\frac{T(X|X')}{T(X'|X)}\right\}. \tag{5.36}$$

For symmetric displacements the proposal ratio is one. Free-bridge proposals instead cancel their corresponding kinetic factors as above.

If $|\psi_T\rangle = \sum_n c_n|\psi_n\rangle$, each excited amplitude relative to the ground-state amplitude is $(c_n/c_0)e^{-\tau(E_n-E_0)}$. A good trial state reduces the coefficients, while small relevant gaps can still require long projection times. Both the projection length and Δ_τ must be checked independently.

5.2.5 Conceptual Difference Between DMC and PIMC

We are in the position now to observe the difference in two algorithms that stems from the same idea: imaginary-time evolution.

In **DMC**, the system is represented by a large population of *walkers*, each corresponding to a single configuration in real space. These walkers are propagated forward in small imaginary-time steps, using a stochastic process. Here algorithmic propagation steps advance the imaginary projection time. They are not physical real-time evolution. Ground-state properties emerge from the long-time asymptotic behavior of the walker distribution.

In contrast, **PIMC** represents a single quantum system as an extended object in imaginary time: a closed *path* or *ring polymer* consisting of P time slices. All imaginary-time configurations are carried along simultaneously as part of the path. In this formulation, imaginary time is treated as an additional coordinate, and the *Monte Carlo simulation time* is distinct from imaginary time. The simulation explores the space of full paths through sampling updates.
This key difference reflects a broader conceptual split: DMC represents imaginary-time projection by an evolving population, whereas PIMC treats it as a statistical ensemble over full imaginary-time trajectories.

5.3 Implementing Symmetries in Real-Space PIMC: Bosons and Permutations

What we described so far treats particles as distinguishable (often called "Boltzmannions"). This is useful, for example, for nuclear quantum effects when exchange is negligible. It is not an exact general description of identical nuclei. The situation becomes more complicated for simulating bosons and fermions. In real-space path integral Monte Carlo, quantum indistinguishability is encoded directly into the configuration space by summing over all permutations of particle labels. For bosons, this implies that the thermal density matrix must be fully symmetric under exchange of any two particles. This fundamental symmetry plays a central role in describing phenomena such as Bose-Einstein condensation and superfluidity, and must be incorporated explicitly in the PIMC sampling.

The symmetrized partition function for N bosons is expressed as:

$$Z_B = \frac{1}{N!}\sum_{\mathcal{P}} \int d\boldsymbol{R}\, \rho(\boldsymbol{R}, \mathcal{P}\boldsymbol{R}; \beta), \tag{5.37}$$

Here the permutations involve identical particles of one species; for a mixture, use a separate permutation sum and factorial for each species. The vector $\boldsymbol{R} = (\boldsymbol{r}_1, \ldots, \boldsymbol{r}_N)$ collects the coordinates of all particles, β is the inverse temperature, and $\mathcal{P}$ is an element of the permutation group of N objects. Notice that here we changed notation from the Sections above, because of the need to explicitly define particles, i.e. they cannot be "hidden" inside the general multi-particle multi-coordinated array $\boldsymbol{r}$ used before. The key feature of this expression is that the density matrix connects each configuration $\boldsymbol{R}$ at imaginary time $\tau = 0$ to a permuted configuration $\mathcal{P}\boldsymbol{R}$ at $\tau = \beta$.

This implies that in the path integral representation, particle worldlines are not required to be closed individually, but rather can form permutation cycles. The endpoint condition is $\boldsymbol{R}(\beta) = \mathcal{P}\boldsymbol{R}(0)$; with $(\mathcal{P}\boldsymbol{R})_i = \boldsymbol{r}_{\mathcal{P}(i)}$, worldline i ends at the initial coordinate of particle $\mathcal{P}(i)$. The set of all worldlines therefore consists of one or more cyclic chains of polymers, corresponding to the decomposition of $\mathcal{P}$ into disjoint cycles.

This topological structure has a direct physical interpretation. In the nondegenerate regime, where the thermal wavelength $\lambda_T = \sqrt{2\pi\beta/m}$ is small compared with the interparticle spacing, the identity permutation dominates, and each particle forms an isolated, closed ring polymer. As the temperature is lowered, permutation cycles involving two or more particles become statistically relevant. At very low temperatures, macroscopic exchange cycles can appear.

In this picture, a path represents not just the thermal fluctuations of a single particle, but an exchange-connected configuration of several indistinguishable particles whose labels are only defined modulo permutation. Exchange cycles alone are not a measure of quantum entanglement. This is a direct reflection of the quantum mechanical principle that the many-body wavefunction for bosons must be symmetric under exchange.

Incorporating these permutations into the Monte Carlo sampling requires moves that change the topology of the path. Since the path is discretized into P time slices, permutations change the worldline connectivity. The trace can be cut at any slice, so permutation updates may act within an arbitrary time window rather than only at a designated boundary.

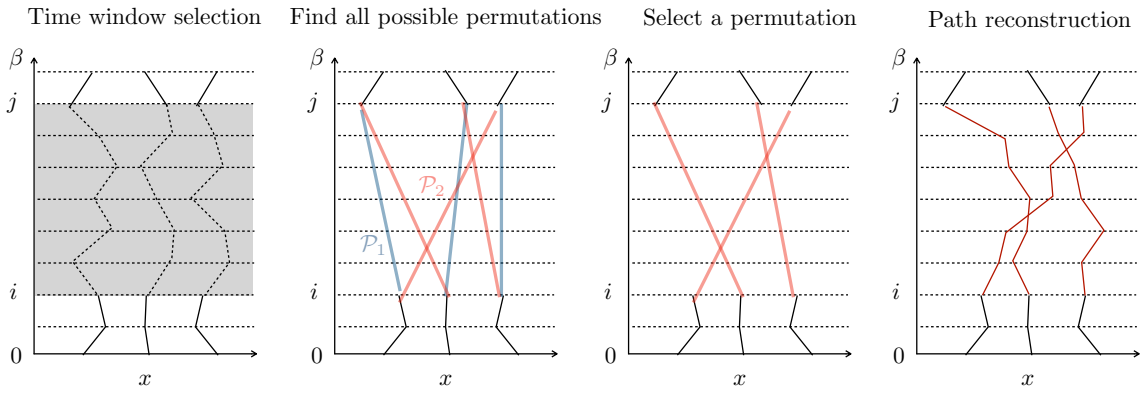


Figure 5.2: Sketch of the permutation sampling algorithm. For sake of visualization we only show two permutations among the all possible ones.

A naive method would simply to sample possible permutation and close the paths accordingly without moving the beads, however these update would suffer from a very-high rejection. A more clever way is the following (schematic):

- Pick an imaginary time window, of say, p slices, from index i to j, $p = j - i$.
- Find all the permutations (restricting a range of $N_{\max}$ particle to exchange, usually it is 2,3 or 4). Let's call this set $\mathcal{S}$.
- Compute the weight of each permutation $\mathcal{P}$ using for instance the free-particle density-matrix
$$w(\mathcal{P}) = \frac{\rho_0(\boldsymbol{R}_i, \mathcal{P}\boldsymbol{R}_j; p\Delta_\tau)}{\sum_{\mathcal{P}' \in \mathcal{S}} \rho_0(\boldsymbol{R}_i, \mathcal{P}'\boldsymbol{R}_j; p\Delta_\tau)}. \tag{5.38}$$
- Select a permutation $\mathcal{P}_k$ using this relative probability table.
- Try to construct paths that connect the chosen endpoints using e.g bisection.
- Accept or reject using the full Metropolis–Hastings ratio, including selection of the particles, the permutation and the intermediate bridge.

Here ρ_0 is the distinguishable free-particle kernel. For a fixed selected set of particles, sample a permutation and all its interior beads from their joint free conditional distribution. If the reverse move has the same candidate endpoint matchings and the same selection normalization, the kinetic and proposal factors cancel and the acceptance reduces to $\min(1, e^{-\Delta S_V})$. If the candidate list or the selection probabilities depend on the current configuration, include their reverse/forward ratio:

$$A(X \to X') = \min\left\{1, \frac{W(X')T(X|X')}{W(X)T(X'|X)}\right\}. \tag{5.39}$$

One must not multiply a free-conditional proposal by a second full kinetic-action Metropolis factor. The proposal step is shown in Fig. 5.2.

5.3.1 The Sign Problem in Continuous Space PIMC

With fermions the partition function is

$$Z_F = \frac{1}{N!} \sum_{\mathcal{P}} \text{sign}[\mathcal{P}] \int d\boldsymbol{R}\, \rho(\boldsymbol{R}, \mathcal{P}\boldsymbol{R}; \beta), \tag{5.40}$$

where $\text{sign}[\mathcal{P}] = +1\,(-1)$ for even (odd) permutations. The alternating signs in the sum cause large cancellations, which are at the core of the so-called *fermion sign problem.*

To perform Monte Carlo sampling, one often defines a modified (bosonic) partition function

$$Z_0 = \int dX\, |W(X)|, \tag{5.41}$$

where $W(X)$ is the configuration weight and X denotes the full set of paths in imaginary time. Configurations are sampled according to $|W(X)|$, and expectation values are reconstructed as

$$\langle \hat{A} \rangle = \frac{\langle A(X) S(X) \rangle_0}{\langle S(X) \rangle_0}, \qquad S(X) = \frac{W(X)}{|W(X)|}, \tag{5.42}$$

where $A(X)$ is the appropriate path estimator and $\langle \cdot \rangle_0$ denotes the average with respect to the normalized absolute-value measure. For the present positive-kernel permutation representation, removing the permutation signs gives the bosonic partition function. In a general representation, the absolute-weight model need not be a physical bosonic system.
The *average sign*

$$S = \langle S(X) \rangle_0 = \frac{Z_F}{Z_0} \tag{5.43}$$

quantifies the severity of the problem. Since Z_F and Z_0 can be written in terms of free energies,

$$Z_F = e^{-\beta F_F}, \qquad Z_0 = e^{-\beta F_0}, \tag{5.44}$$

we obtain

$$S = \frac{Z_F}{Z_0} = e^{-\beta (F_F - F_0)}. \tag{5.45}$$

When the free-energy difference $\Delta F = F_F - F_0$ is extensive with a positive limiting density Δf, one has

$$S \sim e^{-\beta N \Delta f}, \tag{5.46}$$

where $\Delta f = \Delta F / N$ is the free-energy density difference between the fermionic and bosonic systems.

Thus the sign is exponentially small in $\beta N \Delta f$, where Δf itself depends on temperature and the model. A generic consequence for observables with nonvanishing fluctuation prefactors is the scaling

$$\frac{\Delta A}{A} \sim \frac{1}{S\sqrt{M}}, \tag{5.47}$$

unless the number of effectively independent samples M grows correspondingly. This is a typical scaling, not an identity for every observable. Numerator and denominator are correlated. For independent samples and a real estimator $A(X)$, the large-M variance of the ratio estimator is

$$\mathrm{Var}(\widehat{A}) \simeq \frac{\mathrm{Var}_0\Big(S(X)[A(X) - \langle \hat{A} \rangle]\Big)}{M \langle S \rangle_0^2}. \tag{5.48}$$

For a Markov chain, replace the variance in the numerator by the sum of time-lag covariances of this combined random variable. Blocking or a block jackknife should preserve numerator–denominator correlations. This exponential decay of S constitutes the essence of the fermion sign problem: although the sampling scheme is formally exact, the efficiency vanishes exponentially with system size or low temperature.

Restricted paths and nodes. At finite temperature, the relevant nodal constraint is that of a trial fermionic density matrix $\rho_F^T(\boldsymbol{R}, \boldsymbol{R}_0; \tau)$, which depends on the reference configuration and imaginary time. Restrict paths to its allowed nodal region. Exact density-matrix nodes give the exact restricted-path formulation; approximate nodes introduce a

systematic error. This generalizes the fixed-node idea, but is not simply a fixed ground-state wavefunction node imposed at every slice.[4]

5.4 Path Integrals in Spin Models

The derivation of path-integral formula in lattice models is a bit more cumbersome. Let's start with a simple case.

5.4.1 A Single Quantum Spin

Consider a single spin-$\frac{1}{2}$ particle with Hamiltonian

$$\hat{H} = -\gamma\hat{\sigma}^x - h\hat{\sigma}^z, \tag{5.49}$$

where $\hat{\sigma}^x, \hat{\sigma}^z$ are Pauli matrices, and $\gamma \geq 0, h$ are constants. The partition function at inverse temperature β is

$$Z = \mathrm{Tr}\left(e^{-\beta\hat{H}}\right). \tag{5.50}$$

Since $\hat{\sigma}^x$ and $\hat{\sigma}^z$ do not commute, we apply the Trotter decomposition.
Divide the imaginary time β into P slices of size $\Delta_\tau = \beta/P$, so that

$$e^{-\beta\hat{H}} = \left(e^{-\Delta_\tau\hat{H}}\right)^P = \left(e^{\Delta_\tau\gamma\hat{\sigma}^x}e^{\Delta_\tau h\hat{\sigma}^z}\right)^P + \mathcal{O}(\Delta_\tau), \tag{5.51}$$

The error above is the accumulated operator error at fixed β: a one-step $O(\Delta_\tau^2)$ error accumulates over P steps. For the partition function of this finite spin problem there is a useful refinement. With $A = e^{\Delta_\tau\gamma\sigma^x}$ and $B = e^{\Delta_\tau h\sigma^z}$,

$$\mathrm{Tr}(AB)^P = \mathrm{Tr}(B^{1/2}AB^{1/2})^P. \tag{5.52}$$

Indeed, $AB = B^{-1/2}(B^{1/2}AB^{1/2})B^{1/2}$, so the two products are similar. The expression inside the second trace is a symmetric splitting, and consequently $Z_P = Z + O(\Delta_\tau^2)$. Parameter-derivative estimators inherit this second-order convergence for this finite-dimensional problem. This does not make the unsymmetrized propagator itself second order.

To handle the non-commuting operators carefully, we insert resolutions of identity in the $\hat{\sigma}^z$ eigenbasis between every exponential factor.
Each Trotter step consists of two exponentials of non commuting operators, so for P steps, there are $2P$ operators, requiring $2P$ resolutions of identity:

$$\mathbb{I} = \sum_{s=\pm1} |s\rangle\langle s|, \tag{5.53}$$

where $\hat{\sigma}^z|s\rangle = s|s\rangle$, and $s = \pm1$. So the partition function reads,

$$Z_P = \sum_{\{s_1=\pm1; s_2=\pm1,\cdots\}} \langle s_1|e^{\Delta_\tau\gamma\sigma^x}|s_2\rangle\langle s_2|e^{\Delta_\tau h\hat{\sigma}^z}|s_3\rangle \cdots \langle s_k|e^{\Delta_\tau\gamma\sigma^x}|s_{k+1}\rangle\langle s_{k+1}|e^{\Delta_\tau h\hat{\sigma}^z}|s_{k+2}\rangle \tag{5.54}$$

$$\cdots \langle s_{2P-1}|e^{\Delta_\tau\gamma\sigma^x}|s_{2P}\rangle\langle s_{2P}|e^{\Delta_\tau h\hat{\sigma}^z}|s_1\rangle$$

[4] D. M. Ceperley, *Path-integral calculations of normal liquid ^{3}He*, Phys. Rev. Lett. **69**, 331 (1992), https://doi.org/10.1103/PhysRevLett.69.331.

where the sequence of states[5] along imaginary time are $s_1, s_2, \ldots, s_{2P}$.

Since $\hat{\sigma}^z$ is diagonal in the $|s\rangle$ basis, we have

$$\langle s|e^{\Delta_\tau h\hat{\sigma}^z}|s'\rangle = e^{\Delta_\tau hs}\delta_{s,s'}, \tag{5.55}$$

therefore the actual number of slices is P, rather than $2P$. We could have done it at the beginning by inspecting the matrix element of $\langle s|e^{\Delta_\tau\gamma\hat{\sigma}^x}e^{\Delta_\tau h\hat{\sigma}^z}|s'\rangle = \langle s|e^{\Delta_\tau\gamma\hat{\sigma}^x}|s'\rangle e^{\Delta_\tau hs'}$, but this way is more systematic and useful for Hamiltonians beyond the TFIM.

The off-diagonal operator $e^{\Delta_\tau\gamma\hat{\sigma}^x}$ is a bit more cumbersome. The exponential of the Pauli-x matrix, can be computed exactly by Taylor expansion of the exponential and using the special properties of the Pauli matrices[6]

$$e^{\Delta_\tau\gamma\hat{\sigma}^x} = \cosh(\Delta_\tau\gamma)\,\mathbb{I} + \sinh(\Delta_\tau\gamma)\,\hat{\sigma}^x = \begin{pmatrix}\cosh(\Delta_\tau\gamma) & \sinh(\Delta_\tau\gamma)\\ \sinh(\Delta_\tau\gamma) & \cosh(\Delta_\tau\gamma)\end{pmatrix}. \tag{5.56}$$

Therefore,

$$\langle s|e^{\Delta_\tau\gamma\hat{\sigma}^x}|s'\rangle = \begin{cases}\cosh(\Delta_\tau\gamma), & s = s',\\ \sinh(\Delta_\tau\gamma), & s \neq s'.\end{cases} \tag{5.57}$$

To write the two cases compactly, use the product $ss' = \pm 1$ to distinguish equal and opposite spins. We aim to write,

$$\langle s_k|e^{\Delta_\tau\gamma\hat{\sigma}^x}|s_{k+1}\rangle \propto e^{Ks_ks_{k+1}}. \tag{5.58}$$

To determine both constants, write

$$\langle s|e^{\Delta_\tau\gamma\hat{\sigma}^x}|s'\rangle = Ce^{Kss'}, \qquad \cosh(\Delta_\tau\gamma) = Ce^{K}, \qquad \sinh(\Delta_\tau\gamma) = Ce^{-K}. \tag{5.59}$$

Multiplying the last two equations gives

$$C = \sqrt{\cosh(\Delta_\tau\gamma)\sinh(\Delta_\tau\gamma)}. \tag{5.60}$$

The positive solution is understood for $\gamma > 0$. Dividing instead gives

$$\frac{\cosh(\Delta_\tau\gamma)}{\sinh(\Delta_\tau\gamma)} = \coth(\Delta_\tau\gamma) = e^{2K} \implies K = \frac{1}{2}\log\coth(\Delta_\tau\gamma) = -\frac{1}{2}\log\tanh(\Delta_\tau\gamma). \tag{5.61}$$

Putting everything together, the discretized partition function is

$$Z_P = \sum_{\{s_k=\pm1\}}\prod_{k=1}^{P}\langle s_k|e^{\Delta_\tau\gamma\hat{\sigma}^x}|s_{k+1}\rangle e^{\Delta_\tau hs_{k+1}} \tag{5.62}$$

$$= C^P\sum_{\{s_k=\pm1\}}\exp\left\{\Delta_\tau\sum_{k=1}^{P}\left[J_\perp s_ks_{k+1} + hs_k\right]\right\}, \tag{5.63}$$

[5] we use the index k, because we reserve i, j as spatial indexes.

[6] In this case $(\sigma^x)^2 = \mathbb{I}$, $(\sigma^x)^3 = \sigma^x$, etc..

where periodicity $s_{P+1} = s_1$ lets us relabel the longitudinal-field sum. We defined

$$J_\perp = \frac{K}{\Delta_\tau} = -\frac{1}{2\Delta_\tau} \ln \tanh(\Delta_\tau \gamma) > 0. \tag{5.64}$$

Equivalently,

$$\boxed{Z_P = C^P \mathcal{Z}_P, \qquad \mathcal{Z}_P = \sum_{s_1=\pm 1} \cdots \sum_{s_P=\pm 1} e^{-\Delta_\tau H_P(s_1,\ldots,s_P)},} \tag{5.65}$$

with

$$H_P = -\sum_{k=1}^{P} \left(J_\perp s_k s_{k+1} + h s_k \right). \tag{5.66}$$

Define the full weight $W(s) = C^P e^{-\Delta_\tau H_P(s)}$ and its normalized probability $\pi_P(s) = W(s)/Z_P$. At fixed parameters, C^P cancels from normalized sampling probabilities. It must nevertheless be retained when computing Z_P, free energies, or derivatives with respect to β and γ.

The quantum problem has become a classical effective Ising chain of length P with ferromagnetic coupling in imaginary time. The physically relevant dimensionless coupling is $K = \Delta_\tau J_\perp$. In terms of the temperature,

$$J_\perp = -\frac{PT}{2} \ln \tanh\left(\frac{\gamma}{PT}\right), \tag{5.67}$$

since $1/\Delta_\tau = PT$. At fixed P, increasing γ weakens this coupling and favors domain walls; increasing T strengthens it. For $x > 0$ the function $-\ln \tanh x$ is positive, with limits

$$-\ln \tanh x \sim -\ln x \quad (x \to 0), \qquad -\ln \tanh x \sim 2e^{-2x} \quad (x \to \infty). \tag{5.68}$$

Domain walls encode transverse-field events. Their quantitative relation to the transverse magnetization is derived next. At $\gamma = 0$ the original matrix element is simply $\delta_{s,s'}$; the separate limits $C \to 0$ and $K \to \infty$ should not be evaluated numerically. The worldline is then constant.

By sampling the classical chain we calculate finite-temperature quantum properties of a single spin, with a controlled $P \to \infty$ limit. Lowering temperature at fixed P is not itself a controlled continuum limit.

5.4.2 Evaluation of the off-diagonal operator

Diagonal observables can be evaluated on each sampled spin configuration. For $\hat{\sigma}^x$, which is off-diagonal in this basis, we insert the operator in a propagator link.[7] Start from

$$\langle \hat{\sigma}^x \rangle = \frac{\mathrm{Tr}(\hat{\sigma}^x e^{-\beta \hat{H}})}{\mathrm{Tr}(e^{-\beta \hat{H}})}. \tag{5.69}$$

[7] F. Krzakala, A. Rosso, G. Semerjian and F. Zamponi, Phys. Rev. B **78**, 134428 (2008), `https://arxiv.org/abs/0807.2553`.

Repeating the same decomposition replaces one transverse propagator by $\langle s_k|\hat{\sigma}^x e^{\Delta_\tau \gamma \hat{\sigma}^x}|s_{k+1}\rangle$. The longitudinal factor on this link cancels between numerator and denominator. By cyclic invariance we may average the insertion over all P links:

$$\langle \hat{\sigma}^x \rangle_P = \frac{1}{Z_P} \sum_s W(s) \frac{1}{P} \sum_{k=1}^{P} \frac{\langle s_k|\hat{\sigma}^x e^{\Delta_\tau \gamma \hat{\sigma}^x}|s_{k+1}\rangle}{\langle s_k|e^{\Delta_\tau \gamma \hat{\sigma}^x}|s_{k+1}\rangle}. \tag{5.70}$$

With $a = \Delta_\tau \gamma > 0$ and $(\hat{\sigma}^x)^2 = \mathbb{I}$,

$$\hat{\sigma}^x e^{a\hat{\sigma}^x} = \sinh a\, \mathbb{I} + \cosh a\, \hat{\sigma}^x, \tag{5.71}$$

so

$$\langle s_k|\hat{\sigma}^x e^{a\hat{\sigma}^x}|s_{k+1}\rangle = \begin{cases} \sinh a, & s_k = s_{k+1}, \\ \cosh a, & s_k \neq s_{k+1}. \end{cases} \tag{5.72}$$

Dividing by the original link gives

$$\frac{\langle s_k|\hat{\sigma}^x e^{a\hat{\sigma}^x}|s_{k+1}\rangle}{\langle s_k|e^{a\hat{\sigma}^x}|s_{k+1}\rangle} = \begin{cases} \tanh a, & s_k s_{k+1} = +1, \\ \coth a, & s_k s_{k+1} = -1, \end{cases} = \big[\tanh a\big]^{s_k s_{k+1}}. \tag{5.73}$$

The exponent $s_k s_{k+1} = \pm 1$ thus selects $\tanh a$ or its reciprocal. The estimator is

$$\boxed{\langle \hat{\sigma}^x \rangle_P = \left\langle \frac{1}{P} \sum_{k=1}^{P} \big[\tanh(\Delta_\tau \gamma)\big]^{s_k s_{k+1}} \right\rangle_{\pi_P}}. \tag{5.74}$$

Here $\langle\cdot\rangle_{\pi_P}$ is a normalized average. At finite P it is the estimator for the discretized model; the exact quantum expectation is its $P \to \infty$ limit. A particular estimator value can exceed one, since it is a statistical estimator rather than an eigenvalue of $\hat{\sigma}^x$.

Counting domain walls. Define the number of kinks in a periodic worldline by

$$n_{\text{kink}}(s) = \frac{1}{2} \sum_{k=1}^{P} (1 - s_k s_{k+1}). \tag{5.75}$$

There are $P - n_{\text{kink}}$ aligned links and n_{kink} flipped links. Therefore

$$\begin{aligned} \frac{1}{P} \sum_k [\tanh a]^{s_k s_{k+1}} &= \frac{P - n_{\text{kink}}}{P} \tanh a + \frac{n_{\text{kink}}}{P} \coth a \\ &= \tanh a + \frac{n_{\text{kink}}}{P} (\coth a - \tanh a). \end{aligned} \tag{5.76}$$

As $P \to \infty$ at fixed β and $\gamma > 0$, $a = \beta\gamma/P \to 0$. Using $\tanh a \sim a$ and $\coth a \sim 1/a$ gives the continuous-time identity

$$\langle \hat{\sigma}^x \rangle = \frac{\langle n_{\text{kink}} \rangle}{\beta \gamma}. \tag{5.77}$$

Periodic single-spin paths have an even number of kinks. The formula at $\gamma = 0$ is understood by a limit; for this Hamiltonian $\langle \sigma^x \rangle = 0$ at $\gamma = 0$ and finite temperature.

An independent derivation from the partition function. Since $\partial_\gamma \hat{H} = -\hat{\sigma}^x$, the exact thermal identity is $\langle \hat{\sigma}^x \rangle = \beta^{-1} \partial_\gamma \ln Z$. For the discrete weight,

$$\partial_\gamma \ln C = \Delta_\tau \coth(2a), \qquad \partial_\gamma K = -\frac{\Delta_\tau}{\sinh(2a)}. \tag{5.78}$$

Consequently,

$$\begin{aligned} \frac{1}{\beta}\partial_\gamma \ln Z_P &= \left\langle \coth(2a) - \frac{1}{P \sinh(2a)} \sum_k s_k s_{k+1} \right\rangle_{\pi_P} \\ &= \left\langle \frac{1}{P} \sum_k [\tanh a]^{s_k s_{k+1}} \right\rangle_{\pi_P}. \end{aligned} \tag{5.79}$$

The derivative of C^P is essential: dropping it would lose the $\coth(2a)$ term.

Exercise 5.4.1: PIMC of One Spin-1/2

Consider the Hamiltonian of Eq. 5.49. The eigenvalues of the Hamiltonian are:

$$E_\pm = \pm\sqrt{\gamma^2 + h^2} \tag{5.80}$$

The partition function at inverse temperature β is:

$$Z = \mathrm{Tr}\left(e^{-\beta \hat{H}}\right) = 2\cosh\left(\beta\sqrt{\gamma^2 + h^2}\right) \tag{5.81}$$

The thermal average of the energy is:

$$\langle \hat{H} \rangle = -\frac{\partial}{\partial \beta} \log Z = -\sqrt{\gamma^2 + h^2} \tanh\left(\beta\sqrt{\gamma^2 + h^2}\right) \tag{5.82}$$

The thermal expectation values of the spin components are:

$$\langle \hat{\sigma}^x \rangle = \frac{\gamma}{\sqrt{\gamma^2 + h^2}} \tanh\left(\beta\sqrt{\gamma^2 + h^2}\right) \tag{5.83}$$

$$\langle \hat{\sigma}^z \rangle = \frac{h}{\sqrt{\gamma^2 + h^2}} \tanh\left(\beta\sqrt{\gamma^2 + h^2}\right). \tag{5.84}$$

1. Perform a PIMC simulation using the classical isomorphism to an effective 1D classical system, use a value of about $P = 64, 128$.

2. Use a cluster update to improve efficiency. If $h \neq 0$, include the longitudinal field in the cluster orientation probabilities or use a ghost-spin construction; an unconditional zero-field flip rule is not valid.

3. Compare energy and both magnetizations with the exact formulas. Repeat at increasing P and distinguish finite-P bias from statistical uncertainty. Check the transverse estimator against $\beta^{-1}\partial_\gamma \ln Z_P$.

5.4.3 Multidimensional Generalization

The generalization to the case of more than one spin is straightforward. Take the quantum Hamiltonian

$$\hat{H} = -\gamma \sum_i \hat{\sigma}_i^x - h \sum_i \hat{\sigma}_i^z - \sum_{i=1}^{L} \sum_{j<i} J_{ij} \hat{\sigma}_i^z \hat{\sigma}_j^z. \tag{5.85}$$

The interaction is diagonal in the chosen basis, while the transverse factors commute between sites and retain the single-spin form. We can simply add the subscript s_k^i to denote the value of the spin at imaginary-time slice k and spatial position $1 \leq i \leq L$. The *classical* Hamiltonian becomes

$$H_P = -\sum_{k=1}^{P} \left(\sum_{i=1}^{L} J_\perp s_k^i s_{k+1}^i + \sum_{i=1}^{L} h s_k^i + \sum_{i=1}^{L} \sum_{j<i} J_{ij} s_k^i s_k^j \right). \tag{5.86}$$

The full partition function is $Z_P = C^{LP} \sum_s e^{-\Delta_\tau H_P(s)}$. At site i,

$$\langle \hat{\sigma}_i^x \rangle_P = \left\langle \frac{1}{P} \sum_k [\tanh(\Delta_\tau \gamma)]^{s_k^i s_{k+1}^i} \right\rangle_{\pi_P}. \tag{5.87}$$

In continuous time, if n_{kink} counts flips on all sites,

$$\left\langle -\gamma \sum_i \hat{\sigma}_i^x \right\rangle = -\frac{\langle n_{\text{kink}} \rangle}{\beta}. \tag{5.88}$$

For a general lattice model, off-diagonal estimators must be derived from that model's propagator; the single-spin tanh / coth expression is not universal.

Notice that the physical interaction J_{ij} couples the spins at equal-time, while the effective $J_\perp$ only occurs between the same spatial site i but at different imaginary times.

Exercise 5.4.2: PIMC of the 1D TFIM

Consider the Hamiltonian of

$$H = -J \sum_{i=1}^{L} \sigma_i^z \sigma_{i+1}^z - \sum_{i=1}^{L} \Gamma \sigma_i^x, \tag{5.89}$$

with PBC Start with $J, \Gamma = 1$.

1. Set up a PIMC code using the classical isomorphism to an effective 2D classical system, use a value of about $P = 64, 128$.
2. Use the cluster update to improve the efficiency.
3. For $L = 20$, map finite-temperature crossovers as a function of Γ/J and β (a starting range is $0.2 \leq \beta J \leq 5$). The nearest-neighbor 1D TFIM has no finite-temperature ferromagnetic transition; its thermodynamic quantum critical point is at $T = 0$, $\Gamma/J = 1$. A finite chain has smooth crossovers. Check convergence in P at each representative parameter set.
4. Start from $\beta J = 10$ and increase β until the energy and chosen correlations converge. Compare $\langle \sigma_i^z \sigma_{i+1}^z \rangle$ and $\langle \sigma_i^x \rangle$ with VMC and the exact result in Exercise 3.2.1, matching system size, boundary conditions and energy normalization. Increasing β may also require increasing P.
5. Report statistical errors using blocking, finite-P changes, finite-temperature changes and finite-size effects separately. At zero longitudinal field a finite system has $\langle m_z \rangle = 0$ in equilibrium; use $\langle |m_z| \rangle$ or $\langle m_z^2 \rangle$ to describe its ordering crossover without mistaking incomplete tunneling for symmetry breaking.

5.5 The Loop Algorithm

We show now how to generalize the path-integral construction that we did for the Ising model, and directly build-in an efficient cluster update that respect symmetries of a more general Hamiltonian when updating the world-lines.
Indeed, suppose that we are interested in the Heisenberg model instead. In this case, the world lines cannot break in imaginary time. Flipping just one spin in the extended lattice would not work.

5.5.1 The XXZ Model

We start by considering the XXZ model for two spin-$\frac{1}{2}$ particles. The Hamiltonian is given by

$$\hat{H} = -J_{xy} \left(\hat{\sigma}_1^x \hat{\sigma}_2^x + \hat{\sigma}_1^y \hat{\sigma}_2^y \right) - J_z \, \hat{\sigma}_1^z \hat{\sigma}_2^z, \tag{5.90}$$

where $\hat{\sigma}_i^{x,y,z}$ are the Pauli spin operators acting on site $i = 1, 2$. The parameters J_{xy} and J_z control the strength of the spin-spin interaction in the xy plane and along the z axis,

respectively. A positive J_z corresponds to ferromagnetic coupling in the z direction.

In this basis, the non-zero matrix elements of the Hamiltonian $\hat{H}$ are[8]

$$\langle\uparrow\uparrow|\hat{H}|\uparrow\uparrow\rangle = -J_z, \qquad \langle\downarrow\downarrow|\hat{H}|\downarrow\downarrow\rangle = -J_z, \tag{5.91}$$

$$\langle\uparrow\downarrow|\hat{H}|\uparrow\downarrow\rangle = +J_z \qquad \langle\downarrow\uparrow|\hat{H}|\downarrow\uparrow\rangle = +J_z, \tag{5.92}$$

$$\langle\uparrow\downarrow|\hat{H}|\downarrow\uparrow\rangle = -2J_{xy}, \quad \langle\downarrow\uparrow|\hat{H}|\uparrow\downarrow\rangle = -2J_{xy}. \tag{5.93}$$

Now, we expand for small Δ_τ the exponential and we find the corresponding non-zero matrix elements of the operator $1-\Delta_\tau\hat{H}$

$$W_1 = \langle\uparrow\uparrow|1-\Delta_\tau\hat{H}|\uparrow\uparrow\rangle = 1+\Delta_\tau J_z, \quad \langle\downarrow\downarrow|1-\Delta_\tau\hat{H}|\downarrow\downarrow\rangle = 1+\Delta_\tau J_z, \tag{5.94}$$

$$W_2 = \langle\uparrow\downarrow|1-\Delta_\tau\hat{H}|\uparrow\downarrow\rangle = 1-\Delta_\tau J_z, \quad \langle\downarrow\uparrow|1-\Delta_\tau\hat{H}|\downarrow\uparrow\rangle = 1-\Delta_\tau J_z, \tag{5.95}$$

$$W_3 = \langle\uparrow\downarrow|1-\Delta_\tau\hat{H}|\downarrow\uparrow\rangle = 2\Delta_\tau J_{xy}, \quad \langle\downarrow\uparrow|1-\Delta_\tau\hat{H}|\uparrow\downarrow\rangle = 2\Delta_\tau J_{xy}, \tag{5.96}$$

The last row contains the off-diagonal matrix elements. These linearized weights are nonnegative for $J_{xy}\geq 0$ and $\Delta_\tau|J_z|\leq 1$. Uniform negative J_{xy} can be transformed to positive J_{xy} on a bipartite lattice by changing the signs of $\sigma^{x,y}$ on one sublattice. Arbitrary frustrated bond-sign patterns cannot necessarily be removed this way.

For implementation it is useful to retain the exact bond exponential. In the antiparallel subspace,

$$h_{\text{anti}} = J_z\mathbb{I} - 2J_{xy}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \tag{5.97}$$

so exponentiating this 2×2 block gives

$$\boxed{W_1 = e^{\Delta_\tau J_z}, \quad W_2 = e^{-\Delta_\tau J_z}\cosh(2\Delta_\tau J_{xy}), \quad W_3 = e^{-\Delta_\tau J_z}\sinh(2\Delta_\tau J_{xy}).} \tag{5.98}$$

These reduce to the preceding linear weights at first order. They are nonnegative for every $\Delta_\tau > 0$ when $J_{xy}\geq 0$. Below, W_i denotes these exact bond weights unless a linear approximation is explicitly written. The checkerboard splitting still has a finite-Δ_τ error; replacing each exact bond exponential by $1-\Delta_\tau h$ introduces an additional accumulated first-order error.

This last step is needed because we are going to use the path-integral construction. We can directly step-up to a one-dimensional problem. Here we use a different partition for the two non-commuting sets of operators. We consider the spin-$\frac{1}{2}$ XXZ model on a one-dimensional chain of even length L with periodic boundary conditions. The Hamiltonian is given by

$$\hat{H} = -\sum_{i=1}^{L}\left[J_{xy}\left(\hat{\sigma}_i^x\hat{\sigma}_{i+1}^x + \hat{\sigma}_i^y\hat{\sigma}_{i+1}^y\right) + J_z\,\hat{\sigma}_i^z\hat{\sigma}_{i+1}^z\right], \tag{5.99}$$

where $\hat{\sigma}_i^{x,y,z}$ are Pauli matrices acting on site i. To perform the path integral construction using Trotter-Suzuki decomposition, we split the Hamiltonian into two groups, each containing mutually commuting bond terms. The two groups generally do not commute with each

[8] We use Pauli matrices throughout. For spin operators $S^\alpha = \sigma^\alpha/2$, the same numerical exchange constants would give an extra factor 1/4. No factor is omitted in the Hamiltonian written here.

other:

$$\hat{H} = \hat{H}_{\text{even}} + \hat{H}_{\text{odd}}, \tag{5.100}$$

where

$$\hat{H}_{\text{even}} = \sum_{i \in \text{even}} \hat{h}_{i,i+1}, \tag{5.101}$$

$$\hat{H}_{\text{odd}} = \sum_{i \in \text{odd}} \hat{h}_{i,i+1}, \tag{5.102}$$

and the two-site bond Hamiltonian is

$$\hat{h}_{i,i+1} = -J_{xy} \left(\hat{\sigma}_i^x \hat{\sigma}_{i+1}^x + \hat{\sigma}_i^y \hat{\sigma}_{i+1}^y \right) - J_z \, \hat{\sigma}_i^z \hat{\sigma}_{i+1}^z. \tag{5.103}$$

This decomposition ensures that all terms in $\hat{H}_{\text{even}}$ commute with each other, and likewise for $\hat{H}_{\text{odd}}$, since they act on disjoint bonds. The Trotter decomposition of the partition function

$$Z = \text{Tr} \left(e^{-\beta \hat{H}} \right)$$

is performed by alternating the even and odd operators at each time slice:

$$Z \approx \text{Tr} \left[\left(e^{-\Delta_\tau \hat{H}_{\text{even}}} e^{-\Delta_\tau \hat{H}_{\text{odd}}} \right)^P \right], \tag{5.104}$$

with $\beta = P\Delta_\tau$. This results in a path integral representation defined on a $(1+1)$-dimensional checkerboard lattice, where the two-site interactions $\hat{h}_{i,i+1}$ live on alternating plaquettes in spacetime.

We now insert complete sets of states in the $\hat{\sigma}^z$ basis between each exponential factor in the Trotter decomposition. This leads to a sum over spin configurations (replica configurations) defined on a $(1+1)$-dimensional checkerboard lattice.

Let $s_i^t \in \{\uparrow, \downarrow\}$, with $i = 1, \ldots, L$ and $t = 0, \ldots, 2P-1$, and impose $s_i^{2P} = s_i^0$. There are $2P$ intermediate layers because a state must be inserted between the even and odd factors. Choosing the cyclic order $e^{-\Delta_\tau H_{\text{odd}}} e^{-\Delta_\tau H_{\text{even}}}$ gives

$$\begin{aligned} Z_P = \sum_{\{s\}} \prod_{k=0}^{P-1} & \langle s^{2k+2} | e^{-\Delta_\tau H_{\text{odd}}} | s^{2k+1} \rangle \\ & \times \langle s^{2k+1} | e^{-\Delta_\tau H_{\text{even}}} | s^{2k} \rangle. \end{aligned} \tag{5.105}$$

Each matrix element now factorizes over disjoint bonds. A pair of sublayers represents one Trotter step Δ_τ of the full Hamiltonian: the total interval is $\beta = P\Delta_\tau$, not $2P\Delta_\tau$. The sublayer index t is an ordering label for the split operators.

For example,

$$\langle s^{2k+1} | e^{-\Delta_\tau H_{\text{even}}} | s^{2k} \rangle = \prod_{i \in \text{even}} \langle s_i^{2k+1} s_{i+1}^{2k+1} | e^{-\Delta_\tau h_{i,i+1}} | s_i^{2k} s_{i+1}^{2k} \rangle. \tag{5.106}$$

The odd layer has the analogous product. Thus

$$Z_P = \sum_{\{s\}} \prod_{p \in \text{active plaquettes}} W_p(s_p), \tag{5.107}$$

with LP active plaquettes. Conservation of the two-site z magnetization leaves only the six nonzero configurations drawn below, with exact weights given by Eq. 5.98. The quantum partition function is recovered as $P \to \infty$.

Off-diagonal bond-energy estimator. For $\hat{H} = \hat{H}_0 - \lambda \hat{B}$, cyclicity of the trace gives

$$\langle \hat{B} \rangle = \frac{1}{\beta} \partial_\lambda \ln Z, \qquad \langle \hat{B} \rangle_P = \frac{1}{\beta} \left\langle \sum_p \partial_\lambda \ln W_p \right\rangle_P , \tag{5.108}$$

where the second equation defines the thermodynamic estimator for the discretized partition function. All parameter-dependent constants must be differentiated. For the XXZ exchange term, define N_2, N_3 as the numbers of active plaquettes of types 2 and 3, including both spin orientations. With $b = 2\Delta_\tau J_{xy} > 0$,

$$\partial_{J_{xy}} \ln W_1 = 0, \quad \partial_{J_{xy}} \ln W_2 = 2\Delta_\tau \tanh b, \quad \partial_{J_{xy}} \ln W_3 = 2\Delta_\tau \coth b. \tag{5.109}$$

Therefore the exchange-energy estimator is

$$\langle H_{xy} \rangle_P = -\frac{2J_{xy}}{P} \langle N_2 \tanh b + N_3 \coth b \rangle_P . \tag{5.110}$$

In continuous time this reduces to $\langle H_{xy} \rangle = -\langle N_3 \rangle / \beta$. At finite step, the N_2 contribution is necessary. At $J_{xy} = 0$ use the analytic limit rather than evaluating $\coth 0$.

5.5.2 Graph representation and the loop break-ups

Label the six non-zero plaquette (vertex) types by

$$1^\pm,\ 2^\pm,\ 3^\pm,$$

where the index distinguishes the structural class of the vertex and the sign indicates the spin orientation ($+$ for $\uparrow$, $-$ for $\downarrow$). Their local weights are

$$W_{1^\pm} = W_1, \qquad W_{2^\pm} = W_2, \qquad W_{3^\pm} = W_3, \quad \text{with the exact values in Eq. 5.98.}$$

Breakup weights. For a plaquette of class i, introduce a compatible local graph $G_{ij} = G_{ji}$ with nonnegative weight $w_{ij} = w_{ji}$. The graph permits the local spin configurations of classes i and j; it does not specify the type of a neighboring plaquette. The labels $+$ and $-$ have equal weights here. The defining equations are

$$W_i = \sum_j w_{ij}, \qquad w_{ij} \geq 0, \qquad w_{ij} = w_{ji}. \tag{5.111}$$

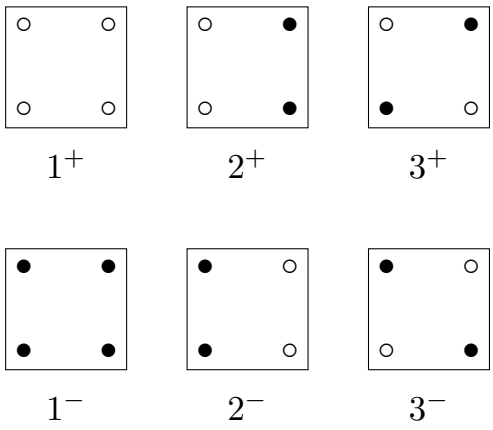


Figure 5.3: The six plaquette (vertex) configurations $1^{\pm}, 2^{\pm}, 3^{\pm}$. Empty circles denote $\uparrow$ spins, filled circles $\downarrow$. The two spins at the bottom/(top) correspond to imaginary time $t/(t+1)$.

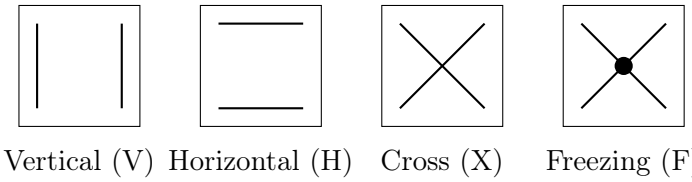


Figure 5.4: Breakup (graph) types used in the loop algorithm: vertical (V), horizontal (H), cross (X), and freezing (F). The freezing breakup is represented like the cross but with a central circle, joining all four legs into one cluster. The joined pieces must flip together; the complete cluster is allowed to flip.

The w_{ij} are unnormalized graph weights. Given the present plaquette class i, choose G_{ij} with probability w_{ij}/W_i. Figure 5.4 displays the allowed connections.[9]

The vertical breakup is compatible with classes 1 and 2 (weight w_{12}); flipping either connected pair changes the class. The horizontal breakup connects antiparallel pairs and is compatible with classes 2 and 3 (w_{23}). The cross breakup connects parallel diagonal pairs and is compatible with classes 1 and 3 (w_{13}). A freezing graph G_{ii} joins all four legs, preserving the relative spin pattern of class i; all four spins must then flip together. Thus freezing can join otherwise separate loops into a larger cluster. Essentially, these are the generalization to a single spin flip for the Ising model case. For the XXZ Hamiltonian the weight of a plaquette of type 4 is exactly zero: it would feature two spins up (down) at the bottom and two spins down(up) at the top. This is not allowed by the specific Hamiltonian Eq. 5.90.
We now assign numerical values to the graph weights by solving Eq. 5.111:

$$w_{11} + w_{12} + w_{13} = W_1, \tag{5.112}$$
$$w_{22} + w_{12} + w_{23} = W_2, \tag{5.113}$$
$$w_{33} + w_{13} + w_{23} = W_3. \tag{5.114}$$

There are many of them and we try to minimize the weight of the freezing to decrease the autocorrelation time. We need to pay attention to not put too many weights to zero to allow for ergodic moves as well. In general, the best solution depends on the values of J_z and J_{xy}. For the ferromagnetic isotropic model $J_z = J_{xy} = J > 0$, the exact weights satisfy $W_1 = W_2 + W_3$. A solution without freezing is

$$w_{12} = W_2 = e^{-\Delta_\tau J}\cosh(2\Delta_\tau J), \qquad w_{13} = W_3 = e^{-\Delta_\tau J}\sinh(2\Delta_\tau J), \tag{5.115}$$

with every other weight zero. Expanding to first order recovers

$$w_{12} \simeq 1 - \Delta_\tau J, \qquad w_{13} \simeq 2\Delta_\tau J. \tag{5.116}$$

For $|J_z| \leq J_{xy}$ the exact weights satisfy all three triangle inequalities and a nonfreezing solution is

$$w_{12} = \frac{W_1 + W_2 - W_3}{2}, \quad w_{13} = \frac{W_1 + W_3 - W_2}{2}, \quad w_{23} = \frac{W_2 + W_3 - W_1}{2}. \tag{5.117}$$

If $J_z > J_{xy}$, choose $w_{12} = W_2$, $w_{13} = W_3$ and $w_{11} = W_1 - W_2 - W_3$; all remaining weights vanish. If $J_z < -J_{xy}$, choose $w_{12} = W_1$, $w_{23} = W_3$ and $w_{22} = W_2 - W_1 - W_3$. These formulas explicitly satisfy positivity and the sum rules. Minimizing freezing often improves sampling, but an ergodicity check is still needed for the chosen ensemble and boundary conditions.

Sampling rule (on-the-fly construction). The following describes a single-cluster version of the graph update:

1. Pick a random spin leg on the checkerboard. Include it in the cluster and initialize a list of legs whose connections remain to be explored.

[9] H. G. Evertz, *The loop algorithm*, Adv. Phys. **52**, 1 (2003), `https://arxiv.org/abs/cond-mat/9707221`.

2. At a newly visited plaquette, inspect its actual spin class i and sample a compatible graph G_{ij} with
$$\Pr(G_{ij} \mid i) = \frac{w_{ij}}{W_i}. \tag{5.118}$$
Store this graph for the remainder of the update. The index j labels a compatible class of this same plaquette, not the neighboring plaquette's type.

3. Follow the graph connection through the plaquette and the fixed worldline links to adjacent plaquettes. Reuse any already assigned graph. At a freezing graph, include all four connected legs and explore every branch.

4. Continue until every leg connected to the seed has been explored. Without freezing the component is a closed loop; with freezing it can consist of several joined loops.

5. Flip every spin in the complete connected component with probability 1/2. For the present zero-field model, the compatible orientations have equal joint graph weight.

Equivalently, assign graphs to all plaquettes, find all connected components, and flip each independently with probability 1/2. Do not redraw a graph when the same plaquette is revisited during one cluster construction. This graph procedure is distinct from a directed-loop update, whose local scattering probabilities require their own balance equations.

The unrestricted trace sums over total-S^z sectors. Clusters winding around imaginary time can change that magnetization; spatial winding sectors can also matter. If a fixed magnetization sector is intended, its constraint must be incorporated into the update. Detailed balance alone is insufficient: verify that the allowed moves connect the desired support. The lack of horizontal breakups in the isotropic ferromagnet is not by itself a proof of ergodicity.

5.5.3 The Kandel–Domany Graph Framework

We now place the loop algorithm construction within the more general and elegant *Kandel–Domany (KD) framework* for cluster updates.

In the KD framework the partition function of a classical model is rewritten as a sum over configurations C and auxiliary "graph" variables G:
$$Z = \sum_C \sum_G W(C, G), \tag{5.119}$$
with the requirement that
$$\sum_G W(C, G) = W(C),$$
where $W(C) = e^{-\beta E[C]}$ is the original Boltzmann weight. One then samples first the graph G given C using probability
$$P_C(G) = \frac{W(C, G)}{W(C)},$$
and subsequently a new configuration C' given the fixed graph G, in a way that satisfies detailed balance.

A key simplification arises if the joint weight factorizes as

$$W(C, G) = \Delta(C, G)\, V(G), \tag{5.120}$$

where $\Delta(C, G) = 1$ if the graph G is compatible with configuration C, and zero otherwise, and $V(G) \geq 0$ is a graph-only weight. When the compatible configurations are obtained by independent binary cluster orientations with equal weight, resample each orientation with probability 1/2. Equivalently, each cluster is flipped independently with probability 1/2. A suitably selected single-cluster proposal can also have unit Metropolis acceptance, but always flipping every cluster would merely reverse every spin and would not provide an ergodic update.

This framework extends naturally to quantum (world-line) systems. One treats the world-line configurations as extended configurations C, and defines allowed local graph elements G (such as vertical, horizontal, cross, and sometimes freezing graphs) over plaquettes, such that the world lines remain intact and valid after applying G, i.e., $\Delta(C, G)$ enforces world-line continuity.

The weights $V(G)$ are chosen so that summing over all compatible graphs reproduces the plaquette weights $W_p(s_p)$, exactly as done above . Thus the KD framework provides a principled derivation of the breakup-decomposition used in loop algorithms: one picks a graph basis G, defines $\Delta(C, G)$, and solves for graph-weights $V(G)$ such that

$$W(C) = \sum_G \Delta(C, G)\, V(G).$$

5.5.4 Complexity of the Sign Problem in Lattice Models

Having introduced the average-sign ratio in Eq. 5.42, we now revisit its statistical cost for lattice models and discuss the generic hardness result of Troyer and Wiese. Exponential sign decay is a common mechanism; it is not a statement that every quantum Monte Carlo problem is hard.

Sampling the sign. Consider a quantum lattice model whose partition function in a path-integral or world-line representation takes the form

$$Z = \sum_{c \in \Omega} W(c), \tag{5.121}$$

where the full (imaginary time) configuration space Ω may include configurations with negative weights $W(c) < 0$. Notice that c denotes the full extended configuration ($L \times P$ for TFIM, or $L \times 2P$ for the checkerboard construction), and W is the weight of that full configuration, not the 4-spins plaquette weights.

Define an auxiliary absolute-weight model, sometimes called the "bosonic version", by replacing each weight with its absolute value. This terminology need not describe literal bosonic particles:

$$Z_b = \sum_{c \in \Omega} |W(c)|. \tag{5.122}$$

Monte Carlo sampling with weights $|W(c)|$ then defines an average sign

$$\langle s(c) \rangle = \frac{Z}{Z_b}, \qquad s(c) = \frac{W(c)}{|W(c)|}. \tag{5.123}$$

The expectation value of an observable O in the original quantum model is

$$\langle O\rangle = \frac{1}{Z}\sum_{c\in\Omega} O(c)\, W(c), \tag{5.124}$$

which can be computed by reweighting the normalized absolute-weight distribution:

$$\langle O\rangle = \frac{\sum_{c\in\Omega} O(c)W(c)}{\sum_{c\in\Omega} W(c)} = \frac{\sum_{c\in\Omega} O(c)s(c)|W(c)|}{\sum_{c\in\Omega} s(c)|W(c)|} \tag{5.125}$$

$$= \frac{Z_b^{-1}\sum_{c\in\Omega} O(c)s(c)|W(c)|}{Z_b^{-1}\sum_{c\in\Omega} s(c)|W(c)|} = \frac{\langle Os\rangle_b}{\langle s\rangle_b}. \tag{5.126}$$

Here $Z_b = \sum_c |W(c)|$, and $\langle\cdot\rangle_b$ denotes an average with probability $|W(c)|/Z_b$.

The problem is in the denominator. The sign is a random variable $s(c) = \pm 1$. Since $s^2 = 1$, its variance is

$$\mathrm{Var}(s) = \langle s^2\rangle - \langle s\rangle^2 = 1 - \langle s\rangle^2. \tag{5.127}$$

For M statistically independent Monte Carlo samples (here we consider non-correlated samples taken along the PIMC simulation), the statistical error on the estimator of $\langle s\rangle$ is therefore

$$\delta s = \sqrt{\frac{\mathrm{Var}(s)}{M}} = \sqrt{\frac{1-\langle s\rangle^2}{M}}. \tag{5.128}$$

The relative error is

$$\frac{\delta s}{\langle s\rangle} = \sqrt{\frac{1-\langle s\rangle^2}{M\,\langle s\rangle^2}} \simeq \frac{1}{\sqrt{M}\,\langle s\rangle}, \tag{5.129}$$

where the last approximation holds when $\langle s\rangle \ll 1$.
Now we can rewrite the ratio of partition functions using their free energies

$$\langle s\rangle = Z/Z_b = e^{-\beta(F-F_b)} = e^{-\beta N\Delta f}, \tag{5.130}$$

where N is the number of physical degrees of freedom and $\Delta f = (F - F_b)/N$. When this free-energy density difference remains positive in the size limit, the following exponential cost applies:

$$\frac{\delta s}{\langle s\rangle} \simeq \frac{1}{\sqrt{M}}\, e^{\beta N\Delta f}. \tag{5.131}$$

Thus, maintaining a fixed relative accuracy requires an exponentially large number of samples,

$$M \sim e^{2\beta N\Delta f}, \tag{5.132}$$

for a fixed small relative error in the average sign. Markov-chain correlations multiply this sampling cost by the appropriate autocorrelation factor. The same statistical obstruction occurs in frustrated spin models; negative weights do not require fermionic particles.

A concrete case for the Heisenberg antiferromagnet. Now, when $J_{xy} < 0$ one would be tempted to claim automatically a sign problem, given that the weight W_3 becomes negative. For the uniform exchange model on a bipartite lattice, the transformation $\hat\sigma^{x,y} \to -\hat\sigma^{x,y}$ on one sublattice makes J_{xy} positive. The corresponding plaquette weights are

then nonnegative. This is not true for a triangular lattice. However also in this case, one can construct a worldline configuration with plus signs: when there is an even number of "exchange" plaquettes. In the triangular lattice, one may as well find negative sign configuration when a world-line loop over a triangular graph. For real factorized weights the precise statement is

$$\operatorname{sign} W(c) = \prod_p \operatorname{sign} W_p(s_p) = (-1)^{N_{\text{negative}}(c)}. \tag{5.133}$$

A negative configuration therefore contains an odd number of negative local factors. On a bipartite lattice, a closed worldline configuration of the uniform antiferromagnetic exchange model has an even number of exchanges; the sublattice transformation removes their signs. On a triangle, moving an up spin around the three sites through three exchanges returns to the initial configuration with an odd number of negative factors. Thus the global compatibility of local events determines whether cancellations are unavoidable.

Basis dependence. The fact that the sign-problem is basis dependent, i.e. it is not present in the basis of the eigenstate of the Hamiltonian, offered a hope to *solve* the sign problem. A change of basis can remove signs for some models. However, finding a useful basis can itself be difficult, and positive weights alone do not guarantee rapid equilibration or polynomial computational cost.

Complexity theory arguments. Troyer and Wiese established that a generic polynomial-cost solution of the sign problem, for models whose corresponding absolute-weight problems are efficiently solvable, would solve an NP-complete problem in polynomial randomized time.[10]

They start from the decision problem for a classical three-dimensional Ising spin glass,

$$H_{\text{glass}} = -\sum_{\langle j,k\rangle} J_{jk}\sigma_j\sigma_k, \qquad \sigma_j = \pm 1, \qquad J_{jk} \in \{0, \pm J\}, \tag{5.134}$$

and ask whether any configuration has energy at most a specified E_0. For the problem family in the reduction this decision is NP-complete. Replacing the classical spins by Pauli-x operators gives

$$\hat{H} = -\sum_{\langle j,k\rangle} J_{jk}\hat{\sigma}_j^x\hat{\sigma}_k^x. \tag{5.135}$$

In the z basis, the random coupling signs appear as signs of off-diagonal matrix elements. The corresponding absolute-weight construction is an unfrustrated ferromagnet, for which efficient cluster sampling is available in the reduction. A generic polynomial-cost algorithm that overcomes the cancellations would therefore yield the spin-glass decision answer at an appropriate temperature and accuracy, both chosen with polynomial resources.

This is a quantum spin example, not a literal fermionic model. Moreover, it is diagonal in the product x basis: removing its off-diagonal signs merely returns the hard classical spin-glass sampling problem. Thus a sign-removing basis change alone is not a solution in the complexity sense used here.

[10] M. Troyer and U.-J. Wiese, Phys. Rev. Lett. **94**, 170201 (2005); preprint (2004), `https://arxiv.org/abs/cond-mat/0408370`. Their footnote on bounded-error Monte Carlo distinguishes BPP from deterministic P.

The conclusion is NP-hardness of a *generic efficient solution*, not NP-completeness of every sign problem. Since Monte Carlo gives bounded-error randomized answers, the precise consequence would be $\mathsf{NP} \subseteq \mathsf{BPP}$. A deterministic polynomial-time algorithm for the same NP-complete decision problem would instead imply $\mathsf{P} = \mathsf{NP}$. Neither statement rules out efficient methods for particular Hamiltonians.

5.6 Real-Time Dynamics vs PIMC Pseudodynamics

PIMC samples an equilibrium path distribution. Its simulation trajectory is a sequence of entire imaginary-time paths. This gives us three distinct notions of time:

Physical time t	Schrödinger evolution or open-system dynamics,
Imaginary time τ	position along a sampled path,
Monte Carlo step n	position along the Markov chain of paths.

In the following we are going to explore if they are related.

5.6.1 Coherent and incoherent tunneling

Consider a symmetric double well with a low-energy doublet separated by $\Delta = E_1 - E_0 > 0$. In a localized basis $|L\rangle, |R\rangle$, and after subtracting the doublet mean energy,

$$\hat{H}_{\text{eff}} = -\frac{\Delta}{2}\sigma^x. \tag{5.136}$$

For an isolated system initially in $|L\rangle$,

$$|\psi(t)\rangle = e^{-it\hat{H}_{\text{eff}}}|L\rangle = \cos(\Delta t/2)|L\rangle + i\sin(\Delta t/2)|R\rangle, \tag{5.137}$$

$$P_{L\to R}(t) = \sin^2(\Delta t/2). \tag{5.138}$$

The first complete transfer occurs at $t = \pi/\Delta$. This is coherent oscillation, with characteristic time proportional to Δ^{-1}.

If an environment destroys the interwell coherence on a time T_2 much shorter than Δ^{-1}, a population-rate description can apply. In the simple resonant pure-dephasing model derived in Appendix F.2,

$$\Gamma_{\text{rel}} \simeq \Delta^2 T_2, \qquad t_{\text{rel}} \simeq \frac{1}{\Delta^2 T_2}. \tag{5.139}$$

For symmetric directional hopping rates, each rate is $\Gamma_{\text{rel}}/2$. These formulas assume that two-level dephasing model; detuning, thermal activation and other bath couplings can change the rates. Restoring units replaces Δt by $\Delta t/\hbar$ and $\Delta^2 T_2$ by $\Delta^2 T_2/\hbar^2$.

5.6.2 Instantons and tunneling in path space

For a particle in a symmetric double well with equal minima $V_{\min}$, define the excess Euclidean action

$$S[x] = \int d\tau \left[\frac{m}{2}\dot{x}^2 + V(x) - V_{\min}\right]. \tag{5.140}$$

A stationary connecting path satisfies

$$m\ddot{x} = V'(x), \qquad x(-\infty) = x_-, \qquad x(+\infty) = x_+. \tag{5.141}$$

This is classical motion in the inverted potential. Multiplication by $\dot{x}$ and integration give

$$\frac{m}{2}\dot{x}^2 = V(x) - V_{\min}, \tag{5.142}$$

and hence the single-instanton action is

$$S_I = \int_{x_-}^{x_+} dx \sqrt{2m[V(x) - V_{\min}]}. \tag{5.143}$$

In the semiclassical regime the splitting has the form

$$\Delta \simeq A e^{-S_I}, \tag{5.144}$$

where the prefactor A includes fluctuations around the saddle. The action alone specifies leading exponential behavior, not the exact splitting. Appendix F.3 gives an explicit quartic example. With $\hbar$ restored the exponential is $e^{-S_I/\hbar}$.

It has beem shown that instantons are the transition state of a PIMC simulations performing tunneling[11] In the barrier-dominated regime, a periodic path crosses the bottleneck through an instanton–anti-instanton pair; an open path can do so through a single instanton. For suitable boundary weights and sampling updates, the mean transition times have the leading exponential scalings

$$\boxed{\xi_{\mathrm{PBC}} \propto e^{2S_I} \propto \Delta^{-2}, \qquad \xi_{\mathrm{OBC}} \propto e^{S_I} \propto \Delta^{-1}.} \tag{5.145}$$

The periodic result shares the gap dependence of incoherent tunneling; opening the paths gives a quadratic improvement in this exponential scaling.

The paired and single crossings can be visualized without identifying either with a physical real-time trajectory:

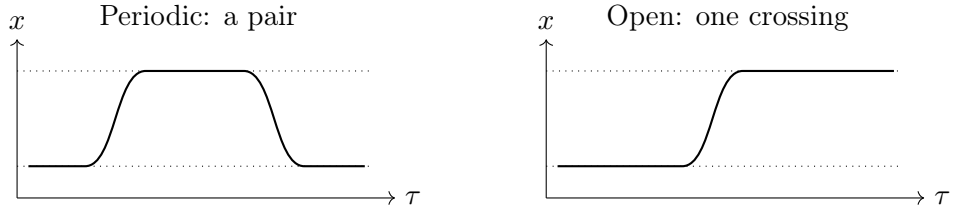


Figure 5.5: Schematic imaginary-time transition paths between two wells. Periodic endpoints require a return crossing. Open endpoint weights can permit a single connecting path.

[11] S. V. Isakov, G. Mazzola, V. N. Smelyanskiy, Z. Jiang, S. Boixo, H. Neven and M. Troyer, *Understanding Quantum Tunneling through Quantum Monte Carlo Simulations*, Phys. Rev. Lett. **117**, 180402 (2016), `https://arxiv.org/abs/1510.08057`.

Exercise 5.6.1: Tunneling in the transverse-field Ising model (**)

Consider the 1D ferromagnetic TFIM of Exercise 5.4.2. Set for example $\Gamma/J = 0.4$ or 0.5. Compare Monte Carlo transitions between opposite magnetizations using periodic (PBC) and open (OBC) boundary conditions in imaginary time.

1. **Calculate the gap.** For increasing small system sizes, diagonalize $\hat{H}$ and obtain $\Delta = E_1 - E_0$.

2. **Construct the paths.** Use the classical mapping derived above, with $\Delta_\tau = \beta/P$ and $K = -\frac{1}{2}\ln\tanh(\Delta_\tau\Gamma)$. Start from $\beta J = 8$ and $P = 128$, checking convergence. Periodic paths have P slices. For open paths use $P+1$ slices, remove the closing time link, and choose the uniform trial amplitude $\psi_T(s) = 2^{-L/2}$. In the symmetric decomposition, the diagonal interaction has half weight at both open endpoints. The total imaginary-time length is β in both cases.

3. **Specify the updates.** Use cluster updates along imaginary time at one physical site at a time, including the longitudinal field generated by the other sites. Define one sweep as L site-worldline updates. Use the same convention for PBC and OBC, without global magnetization-reversal moves.

4. **Measure transition times.** Initialize every spin on every slice to $+1$. Monitor $m_k = L^{-1}\sum_i s_k^i$ and record the first sweep at which at least one quarter of the slices have $m_k = -1$. Repeat over independent runs and estimate the mean time ξ and its uncertainty. Check sensitivity to replacing $m_k = -1$ by $m_k < 0$; do not discard unfinished runs without accounting for their time limits.

5. **Compare the scaling.** At fixed Γ/J, vary L and test

$$\boxed{\xi_{\mathrm{PBC}} \propto (J/\Delta)^2, \qquad \xi_{\mathrm{OBC}} \propto J/\Delta.} \tag{5.146}$$

   Plot $\ln\xi$ against $\ln(\Delta/J)$. Explain the expected slopes using an instanton–anti-instanton pair for periodic paths and a single instanton for open paths.

6. **Interpret the result.** Check time-step convergence and vary β to identify the regime where tunneling dominates thermal activation. Explain why these algorithm-dependent transition times, measured in sweeps, are not physical real-time trajectories.

Exercise 5.6.2: Instantons and Monte Carlo transition times (***)

Here we want to show that the instanton formalism works also in continuos space models[a] Consider $\hat{H} = -\frac{1}{2m}d^2/dx^2 + \lambda x^4 - x^2$, with $m = 1$ in chosen units.

1. Diagonalize a spatial-grid Hamiltonian to obtain $\Delta = E_1 - E_0$. Check convergence in grid spacing and box size as λ is varied. See Appendix A.4.

2. Implement periodic PIMC and open-path PIGS with a flat trial state in a sufficiently large box. For the open action include both half-weight endpoint potentials. Check box effects; the trial state must support both wells.

3. Use a documented local Metropolis update protocol. Define one sweep as one attempted displacement per bead, including the endpoints for open paths. State and keep a consistent rule for choosing displacement sizes. Do not include a global reflection move in the scaling measurement.

4. Start independent runs with all beads in the right well. Define a transition, for example, as the first sweep at which at least one quarter of the beads have $x < -x_{\min}/2$, where $x_{\min} = 1/\sqrt{2\lambda}$. Record the first-passage time, repeat over independent runs, and report uncertainty in its mean. Check that changing the threshold does not change the inferred leading scaling.

5. Check convergence in Δ_τ and in the relevant path lengths, and work where thermal activation does not dominate. In the dilute-instanton regime compare $\log \xi$ with $\log \Delta$: the expected leading slopes are -2 for periodic and -1 for open paths. Varying path lengths or the update protocol can change prefactors and even the applicable regime. Do not assume that the condition for suppressing intrawell excitations is necessarily $\beta\Delta \gg 1$.

[a] G Mazzola, VN Smelyanskiy, M Troyer, *Quantum monte carlo tunneling from quantum chemistry to quantum annealing* Physical Review B 96 (13), 134305 (2017)

Appendix A

Representations, Discretization, and Error Estimates

This appendix collects the formal correspondence between coordinate and occupation representations, two continuum discretization exercises, and useful variational error identities. These details support the main chapter but are not prerequisites for its two model examples.

A.1 Coordinates, Spin, and Exchange Symmetry

For particles in continuous space, first quantization describes a state by a many-particle wavefunction. When spin is present, write $x = (\mathbf{r}, s)$ and use

$$\int dx \equiv \sum_s \int d\mathbf{r}. \tag{A.1}$$

Exchange symmetry acts on the complete coordinates x_i, including spin. The spatial part alone need not have the same symmetry as the full state.

For N_e electrons and N_n nuclei, the nonrelativistic Coulomb Hamiltonian in atomic units ($\hbar = m_e = e = 4\pi\epsilon_0 = 1$) is

$$\begin{aligned} H = & -\frac{1}{2}\sum_{i=1}^{N_e} \nabla_{\mathbf{r}_i}^2 - \sum_{I=1}^{N_n} \frac{1}{2M_I} \nabla_{\mathbf{R}_I}^2 + \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} \\ & + \sum_{I<J} \frac{Z_I Z_J}{|\mathbf{R}_I - \mathbf{R}_J|} - \sum_{i,I} \frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|}. \end{aligned} \tag{A.2}$$

Here M_I is the nuclear mass in electron-mass units. For a proton, the kinetic coefficient is $1/(2m_p)$, not $1/2$. In a clamped-nuclei electronic calculation, the $\mathbf{R}_I$ are fixed parameters: nuclear kinetic energy is omitted and nuclear repulsion contributes a constant to the electronic problem. Nuclear motion on the resulting electronic energy surface is a further calculation within the Born–Oppenheimer approximation.

Symmetrized and antisymmetrized basis states. Let $\{\phi_p(x)\}$ be orthonormal spin-orbitals. For two distinct orbitals,

$$\Psi_\pm(x_1, x_2) = \frac{\phi_a(x_1)\phi_b(x_2) \pm \phi_b(x_1)\phi_a(x_2)}{\sqrt{2}} \tag{A.3}$$

is normalized, with + for bosons and − for fermions. If both bosons occupy the same normalized orbital, their normalized wavefunction is simply $\phi_a(x_1)\phi_a(x_2)$. Two identical fermions cannot occupy the same spin-orbital.

For N fermions in distinct orthonormal spin-orbitals, one obtains a Slater determinant,

$$\Phi_{i_1\dots i_N}(x_1,\dots,x_N) = \frac{1}{\sqrt{N!}}\det[\phi_{i_k}(x_j)]_{j,k=1}^{N}. \tag{A.4}$$

For bosons in orthonormal orbitals with occupations n_p, $\sum_p n_p = N$, the corresponding state is

$$\Phi_{\{n_p\}}(x_1,\dots,x_N) = \frac{1}{\sqrt{N!\prod_p n_p!}}\sum_{P\in S_N} P\left[\prod_{j=1}^{N}\phi_{i_j}(x_j)\right], \tag{A.5}$$

where the list $i_1,\dots,i_N$ contains n_p copies of orbital p, and the sum includes all $N!$ permutations, including repetitions of identical terms. The numerator is a permanent.

These are basis states, not the most general many-body wavefunctions. A correlated state is generally a superposition of determinants or permanents. With a complete one-particle basis these expansions are complete in the corresponding fixed-particle Hilbert space; a finite orbital set defines a truncation.

For example, two electrons in the same normalized spatial orbital u_b form the normalized singlet

$$\Psi(x_1,x_2) = u_b(\mathbf{r}_1)u_b(\mathbf{r}_2)\frac{\chi_\uparrow(s_1)\chi_\downarrow(s_2)-\chi_\downarrow(s_1)\chi_\uparrow(s_2)}{\sqrt{2}}. \tag{A.6}$$

The spatial factor is symmetric and has no extra $1/\sqrt{2}$ prefactor. The full spin-space wavefunction is the Slater determinant of $u_b\chi_\uparrow$ and $u_b\chi_\downarrow$.

A.2 From First to Second Quantization

Second quantization labels a basis state by orbital occupations. It automatically implements exchange symmetry and is equally useful at fixed or variable particle number.

A.2.1 Occupation states and operator ordering

Fix an ordering of M fermionic modes, labeled $1,\dots,M$, and define

$$|f_1,\dots,f_M\rangle = (c_1^\dagger)^{f_1}(c_2^\dagger)^{f_2}\cdots(c_M^\dagger)^{f_M}|\varnothing\rangle, \qquad f_p\in\{0,1\}. \tag{A.7}$$

Operators act from right to left. The vacuum has norm one and is annihilated by every c_p. The anticommutation relations are

$$\{c_p,c_q\} = \{c_p^\dagger,c_q^\dagger\} = 0, \qquad \{c_p,c_q^\dagger\} = \delta_{pq}. \tag{A.8}$$

With the convention above,

$$c_p^\dagger|f_1,\dots,f_M\rangle = \delta_{f_p,0}(-1)^{\sum_{q<p}f_q}|f_1,\dots,1_p,\dots,f_M\rangle, \tag{A.9}$$

$$c_p|f_1,\dots,f_M\rangle = \delta_{f_p,1}(-1)^{\sum_{q<p}f_q}|f_1,\dots,0_p,\dots,f_M\rangle. \tag{A.10}$$

The sign counts the occupied modes crossed when restoring the chosen order. For two modes, $c_1^\dagger c_2^\dagger |\varnothing\rangle = |1,1\rangle$ and $c_2^\dagger c_1^\dagger |\varnothing\rangle = -|1,1\rangle$. On different sites, spin raising operators commute instead, so they cannot be substituted directly for fermionic creation operators.

In the main chapter the dimer basis is defined by explicit creation-operator products. For the canonical mode order $(1\uparrow, 1\downarrow, 2\uparrow, 2\downarrow)$, its state $|3\rangle = c_{2\uparrow}^\dagger c_{1\downarrow}^\dagger |\varnothing\rangle$ is minus the corresponding canonically ordered occupation state. Keeping this phase is necessary to reproduce Eq. 1.15.

For bosons,

$$[b_p, b_q] = [b_p^\dagger, b_q^\dagger] = 0, \qquad [b_p, b_q^\dagger] = \delta_{pq}, \tag{A.11}$$

and normalized occupation states are

$$|n_1, \ldots, n_M\rangle = \prod_{p=1}^{M} \frac{(b_p^\dagger)^{n_p}}{\sqrt{n_p!}} |\varnothing\rangle . \tag{A.12}$$

The creation and annihilation amplitudes are $\sqrt{n_p+1}$ and $\sqrt{n_p}$, respectively. Unlike fermions, a bosonic mode has no intrinsic occupation cutoff. For fixed total particle number N, the dimensions of M-mode spaces are $\binom{M}{N}$ for fermions and $\binom{M+N-1}{N}$ for bosons.

A.2.2 Field operators and Hamiltonian matrix elements

For a complete orthonormal one-particle basis, the fermionic field operators are

$$\hat\Psi(x) = \sum_p \phi_p(x) c_p, \qquad \hat\Psi^\dagger(x) = \sum_p \phi_p^*(x) c_p^\dagger, \qquad c_p^\dagger = \int dx\, \phi_p(x) \hat\Psi^\dagger(x). \tag{A.13}$$

They obey $\{\hat\Psi(x), \hat\Psi^\dagger(x')\} = \delta(x-x')$, with the delta understood as a spatial Dirac delta times a spin Kronecker delta. For a finite orbital set, the sum defines a projected field and the delta is replaced by the basis projector $\sum_p \phi_p(x)\phi_p^*(x')$. The bosonic construction uses commutators instead.

For a one-body operator h and a symmetric, spin-independent pair potential V, the first-quantized Hamiltonian is

$$H = \sum_{i=1}^{N} h(i) + \frac{1}{2} \sum_{i \neq j} V(i,j). \tag{A.14}$$

Its second-quantized form is

$$\begin{aligned} H = & \int dx\, \hat\Psi^\dagger(x) h \hat\Psi(x) \\ & + \frac{1}{2} \int dx\, dx'\, \hat\Psi^\dagger(x) \hat\Psi^\dagger(x') V(x,x') \hat\Psi(x') \hat\Psi(x). \end{aligned} \tag{A.15}$$

Expanding in orbitals gives

$$H = \sum_{pq} h_{pq} c_p^\dagger c_q + \frac{1}{2} \sum_{pqrs} v_{pqrs} c_p^\dagger c_q^\dagger c_s c_r, \tag{A.16}$$

where

$$h_{pq} = \int dx\, \phi_p^*(x) h \phi_q(x), \tag{A.17}$$

$$v_{pqrs} = \int dx\, dx'\, \phi_p^*(x) \phi_q^*(x') V(x,x') \phi_r(x) \phi_s(x'). \tag{A.18}$$

These v_{pqrs} are unsymmetrized interaction integrals. With antisymmetrized fermionic integrals $\bar{v}_{pqrs} = v_{pqrs} - v_{pqsr}$, the two-body prefactor becomes $1/4$. Mixing these conventions changes the interaction strength or sign. The same unsymmetrized formula applies to bosons with c replaced by b.

A.3 A Minimal Molecular Model and Its Approximations

To connect a molecule with the Hubbard dimer, first fix two proton positions and choose two localized, orthonormal spatial orbitals u_1, u_2. Raw atomic orbitals on different centers generally overlap; they must be orthogonalized before using canonical fermionic operators in this basis. The resulting localized orbitals need not remain simple translated copies of an isolated atomic orbital.

The electronic one-body operator is

$$h = -\frac{1}{2}\nabla^2 - \frac{1}{|\mathbf{r} - \mathbf{R}_1|} - \frac{1}{|\mathbf{r} - \mathbf{R}_2|}. \tag{A.19}$$

For a symmetric real basis define

$$-t = \int d\mathbf{r}\, u_1^*(\mathbf{r}) h u_2(\mathbf{r}), \tag{A.20}$$

$$U = \int d\mathbf{r}\, d\mathbf{r}'\, \frac{|u_i(\mathbf{r})|^2 |u_i(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|}, \tag{A.21}$$

$$V = \int d\mathbf{r}\, d\mathbf{r}'\, \frac{|u_1(\mathbf{r})|^2 |u_2(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|}. \tag{A.22}$$

Symmetry gives equal onsite energies $h_{11} = h_{22} = \varepsilon$ and equal onsite repulsions. The phase of one orbital can be chosen so that $t \geq 0$.

Keeping only hopping and density interactions yields the extended dimer

$$H_{\text{ext}} = -t \sum_\sigma \left(c_{1\sigma}^\dagger c_{2\sigma} + c_{2\sigma}^\dagger c_{1\sigma} \right) + U \sum_{i=1}^{2} n_{i\uparrow} n_{i\downarrow} + V n_1 n_2, \tag{A.23}$$

where $n_i = n_{i\uparrow} + n_{i\downarrow}$. At fixed electron number $N = 2$, we have omitted the scalar $2\varepsilon + 1/R$, with $R = |\mathbf{R}_1 - \mathbf{R}_2|$. It must be restored when comparing total molecular energies or bond lengths.

Projection into two orbitals alone does *not* give Eq. A.23: the general projected Hamiltonian also contains exchange, pair hopping, and other off-diagonal Coulomb integrals. Dropping them is a further model approximation. Setting $V = 0$ gives the Hubbard dimer used

in the main chapter. This is useful for isolating the effect of onsite repulsion, but is not justified solely by small orbital overlap: for well-separated localized orbitals, $V \sim 1/R$, whereas hopping and overlap-dependent integrals decay much faster.

Within Eq. A.23, the $(|C\rangle, |I\rangle)$ block and its lower eigenvalue are

$$H_{CI} = \begin{pmatrix} V & -2t \\ -2t & U \end{pmatrix}, \qquad E_- = \frac{U + V - \sqrt{(U-V)^2 + 16t^2}}{2}. \tag{A.24}$$

The correlated state of Eq. 1.29 remains exact for this block after optimizing λ. When $U \geq V$ and $t > 0$, the optimum lies in $0 < \lambda \leq 1$. If $V > U$, ionic configurations are favored and $\lambda > 1$ is needed; a correlator restricted to $g \geq 0$ would exclude that regime.

A.4 Discretizing a Continuous Coordinate

An orbital expansion is not the only route from a differential operator to a matrix. A real-space grid provides an alternative, with boundary conditions and grid spacing as explicit convergence parameters.

A.4.1 The harmonic oscillator as a controlled test

Use oscillator units: energies in $\hbar\omega$ and lengths in $\sqrt{\hbar/(m\omega)}$. Then

$$H = -\frac{1}{2}\frac{d^2}{dx^2} + \frac{1}{2}x^2, \qquad E_0 = \frac{1}{2}, \qquad \psi_0(x) = \pi^{-1/4} e^{-x^2/2}. \tag{A.25}$$

These units additionally set the oscillator frequency to one; atomic units alone would not do so for a general oscillator.

Restrict the coordinate to $[-R, R]$ with Dirichlet boundary conditions $\psi(-R) = \psi(R) = 0$. Use M *interior* grid points,

$$x_j = -R + j\Delta x, \qquad j = 1, \ldots, M, \qquad \Delta x = \frac{2R}{M+1}. \tag{A.26}$$

The centered finite difference gives

$$\psi''(x_j) = \frac{\psi(x_{j+1}) - 2\psi(x_j) + \psi(x_{j-1})}{(\Delta x)^2} + O((\Delta x)^2), \tag{A.27}$$

with the boundary values used for $j = 1$ and $j = M$. The matrix is

$$H_{jk} = \left[\frac{1}{(\Delta x)^2} + \frac{1}{2}x_j^2\right]\delta_{jk} - \frac{\delta_{j,k+1} + \delta_{j,k-1}}{2(\Delta x)^2}. \tag{A.28}$$

This is a second-order finite-difference scheme. It should be distinguished from the sinc discrete variable representation (DVR) below.

If the eigensolver returns $\sum_j |v_j|^2 = 1$, continuum normalization is recovered approximately through

$$\psi(x_j) \simeq \frac{v_j}{\sqrt{\Delta x}}, \qquad \Delta x \sum_j |\psi(x_j)|^2 \simeq 1. \tag{A.29}$$

When comparing with an analytical wavefunction, also align the arbitrary overall phase or sign.

There are two independent errors: the finite box and the finite spacing. Refine Δx at fixed R, and enlarge R while keeping Δx fixed or smaller. Increasing R at fixed M coarsens the grid and need not improve the result. Finite-difference eigenvalues are not generally variational upper bounds to continuum eigenvalues; they can approach the continuum value from below.

Exercise A.4.1: A continuum convergence study

Use the harmonic oscillator of Eq. A.25.

1. Construct Eq. A.28, starting with $R = 6$ and several values of M.
2. Compare the ground-state energy and normalized wavefunction with Eq. A.25. Check the expected second-order energy convergence once the box error is negligible.
3. Vary R at approximately fixed spacing to isolate the boundary error. Explain why varying M alone cannot test both errors.

Many-particle grids. With M points per Cartesian coordinate in d spatial dimensions, an unsymmetrized N-particle grid has M^{dN} entries. Exchange symmetry changes the counting, but does not generally remove the rapid growth. If the one-particle discretization has $K = M^d$ spinless modes, the fixed-N fermionic and bosonic dimensions are $\binom{K}{N}$ and $\binom{K+N-1}{N}$, respectively. Spin enlarges the one-particle mode count.

A.4.2 The sinc kinetic-energy matrix

For a coordinate of mass μ, in atomic units, the sinc-DVR kinetic matrix on an infinite uniform grid of spacing Δx is

$$T_{jk} = \frac{1}{2\mu(\Delta x)^2} \begin{cases} \pi^2/3, & j = k, \\ 2(-1)^{j-k}/(j-k)^2, & j \neq k. \end{cases} \tag{A.30}$$

Unlike the three-point finite difference, this matrix is dense. In the usual DVR approximation the potential is diagonal, $V_{jk} = V(x_j)\delta_{jk}$. A calculation retains a finite set of grid functions; the represented interval and spacing must both be converged. This truncation should not be confused with imposing the exact finite-box Dirichlet discretization used above.[1]

A.4.3 Optional application: a Morse model for deuterium vibration

For rotationless nuclear motion, take

$$H_{\text{vib}} = -\frac{1}{2\mu}\frac{d^2}{dr^2} + V(r), \qquad V(r) = D_e\left(1 - e^{-a(r-r_e)}\right)^2 + E_{\text{diss}} - D_e. \tag{A.31}$$

[1] D. T. Colbert and W. H. Miller, *A novel discrete variable representation for quantum mechanical reactive scattering via the S-matrix Kohn method*, J. Chem. Phys. **96**, 1982–1991 (1992), doi:10.1063/1.462100.

Use the parameters from a molecular fit[2],

$$\begin{aligned} D_e &= 0.143783 \text{ Ha}, & a &= 1.164822 \text{ Bohr}^{-1}, \\ r_e &= 1.415847 \text{ Bohr}, & E_{\text{diss}} &= -1.025862 \text{ Ha}, \\ \mu &= 1837.4711\, m_e. \end{aligned}$$

This reduced mass is appropriate to D_2; H_2 requires a different mass and has different vibrational levels. At the Born–Oppenheimer level the electronic potential is isotope independent. Here the parameters are retained as a numerical benchmark for a fitted model, not as an independently validated spectroscopic potential. A fit near the minimum need not predict dissociation correctly.

The analytical Morse levels on the full line are

$$E_v = E_{\text{diss}} - D_e + \omega_e \left(v + \frac{1}{2}\right) - \frac{a^2}{2\mu}\left(v + \frac{1}{2}\right)^2, \qquad \omega_e = a\sqrt{\frac{2D_e}{\mu}}, \tag{A.32}$$

for integers $v \geq 0$ satisfying $v + 1/2 < \sqrt{2\mu D_e}/a$. For the stated parameters,

$$\begin{aligned} E_{\text{min}} &= -1.169645 \text{ Ha}, \\ E_0 - E_{\text{min}} &= 0.0071936803 \text{ Ha}, \\ E_0 &= -1.1624513197 \text{ Ha}, \\ E_1 - E_0 &= 0.0138335517 \text{ Ha}. \end{aligned}$$

The zero-point energy $E_0 - E_{\text{min}}$ and a transition energy such as $E_1 - E_0$ are different observables. Molecular vibration physically has $r > 0$ and a boundary condition at the origin. The full-line Morse formula is an excellent low-lying benchmark when the wavefunction is negligible near that boundary, but the two boundary-value problems should not be identified exactly.

Exercise A.4.2: Morse potential and zero-point motion

Use the D_2 parameters above.

1. Discretize the vibrational Hamiltonian with the stated reduced mass. Begin with an interval containing the potential minimum and extend it until the first two levels are stable.
2. Compare finite differences and sinc DVR while refining the spacing. Compare the converged energies with the analytical Morse benchmark, keeping track of the boundary convention.
3. Plot the two lowest normalized wavefunctions and calculate the zero-point and transition energies. Explain which quantities could be compared with spectroscopy and which errors would remain from the fitted potential.

[2] C Cozza, K Nakano, S Howard, H Xie, R Helled, G Mazzola, *Denser hydrogen inferred from first-principles simulations challenges Jupiter's interior models*, Physical Review Research 8 (1), 013089 11 2026

A.5 Energy, Fidelity, and Variance

Assume a normalized trial state and a nondegenerate ground state $|\phi_0\rangle$ separated by a gap $\Delta > 0$. Choose the overall phase so that the ground-state overlap is real and nonnegative, and write

$$|\psi\rangle = \sqrt{1-\eta^2}\,|\phi_0\rangle + \eta\,|\chi\rangle\,, \qquad \langle\phi_0|\chi\rangle = 0, \qquad \langle\chi|\chi\rangle = 1. \tag{A.33}$$

For a finite-dimensional Hamiltonian, define

$$A = \langle\chi|\,(H - E_0)\,|\chi\rangle\,, \qquad B = \langle\chi|\,(H - E_0)^2\,|\chi\rangle\,. \tag{A.34}$$

Orthogonality and the eigenvalue equation give

$$\epsilon \equiv \langle\psi|\,H\,|\psi\rangle - E_0 = \eta^2 A, \qquad A \geq \Delta, \qquad \eta^2 \leq \frac{\epsilon}{\Delta}. \tag{A.35}$$

This proves Eq. 1.23. The phase-aligned vector distance satisfies

$$\|\psi - \phi_0\|^2 = 2 - 2\sqrt{1-\eta^2} = \eta^2 + O(\eta^4). \tag{A.36}$$

For a fixed finite Hamiltonian the energy error is therefore second order in a small wavefunction deviation. For an unbounded continuum operator, small Hilbert-space norm error alone is insufficient; control of the relevant energy moments is also required.

The energy variance is exactly

$$\mathrm{Var}_\psi(H) = \langle\psi|\,H^2\,|\psi\rangle - \langle\psi|\,H\,|\psi\rangle^2 = \eta^2 B - \eta^4 A^2. \tag{A.37}$$

For a fixed orthogonal direction $|\chi\rangle$, elimination of η gives

$$\mathrm{Var}_\psi(H) = \frac{B}{A}\epsilon - \epsilon^2. \tag{A.38}$$

An approximately linear energy–variance relation can consequently emerge near an eigenstate when B/A varies little along the trial family. It is not a universal relation with a fixed slope for arbitrary trial states.

Finally,

$$\mathrm{Var}_\psi(H) = \|(H - \langle H\rangle_\psi)\,|\psi\rangle\,\|^2. \tag{A.39}$$

Zero variance identifies an eigenstate, including a possible excited eigenstate. Ground-state identification needs additional information. If the ground space is degenerate, fidelity should instead mean the total weight in that space, and the gap is measured to the first level outside it.

Appendix B

Details of the Fokker–Planck calculations

Transition probability $K(x'|x)$

We want to compute the integral, with $\Delta > 0$ and $k_B T > 0$,

$$\int \frac{dz}{\sqrt{2\pi}}\, e^{-z^2/2}\, \delta\left(x' - x - \Delta f(x) - \sqrt{2\Delta k_B T}\, z\right). \tag{B.1}$$

Let us define

$$y = x' - x - \Delta f(x), \tag{B.2}$$

so that the delta function becomes

$$\delta\left(y - \sqrt{2\Delta k_B T}\, z\right). \tag{B.3}$$

Using the change-of-variable identity for the delta function, for $a \neq 0$,

$$\delta(az - b) = \frac{1}{|a|}\, \delta\left(z - \frac{b}{a}\right), \tag{B.4}$$

we obtain

$$\delta\left(y - \sqrt{2\Delta k_B T}\, z\right) = \frac{1}{\sqrt{2\Delta k_B T}}\, \delta\left(z - \frac{y}{\sqrt{2\Delta k_B T}}\right). \tag{B.5}$$

Substituting back into the integral, we have

$$\int \frac{dz}{\sqrt{2\pi}}\, e^{-z^2/2} \frac{1}{\sqrt{2\Delta k_B T}}\, \delta\left(z - \frac{y}{\sqrt{2\Delta k_B T}}\right). \tag{B.6}$$

Using the sifting property of the delta function, the integral evaluates to

$$\frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2\Delta k_B T}} \exp\left[-\frac{1}{2}\left(\frac{y}{\sqrt{2\Delta k_B T}}\right)^2\right]. \tag{B.7}$$

Substituting $y = x' - x - \Delta f(x)$, we obtain the final result

$$\begin{aligned}\int \frac{dz}{\sqrt{2\pi}}\, e^{-z^2/2}\, \delta\left(x' - x - \Delta f(x) - \sqrt{2\Delta k_B T}\, z\right) \\ = \frac{1}{\sqrt{4\pi\Delta k_B T}} \exp\left[-\frac{(x' - x - \Delta f(x))^2}{4\Delta k_B T}\right] = K(x'|x).\end{aligned} \tag{B.8}$$

This is the exact one-step transition density of the Euler-discretized Langevin equation. At $T = 0$ the Gaussian expression is replaced by a deterministic delta kernel.

From the Master equation to Fokker–Planck

We start from the Master equation for the probability distribution,

$$P_{n+1}(x') = \int dx\, K(x'|x) P_n(x). \tag{B.9}$$

We now use the symbol $P(x,t)$ rather than $\rho_n(x)$, with $t = n\Delta$, and set $k_B = 1$ in this section. We therefore have two integrals:

$$\begin{aligned}P(x', t+\Delta) &= \int dx\, K(x'|x) P(x,t) \\ &= \int dx \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} \delta\left(x' - x - \Delta f(x) - \sqrt{2\Delta T}\, z\right) P(x,t).\end{aligned} \tag{B.10}$$

We perform the integral over x. The delta-function identity must include the roots and their Jacobians: if F has simple roots x_r, then

$$\int dx\, \delta(F(x)) P(x,t) = \sum_{r: F(x_r)=0} \frac{P(x_r,t)}{|F'(x_r)|}. \tag{B.11}$$

In the present case the roots satisfy

$$x_r + \Delta f(x_r) = x' - \sqrt{2\Delta T}\, z. \tag{B.12}$$

Thus the exact result of the x integration is

$$P(x', t+\Delta) = \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} \sum_r \frac{P(x_r,t)}{|1 + \Delta f'(x_r)|}. \tag{B.13}$$

The force and the Jacobian are evaluated at the root x_r, not at x'. In particular, replacing x_r by $x' - \Delta f(x') - \sqrt{2\Delta T}\, z$ is a small-Δ approximation, not an exact change of variables.

To make this expansion straightforward, assume that f is sufficiently smooth and that $x \mapsto x + \Delta f(x)$ is one-to-one with positive derivative for sufficiently small Δ. A bounded f' is a sufficient condition for this monotonicity at small step size. Denote the unique root by x_*, and assume sufficient regularity and decay of the density to justify expansion and Gaussian integration. More general cases require handling all roots or using a weak expansion of the transition operator; the single-root step should not be assumed without this condition.

Denote spatial derivatives by $\partial_x f = f'$. Iterating the implicit root equation gives, for fixed z,

$$x_* = x' - \sqrt{2\Delta T}\, z - \Delta f(x_*), \tag{B.14}$$

$$= x' - \sqrt{2\Delta T}\, z - \Delta f\left(x' - \sqrt{2\Delta T}\, z\right) + O(\Delta^2), \tag{B.15}$$

$$= x' - \sqrt{2\Delta T}\, z - \Delta f(x') + O(\Delta^{3/2}). \tag{B.16}$$

Similarly, the Jacobian becomes

$$\frac{1}{|1 + \Delta f'(x_*)|} = 1 - \Delta f'(x') + O(\Delta^{3/2}). \tag{B.17}$$

Here the remainder coefficients can depend on z. Under the stated regularity and integrability assumptions, the omitted terms give $o(\Delta)$ after Gaussian averaging. We may therefore write, to the order needed for the continuous-time limit,

$$\begin{aligned} P(x', t+\Delta) = \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} \big[1 - \Delta f'(x')\big] \\ \times P\left(x' - \Delta f(x') - \sqrt{2\Delta T}\, z, t\right) + o(\Delta). \end{aligned} \tag{B.18}$$

Now expand P around x' through second order in the displacement, because the noise is of order $\sqrt{\Delta}$. In the following equations P, P' and P'' are evaluated at (x', t), while f and f' are evaluated at x':

$$\begin{aligned} P\left(x' - \Delta f - \sqrt{2\Delta T}\, z, t\right) = P - P'\left[\Delta f + \sqrt{2\Delta T}\, z\right] \\ + \frac{P''}{2}\left[(\Delta f)^2 + 2\Delta f\sqrt{2\Delta T}\, z + 2\Delta T z^2\right] + \cdots . \end{aligned} \tag{B.19}$$

Both terms in the displacement have a minus sign. The mixed term in its square is present, but is of order $\Delta^{3/2}$; the drift-squared term is of order Δ^2. Neither contributes at order Δ.

Using the Jacobian expansion and putting everything together gives

$$\begin{aligned} P(x', t+\Delta) = \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} (1 - \Delta f') \Bigg\{ P - P'\left[\Delta f + \sqrt{2\Delta T}\, z\right] \\ + \frac{P''}{2}\left[(\Delta f)^2 + 2\Delta f\sqrt{2\Delta T}\, z + 2\Delta T z^2\right] \Bigg\} + o(\Delta). \end{aligned} \tag{B.20}$$

Retaining terms through linear order in Δ, including the noise term of order $\sqrt{\Delta}$ before integration, we obtain

$$\begin{aligned} P(x', t+\Delta) = \int \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} \Big[P - \Delta f P' - \Delta f' P \\ - P'\sqrt{2\Delta T}\, z + P''\Delta T z^2\Big] + o(\Delta). \end{aligned} \tag{B.21}$$

Only a few terms depend on z. We use the Gaussian moments

$$\int_{-\infty}^{\infty} \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} = 1, \tag{B.22}$$

$$\int_{-\infty}^{\infty} \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} z = 0, \tag{B.23}$$

$$\int_{-\infty}^{\infty} \frac{dz}{\sqrt{2\pi}} e^{-z^2/2} z^2 = 1. \tag{B.24}$$

The term proportional to $\sqrt{\Delta}$ therefore vanishes after averaging, while the remaining displayed terms give

$$P(x', t+\Delta) = P - \Delta f P' - \Delta f' P + \Delta T P'' + o(\Delta) \tag{B.25}$$

$$= P + \Delta \left[-(fP)' + TP'' \right] + o(\Delta). \tag{B.26}$$

Consequently,

$$\frac{P(x, t+\Delta) - P(x,t)}{\Delta} = -\frac{\partial}{\partial x}[f(x)P(x,t)] + T\frac{\partial^2 P(x,t)}{\partial x^2} + o(1), \tag{B.27}$$

where we renamed the remaining spatial variable x' as x. Taking $\Delta \to 0$ leads to the continuous-time Fokker–Planck equation,

$$\boxed{\frac{\partial P(x,t)}{\partial t} = -\frac{\partial}{\partial x}[f(x)P(x,t)] + T\frac{\partial^2 P(x,t)}{\partial x^2}.} \tag{B.28}$$

Restoring k_B replaces the diffusion coefficient T by $k_B T$, as in the main chapter. The differential equation describes the continuous-time limit; its stationary solution need not be the exact stationary distribution of the finite-step Euler chain.

Appendix C

Cusp Conditions

We use three-dimensional atomic units, $\hbar = m_e = e^2/(4\pi\epsilon_0) = 1$, and fixed nuclei. Wavefunctions and their smooth prefactors may be complex. The cusp conditions follow from cancellation of the singular terms in the Schrödinger equation; they do not require a real wavefunction. The other particles are held away from additional coalescences, so their contributions are regular in the relative coordinate under consideration.

Electron-ion cusp

Consider the singular part of the Hamiltonian for an electron near a nucleus at $\boldsymbol{R}_I$:

$$\hat{H} = -\frac{1}{2}\nabla^2 - \frac{Z_I}{r}, \qquad r = |\boldsymbol{r} - \boldsymbol{R}_I|. \tag{C.1}$$

The remaining terms in the full many-electron Hamiltonian do not alter the leading $1/r$ cancellation. We use radial coordinates centered on the nucleus. A general wavefunction need not be spherically symmetric. The usual electron–nucleus cusp condition concerns its spherical average

$$\overline{\psi}(r) = \frac{1}{4\pi}\int d\Omega\, \psi(\boldsymbol{R}_I + r\hat{\boldsymbol{n}}, \text{other coordinates}). \tag{C.2}$$

The angular part of the Laplacian integrates to zero, so

$$\overline{\nabla^2\psi} = \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\overline{\psi}}{dr}\right). \tag{C.3}$$

The singular radial equation is therefore

$$-\frac{1}{2}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\overline{\psi}}{dr}\right) - \frac{Z_I}{r}\overline{\psi} + \text{regular terms} = E\overline{\psi}. \tag{C.4}$$

Assume a regular one-sided radial expansion,

$$\overline{\psi}(r) = \overline{\psi}(0) + r\overline{\psi}'(0) + \mathcal{O}(r^2), \tag{C.5}$$

whose coefficients may be complex. Inserting this into the Laplacian gives

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\overline{\psi}'(r)\right) = \frac{1}{r^2}\frac{d}{dr}\left[r^2\overline{\psi}'(0) + \mathcal{O}(r^3)\right] \tag{C.6}$$

$$= \frac{2\overline{\psi}'(0)}{r} + \mathcal{O}(1). \tag{C.7}$$

Collecting singular terms in the Schrödinger equation,

$$-\frac{\overline{\psi}'(0)}{r} - \frac{Z_I\overline{\psi}(0)}{r} + \mathcal{O}(1) = E\overline{\psi}(0) + \mathcal{O}(r). \tag{C.8}$$

Thus

$$\boxed{\overline{\psi}'(0) = -Z_I\overline{\psi}(0).} \tag{C.9}$$

For a radial wavefunction, $\overline{\psi} = \psi$ and this reduces to the familiar expression. When $\overline{\psi}(0) \neq 0$, its logarithmic radial derivative is the real number $-Z_I$, even if $\overline{\psi}$ is complex. If the leading component has nonzero angular momentum, the appropriate condition concerns its radial coefficient rather than a nonzero s-wave value at the origin.

Antiparallel-spin cusp condition

Near electron–electron coalescence, isolate the relative coordinate $\boldsymbol{r} = \boldsymbol{r}_i - \boldsymbol{r}_j$ and the center of mass $\boldsymbol{R} = (\boldsymbol{r}_i + \boldsymbol{r}_j)/2$. The kinetic energy separates as

$$-\frac{1}{2}(\nabla_i^2 + \nabla_j^2) = -\frac{1}{4}\nabla_R^2 - \nabla_r^2. \tag{C.10}$$

The reduced mass is $\mu = 1/2$, and the singular relative Hamiltonian is

$$\hat{H}_{\mathrm{rel}} = -\nabla_r^2 + \frac{1}{r}. \tag{C.11}$$

Center-of-mass and other interaction terms do not change the leading relative cusp.

We impose cancellation of the Coulomb divergence in the local energy, following the standard argument reviewed by Foulkes et al.[1] Write

$$\psi(\boldsymbol{r}) = e^{-u(r)}f(\boldsymbol{r}), \tag{C.12}$$

where u is the radial Jastrow exponent and f is smooth and possibly complex. For the usual antiparallel-spin cusp, assume $f(0) \neq 0$ and bounded derivatives near the origin. Opposite spins allow such a component but do not force every state to have one.

The local energy of the relative problem is

$$E_L(\boldsymbol{r}) = \frac{H_{\mathrm{rel}}e^{-u(r)}f(\boldsymbol{r})}{e^{-u(r)}f(\boldsymbol{r})}. \tag{C.13}$$

[1] W. M. C. Foulkes, L. Mitas, R. J. Needs and G. Rajagopal, *Quantum Monte Carlo simulations of solids*, Reviews of Modern Physics **73**, 33–83 (2001), doi:10.1103/RevModPhys.73.33.

First calculate

$$\nabla e^{-u} = -u' e^{-u} \hat{\boldsymbol{r}}, \qquad \nabla^2 e^{-u} = e^{-u} \left(u'^2 - u'' - \frac{2u'}{r} \right). \tag{C.14}$$

The product rule gives

$$\nabla^2 \psi = f \nabla^2 e^{-u} + 2 \nabla e^{-u} \cdot \nabla f + e^{-u} \nabla^2 f \tag{C.15}$$

$$= e^{-u} \left[\left(u'^2 - u'' - \frac{2u'}{r} \right) f - 2u' \hat{\boldsymbol{r}} \cdot \nabla f + \nabla^2 f \right]. \tag{C.16}$$

There is no division by f until we divide by ψ. No complex conjugation enters this differential identity. Thus

$$E_L = -\frac{\nabla^2 \psi}{\psi} + \frac{1}{r} \tag{C.17}$$

$$= -u'^2 + u'' + \frac{2u'}{r} + 2u' \frac{\hat{\boldsymbol{r}} \cdot \nabla f}{f} - \frac{\nabla^2 f}{f} + \frac{1}{r}. \tag{C.18}$$

For nonzero $f(0)$, bounded derivatives of f, and finite $u'(0), u''(0)$, the singular terms are

$$\frac{2u'(r) + 1}{r}. \tag{C.19}$$

Canceling their leading coefficient gives

$$2u'(0) + 1 = 0, \qquad \boxed{u'(0) = -\frac{1}{2}.} \tag{C.20}$$

This is the standard antiparallel-spin condition for the convention e^{-u}. The spherical average of the relative wavefunction correspondingly satisfies $\overline{\psi}'(0) = \overline{\psi}(0)/2$. The sign would reverse if one defined the Jastrow factor as e^{+u} instead.

Parallel-spin cusp condition

For parallel spins the spatial state is antisymmetric under exchange of the two electrons. At fixed center of mass, this sends $\boldsymbol{r} \to -\boldsymbol{r}$, so the smooth factor is odd. In the generic case with a nonzero linear term,

$$f(\boldsymbol{r}) = \boldsymbol{a} \cdot \boldsymbol{r} + \mathcal{O}(r^3), \tag{C.21}$$

where $\boldsymbol{a}$ may be complex and may depend on the other coordinates. At leading order,

$$\nabla f = \boldsymbol{a}, \qquad \frac{\hat{\boldsymbol{r}} \cdot \nabla f}{f} = \frac{\hat{\boldsymbol{r}} \cdot \boldsymbol{a}}{\boldsymbol{r} \cdot \boldsymbol{a}} = \frac{1}{r}, \tag{C.22}$$

away from directions in which the leading factor vanishes. The Laplacian of the linear term is zero. A general odd function need not have zero Laplacian: for example, $\nabla^2 x^3 = 6x$. Here the cubic and higher terms give $\nabla^2 f = \mathcal{O}(r)$, so $\nabla^2 f / f$ is of order one along directions with a nonzero leading linear coefficient.

Substituting into Eq. (C.18), the leading singular terms become

$$\frac{2u'(r)}{r} + 2u'(r)\frac{\hat{\boldsymbol{r}}\cdot\boldsymbol{a}}{\boldsymbol{r}\cdot\boldsymbol{a}} + \frac{1}{r} = \frac{4u'(r)+1}{r}. \tag{C.23}$$

Their cancellation requires

$$4u'(0) + 1 = 0, \qquad \boxed{u'(0) = -\frac{1}{4}.} \tag{C.24}$$

Thus the same cusp coefficient holds for complex f; it is the relative angular-momentum channel, not a choice of real phase, that changes the coefficient. This argument concerns the Coulomb coalescence singularity, not arbitrary divergences of a trial local energy at other nodes.

For completeness, if the leading smooth factor is a solid harmonic $r^\ell Y_{\ell m}$, then $\hat{\boldsymbol{r}}\cdot\nabla f/f = \ell/r$ and $\nabla^2 f = 0$ at leading order. The relative electron–electron cusp is therefore $u'(0) = -1/[2(\ell+1)]$. The usual antiparallel and parallel results correspond to $\ell = 0$ and $\ell = 1$, respectively.

Appendix D

Details of Quantum State Tomography

This appendix derives the formulas used in Sect. 3.4. We first treat the positive-state reconstruction used in the exercise, then state the likelihood gradient for a general complex wavefunction measured in several bases. The statistical data are complete measurement configurations.

D.1 From the KL divergence to a data likelihood

Let $p_*(\boldsymbol{\sigma})$ be the target probability in a fixed basis, and let $q_\theta(\boldsymbol{\sigma})$ be a normalized, strictly positive model. With natural logarithms,

$$D_{\mathrm{KL}}(p_*\|q_\theta) = \sum_{\boldsymbol{\sigma}} p_*(\boldsymbol{\sigma}) \log p_*(\boldsymbol{\sigma}) - \sum_{\boldsymbol{\sigma}} p_*(\boldsymbol{\sigma}) \log q_\theta(\boldsymbol{\sigma}) \tag{D.1}$$

$$= -\mathcal{H}(p_*) + \mathbb{E}_{p_*}[-\log q_\theta(\boldsymbol{\sigma})], \tag{D.2}$$

where $\mathcal{H}(p_*) = -\sum_{\boldsymbol{\sigma}} p_*(\boldsymbol{\sigma}) \log p_*(\boldsymbol{\sigma})$ is independent of θ. For a dataset $\mathcal{D} = \{\boldsymbol{\sigma}^{(n)}\}_{n=1}^{N_\mathcal{D}}$, the empirical probability is

$$\widehat{p}_\mathcal{D}(\boldsymbol{\sigma}) = \frac{1}{N_\mathcal{D}} \sum_{n=1}^{N_\mathcal{D}} \mathbf{1}_{\boldsymbol{\sigma}=\boldsymbol{\sigma}^{(n)}}. \tag{D.3}$$

The corresponding loss satisfies the exact finite-data relation

$$\mathcal{L}_\mathcal{D}(\theta) = -\sum_{\boldsymbol{\sigma}} \widehat{p}_\mathcal{D}(\boldsymbol{\sigma}) \log q_\theta(\boldsymbol{\sigma}) \tag{D.4}$$

$$= -\frac{1}{N_\mathcal{D}} \sum_n \log q_\theta(\boldsymbol{\sigma}^{(n)}) \tag{D.5}$$

$$= D_{\mathrm{KL}}(\widehat{p}_\mathcal{D}\|q_\theta) + \mathcal{H}(\widehat{p}_\mathcal{D}). \tag{D.6}$$

For independent snapshots, maximizing $\prod_n q_\theta(\boldsymbol{\sigma}^{(n)})$ is therefore equivalent to minimizing this loss. It estimates the population objective rather than giving exact access to the unknown p_*.

D.2 Marginalizing a binary RBM

Use $\sigma_j, h_k \in \{-1, +1\}$ and real a_j, b_k, W_{jk}. Define the unnormalized joint weight

$$\widetilde{q}_\theta(\boldsymbol{\sigma}, \boldsymbol{h}) = \exp\left(\sum_j a_j\sigma_j + \sum_k b_k h_k + \sum_{jk} W_{jk}\sigma_j h_k\right), \qquad Z_\theta = \sum_{\boldsymbol{\sigma},\boldsymbol{h}} \widetilde{q}_\theta(\boldsymbol{\sigma}, \boldsymbol{h}). \tag{D.7}$$

With $\theta_k(\boldsymbol{\sigma}) = b_k + \sum_j W_{jk}\sigma_j$, summing over hidden spins gives

$$\widetilde{q}_\theta(\boldsymbol{\sigma}) = e^{\sum_j a_j\sigma_j} \sum_{h_1=\pm1} \cdots \sum_{h_M=\pm1} \prod_{k=1}^{M} e^{h_k\theta_k(\boldsymbol{\sigma})} \tag{D.8}$$

$$= e^{\sum_j a_j\sigma_j} \prod_{k=1}^{M} \left(\sum_{h_k=\pm1} e^{h_k\theta_k(\boldsymbol{\sigma})}\right) \tag{D.9}$$

$$= e^{\sum_j a_j\sigma_j} \prod_k \left[e^{\theta_k(\boldsymbol{\sigma})} + e^{-\theta_k(\boldsymbol{\sigma})}\right] \tag{D.10}$$

$$= e^{\sum_j a_j\sigma_j} \prod_k 2\cosh\theta_k(\boldsymbol{\sigma}) = e^{A_\theta(\boldsymbol{\sigma})}, \tag{D.11}$$

where

$$A_\theta(\boldsymbol{\sigma}) = \sum_j a_j\sigma_j + \sum_k \log\left[2\cosh\theta_k(\boldsymbol{\sigma})\right]. \tag{D.12}$$

Thus $q_\theta(\boldsymbol{\sigma}) = e^{A_\theta(\boldsymbol{\sigma})}/Z_\theta$, and the normalized positive state is

$$\psi_\theta(\boldsymbol{\sigma}) = \sqrt{q_\theta(\boldsymbol{\sigma})} = \frac{e^{A_\theta(\boldsymbol{\sigma})/2}}{\sqrt{Z_\theta}}. \tag{D.13}$$

The unnormalized amplitude is $e^{A/2}$. In this convention, for any parameter coordinate θ_μ,

$$O_\mu^{\text{unnorm}}(\boldsymbol{\sigma}) = \frac{1}{2}\partial_\mu A_\theta(\boldsymbol{\sigma}), \tag{D.14}$$

$$\partial_\mu \log\psi_\theta(\boldsymbol{\sigma}) = \frac{1}{2}\left[\partial_\mu A_\theta(\boldsymbol{\sigma}) - \partial_\mu \log Z_\theta\right]. \tag{D.15}$$

Keeping normalized and unnormalized quantities separate avoids an extra factor of two in gradients or sampling probabilities.

D.3 Differentiating the partition function and the loss

For finite RBM parameters on a finite configuration space, differentiation and the sum commute. We have

$$\partial_\mu \log Z_\theta = \frac{1}{Z_\theta}\partial_\mu \sum_{\boldsymbol{\sigma}} e^{A_\theta(\boldsymbol{\sigma})} \tag{D.16}$$

$$= \frac{1}{Z_\theta}\sum_{\boldsymbol{\sigma}} e^{A_\theta(\boldsymbol{\sigma})}\partial_\mu A_\theta(\boldsymbol{\sigma}) \tag{D.17}$$

$$= \sum_{\boldsymbol{\sigma}} q_\theta(\boldsymbol{\sigma})\partial_\mu A_\theta(\boldsymbol{\sigma}) = \langle\partial_\mu A_\theta\rangle_{q_\theta}. \tag{D.18}$$

Since $\log q_\theta = A_\theta - \log Z_\theta$, its score is

$$\partial_\mu \log q_\theta(\boldsymbol{\sigma}) = \partial_\mu A_\theta(\boldsymbol{\sigma}) - \langle \partial_\mu A_\theta \rangle_{q_\theta}, \qquad \langle \partial_\mu \log q_\theta \rangle_{q_\theta} = 0. \tag{D.19}$$

The data are fixed when differentiating, so

$$\partial_\mu \mathcal{L}_\mathcal{D} = -\frac{1}{N_\mathcal{D}} \sum_n \partial_\mu \log q_\theta(\boldsymbol{\sigma}^{(n)}) \tag{D.20}$$

$$= \langle \partial_\mu A_\theta \rangle_{q_\theta} - \frac{1}{N_\mathcal{D}} \sum_n \partial_\mu A_\theta(\boldsymbol{\sigma}^{(n)}). \tag{D.21}$$

The model contribution comes precisely from the unknown normalization. Sampling the model can estimate this term without computing Z_θ itself.

For the individual parameters,

$$\partial_{a_j} A_\theta(\boldsymbol{\sigma}) = \sigma_j, \tag{D.22}$$

$$\partial_{b_k} A_\theta(\boldsymbol{\sigma}) = \frac{2\sinh\theta_k(\boldsymbol{\sigma})}{2\cosh\theta_k(\boldsymbol{\sigma})} = \tanh\theta_k(\boldsymbol{\sigma}), \tag{D.23}$$

$$\partial_{W_{jk}} A_\theta(\boldsymbol{\sigma}) = \tanh\theta_k(\boldsymbol{\sigma})\, \partial_{W_{jk}} \theta_k(\boldsymbol{\sigma}) = \sigma_j \tanh\theta_k(\boldsymbol{\sigma}). \tag{D.24}$$

These are derivatives of $A = \log \widetilde{q}$, not of $\widetilde{q}$ itself. Substituting them gives the three gradient formulas in the main text.

D.4 The conditional distributions and block Gibbs chain

At fixed $\boldsymbol{\sigma}$, all terms involving different hidden spins factorize:

$$q_\theta(\boldsymbol{h} \mid \boldsymbol{\sigma}) = \frac{\exp[\sum_k h_k \theta_k(\boldsymbol{\sigma})]}{\prod_k 2\cosh\theta_k(\boldsymbol{\sigma})} \tag{D.25}$$

$$= \prod_k \frac{e^{h_k \theta_k(\boldsymbol{\sigma})}}{2\cosh\theta_k(\boldsymbol{\sigma})}. \tag{D.26}$$

Consequently,

$$q(h_k = +1 \mid \boldsymbol{\sigma}) = \frac{e^{\theta_k(\boldsymbol{\sigma})}}{e^{\theta_k(\boldsymbol{\sigma})} + e^{-\theta_k(\boldsymbol{\sigma})}} = \frac{1 + \tanh\theta_k(\boldsymbol{\sigma})}{2}. \tag{D.27}$$

The same calculation at fixed $\boldsymbol{h}$, with $r_j(\boldsymbol{h}) = a_j + \sum_k W_{jk} h_k$, gives

$$q_\theta(\boldsymbol{\sigma} \mid \boldsymbol{h}) = \prod_j \frac{e^{\sigma_j r_j(\boldsymbol{h})}}{2\cosh r_j(\boldsymbol{h})}, \tag{D.28}$$

$$q(\sigma_j = +1 \mid \boldsymbol{h}) = \frac{1 + \tanh r_j(\boldsymbol{h})}{2}. \tag{D.29}$$

One visible-to-visible block Gibbs step has transition probability

$$P^{(\theta)}_{\boldsymbol{\sigma}\boldsymbol{\sigma}'} = \sum_{\boldsymbol{h}} q_\theta(\boldsymbol{h} \mid \boldsymbol{\sigma}) q_\theta(\boldsymbol{\sigma}' \mid \boldsymbol{h}). \tag{D.30}$$

It satisfies detailed balance for the visible marginal:

$$q_\theta(\boldsymbol{\sigma})P^{(\theta)}_{\boldsymbol{\sigma}\boldsymbol{\sigma}'} = \sum_{\boldsymbol{h}} q_\theta(\boldsymbol{\sigma},\boldsymbol{h})q_\theta(\boldsymbol{\sigma}' \mid \boldsymbol{h}) \tag{D.31}$$

$$= \sum_{\boldsymbol{h}} q_\theta(\boldsymbol{h})q_\theta(\boldsymbol{\sigma} \mid \boldsymbol{h})q_\theta(\boldsymbol{\sigma}' \mid \boldsymbol{h}) \tag{D.32}$$

$$= q_\theta(\boldsymbol{\sigma}')P^{(\theta)}_{\boldsymbol{\sigma}'\boldsymbol{\sigma}}. \tag{D.33}$$

For finite real parameters, all conditional probabilities are strictly between zero and one, so the finite chain is irreducible and aperiodic. It therefore converges to q_θ, though its mixing can still be slow.

The hidden conditional expectation is

$$\mathbb{E}[h_k \mid \boldsymbol{\sigma}] = (+1)q(h_k = +1 \mid \boldsymbol{\sigma}) + (-1)q(h_k = -1 \mid \boldsymbol{\sigma}) \tag{D.34}$$

$$= 2q(h_k = +1 \mid \boldsymbol{\sigma}) - 1 = \tanh\theta_k(\boldsymbol{\sigma}). \tag{D.35}$$

Therefore

$$\mathbb{E}_{q(\boldsymbol{\sigma},\boldsymbol{h})}[\sigma_j h_k] = \mathbb{E}_{q(\boldsymbol{\sigma})}[\sigma_j \tanh\theta_k(\boldsymbol{\sigma})]. \tag{D.36}$$

Using these conditional means in gradient measurements avoids unnecessary hidden-spin noise. The mean is $\tanh\theta_k$, not the probability $(1+\tanh\theta_k)/2$. This averaging does not replace the random hidden draws required to move the Gibbs chain.

D.5 What contrastive divergence approximates

Let $p_0 = \widehat{p}_{\mathcal{D}}$ and let $p_m = p_0(P^{(\theta)})^m$ be the distribution after m block Gibbs steps, with parameters held fixed. For $B_i(\boldsymbol{\sigma}) = \partial_{\theta_i} A_\theta(\boldsymbol{\sigma})$, the exact likelihood gradient is

$$g_i^{\mathrm{ML}} = \langle B_i\rangle_{q_\theta} - \langle B_i\rangle_{p_0}. \tag{D.37}$$

The expected CD-m update uses

$$g_i^{\text{CD-}m} = \langle B_i\rangle_{p_m} - \langle B_i\rangle_{p_0}. \tag{D.38}$$

Its bias relative to maximum likelihood is exactly

$$g_i^{\text{CD-}m} - g_i^{\mathrm{ML}} = \langle B_i\rangle_{p_m} - \langle B_i\rangle_{q_\theta}. \tag{D.39}$$

Increasing m reduces this discrepancy in the limit of convergence at fixed parameters, but does not guarantee monotonic improvement in each finite stochastic training run. The dataset distribution is not assumed to equal the model distribution.

The name contrastive divergence is associated with the functional

$$\mathrm{CD}_m(\theta) = D_{\mathrm{KL}}(p_0\|q_\theta) - D_{\mathrm{KL}}(p_m\|q_\theta). \tag{D.40}$$

Since p_m also depends on θ through the transition kernel, differentiating both terms gives

$$\partial_{\theta_i}\mathrm{CD}_m = \langle B_i\rangle_{p_m} - \langle B_i\rangle_{p_0} - \sum_{\boldsymbol{\sigma}} \partial_{\theta_i} p_m(\boldsymbol{\sigma}) \log\frac{p_m(\boldsymbol{\sigma})}{q_\theta(\boldsymbol{\sigma})}. \tag{D.41}$$

Normalization removes the additional term $\sum_{\boldsymbol{\sigma}} \partial_{\theta_i} p_m(\boldsymbol{\sigma}) = 0$. The usual CD update omits the last term in Eq. (D.41); it is generally neither the exact likelihood gradient nor the full derivative of this functional. At equilibrium $p_m = q_\theta$, the discrepancy vanishes.

In an implementation, ordinary CD does not backpropagate through sampled Gibbs transitions. It evaluates the two sets of sufficient statistics at the current parameters and subtracts them. Persistent contrastive divergence starts model chains from their previous states rather than resetting them to data; it is an alternative approximation to the model expectation, not a guarantee of equilibrium.

D.6 Fisher geometry for the positive-state model

Define $B_i = \partial_i A_\theta$. The model Fisher information is

$$I_{ij} = \langle \partial_i \log q_\theta \, \partial_j \log q_\theta \rangle_q \tag{D.42}$$

$$= \langle B_i B_j \rangle_q - \langle B_i \rangle_q \langle B_j \rangle_q. \tag{D.43}$$

For the unnormalized amplitude $e^{A/2}$, $O_i = B_i/2$. Thus

$$S_{ij} = \frac{1}{4} \left[\langle B_i B_j \rangle_q - \langle B_i \rangle_q \langle B_j \rangle_q \right], \qquad I = 4S. \tag{D.44}$$

An optional classical natural-gradient likelihood update solves

$$(I + \lambda \mathbb{I}) \, \delta\theta = -\eta \nabla_\theta \mathcal{L}, \tag{D.45}$$

with a suitable regularization. The covariance must estimate the stated model scores; the covariance of data-minus-short-chain statistics is not automatically this Fisher matrix. The vector applied to the parameters must be the solution of the linear system, not the unpreconditioned gradient.

D.7 Likelihood gradients for a general complex wavefunction

Finally consider a complex, unnormalized $\psi_\theta(x)$ with real parameter coordinates θ. For a fixed measurement basis b, define

$$U_b(y, x) = \langle y; b | x \rangle, \qquad \psi_\theta^{[b]}(y) = \sum_x U_b(y, x) \psi_\theta(x), \qquad q_\theta^{[b]}(y) = \frac{|\psi_\theta^{[b]}(y)|^2}{\mathcal{N}_\theta}. \tag{D.46}$$

At outcomes of nonzero model probability, define the rotated logarithmic derivative

$$O_i^{[b]}(y) = \frac{\partial_i \psi_\theta^{[b]}(y)}{\psi_\theta^{[b]}(y)} = \frac{\sum_x U_b(y, x) \psi_\theta(x) O_i(x)}{\sum_x U_b(y, x) \psi_\theta(x)}. \tag{D.47}$$

The basis transformation is fixed, so its parameter derivative is zero. Since

$$\partial_i \log \mathcal{N}_\theta = 2 \, \mathrm{Re} \langle O_i \rangle_{P_\theta}, \qquad P_\theta(x) = |\psi_\theta(x)|^2 / \mathcal{N}_\theta, \tag{D.48}$$

we find

$$\partial_i \log q_\theta^{[b]}(y) = \partial_i \log \psi_\theta^{[b]}(y) + \partial_i \log \psi_\theta^{[b]*}(y) - \partial_i \log \mathcal{N}_\theta \tag{D.49}$$

$$= 2\,\mathrm{Re}\left[O_i^{[b]}(y) - \langle O_i \rangle_{P_\theta}\right]. \tag{D.50}$$

Therefore, for datasets $\mathcal{D}_b$ and fixed positive weights ω_b,

$$\mathcal{L}(\theta) = -\sum_b \omega_b \langle \log q_\theta^{[b]} \rangle_{\mathcal{D}_b}, \tag{D.51}$$

$$\partial_i \mathcal{L} = 2\sum_b \omega_b\,\mathrm{Re}\left[\langle O_i \rangle_{P_\theta} - \langle O_i^{[b]} \rangle_{\mathcal{D}_b}\right]. \tag{D.52}$$

At a data outcome with zero model probability, the negative log-likelihood is infinite; the displayed ratios assume nonzero amplitudes at observed outcomes. Rotated-basis sums can also be expensive, so these general equations are not by themselves an efficient algorithm for arbitrary bases.

For the positive RBM with only the reference basis, $O_i = \frac{1}{2}\partial_i A$ and the factor of two cancels, recovering the model-minus-data gradient derived above. With a general complex state and only the reference basis, configuration-dependent phases are invisible to the likelihood. Measurements in sufficiently informative additional bases, or justified prior restrictions on the state, are necessary to learn them.

Appendix E

Derivation of Importance Sampling DMC

We consider the imaginary-time Schrödinger equation

$$-\frac{\partial}{\partial\tau}\psi(\boldsymbol{r},\tau) = (\hat{H} - E_T)\psi(\boldsymbol{r},\tau), \qquad \hat{H} = -\frac{1}{2}\nabla^2 + V(\boldsymbol{r}). \tag{E.1}$$

Here ∇ and ∇^2 act on all $3N$ spatial coordinates.

Let $\psi_T(\boldsymbol{r})$ be a real, time-independent guiding wavefunction. We define

$$f(\boldsymbol{r},\tau) = \psi_T(\boldsymbol{r})\psi(\boldsymbol{r},\tau), \tag{E.2}$$

together with the drift velocity and the local energy,

$$\mathbf{v}(\boldsymbol{r}) = \frac{\nabla\psi_T(\boldsymbol{r})}{\psi_T(\boldsymbol{r})}, \qquad E_L(\boldsymbol{r}) = \frac{\hat{H}\psi_T(\boldsymbol{r})}{\psi_T(\boldsymbol{r})} = -\frac{1}{2}\frac{\nabla^2\psi_T(\boldsymbol{r})}{\psi_T(\boldsymbol{r})} + V(\boldsymbol{r}). \tag{E.3}$$

The following identities hold wherever $\psi_T \neq 0$. For fixed-node DMC, they apply inside each nodal pocket; the nodal boundary condition is imposed separately. We suppress the arguments to simplify the notation.

Since ψ_T does not depend on imaginary time,

$$-\frac{\partial f}{\partial\tau} = -\psi_T\frac{\partial\psi}{\partial\tau} = \psi_T(\hat{H} - E_T)\psi. \tag{E.4}$$

We therefore need to express the right-hand side in terms of f.

First, apply the product rule to $f = \psi_T\psi$:

$$\nabla f = \psi_T\nabla\psi + \psi\nabla\psi_T, \tag{E.5}$$

$$\nabla^2 f = \psi_T\nabla^2\psi + 2\nabla\psi_T\cdot\nabla\psi + \psi\nabla^2\psi_T. \tag{E.6}$$

Next, notice that the product of the drift velocity and f simplifies:

$$\mathbf{v}f = \frac{\nabla\psi_T}{\psi_T}\,\psi_T\psi = \psi\nabla\psi_T. \tag{E.7}$$

Taking its divergence gives

$$\nabla \cdot (\mathbf{v}f) = \nabla\psi \cdot \nabla\psi_T + \psi\nabla^2\psi_T. \tag{E.8}$$

Combining Eqs. (E.6) and (E.8), we obtain

$$\begin{aligned} -\frac{1}{2}\nabla^2 f + \nabla \cdot (\mathbf{v}f) &= -\frac{1}{2}\psi_T\nabla^2\psi - \nabla\psi_T \cdot \nabla\psi - \frac{1}{2}\psi\nabla^2\psi_T \\ &\quad + \nabla\psi \cdot \nabla\psi_T + \psi\nabla^2\psi_T \qquad \text{(E.9)} \\ &= -\frac{1}{2}\psi_T\nabla^2\psi + \frac{1}{2}\psi\nabla^2\psi_T. \qquad \text{(E.10)} \end{aligned}$$

The mixed-gradient terms cancel.

Finally, multiplying the local energy by f gives

$$\begin{aligned} E_L f &= \left(-\frac{1}{2}\frac{\nabla^2\psi_T}{\psi_T} + V\right)\psi_T\psi \qquad \text{(E.11)} \\ &= -\frac{1}{2}\psi\nabla^2\psi_T + V\psi_T\psi. \qquad \text{(E.12)} \end{aligned}$$

Adding this expression cancels the remaining $\nabla^2\psi_T$ term:

$$\begin{aligned} -\frac{1}{2}\nabla^2 f + \nabla \cdot (\mathbf{v}f) + E_L f &= -\frac{1}{2}\psi_T\nabla^2\psi + V\psi_T\psi \qquad \text{(E.13)} \\ &= \psi_T\hat{H}\psi. \qquad \text{(E.14)} \end{aligned}$$

Substituting into Eq. (E.4) therefore yields

$$\boxed{-\frac{\partial f}{\partial \tau} = -\frac{1}{2}\nabla^2 f + \nabla \cdot (\mathbf{v}f) + (E_L - E_T)f.} \tag{E.15}$$

Equivalently,

$$\frac{\partial f}{\partial \tau} = \frac{1}{2}\nabla^2 f - \nabla \cdot (\mathbf{v}f) - (E_L - E_T)f. \tag{E.16}$$

The first two terms describe diffusion with drift velocity $\mathbf{v}$, while the last term changes the local weight. For $E_L < E_T$ the weight grows; for $E_L > E_T$ it decreases.

Appendix F

Supplementary Calculations for PIMC

F.1 Derivation of the Centroid-Virial Estimator

We consider the distinguishable-particle primitive path integral of the main chapter. Write $D = dN$ for the number of Cartesian coordinates and use $\hbar = k_B = 1$. At fixed bead number P,

$$Z_P = \left(\frac{mP}{2\pi\beta}\right)^{DP/2} \int \prod_{j=0}^{P-1} d\boldsymbol{r}_j \, \exp\left[-\frac{mP}{2\beta}\sum_j |\boldsymbol{r}_{j+1}-\boldsymbol{r}_j|^2 - \frac{\beta}{P}\sum_j V(\boldsymbol{r}_j)\right], \tag{F.1}$$

with $\boldsymbol{r}_P = \boldsymbol{r}_0$. We assume a sufficiently regular confining potential so that these integrals and the derivatives used below exist. Hard boundaries, periodic winding sectors and permutation cycles require extra care and are not covered by this simple coordinate change.

Introduce the centroid and internal coordinates,

$$\boldsymbol{r}_c = \frac{1}{P}\sum_j \boldsymbol{r}_j, \qquad \boldsymbol{u}_j = \boldsymbol{r}_j - \boldsymbol{r}_c, \qquad \sum_j \boldsymbol{u}_j = 0. \tag{F.2}$$

There are $D(P-1)$ independent internal coordinates. Scale them as

$$\boldsymbol{u}_j = \sqrt{\beta}\,\boldsymbol{q}_j, \qquad \sum_j \boldsymbol{q}_j = 0. \tag{F.3}$$

The transformation from $\{\boldsymbol{r}_j\}$ to a centroid and independent internal coordinates has a constant Jacobian. The subsequent scaling contributes $\beta^{D(P-1)/2}$. Consequently,

$$\left(\frac{mP}{2\pi\beta}\right)^{DP/2} \prod_j d\boldsymbol{r}_j = \mathcal{C}_P \beta^{-D/2}\, d\boldsymbol{r}_c\, d\boldsymbol{q}, \tag{F.4}$$

where $d\boldsymbol{q}$ denotes the $D(P-1)$ independent coordinates and $\mathcal{C}_P$ is independent of β.

The kinetic action becomes independent of β:

$$\frac{mP}{2\beta}\sum_j |\boldsymbol{r}_{j+1}-\boldsymbol{r}_j|^2 = \frac{mP}{2}\sum_j |\boldsymbol{q}_{j+1}-\boldsymbol{q}_j|^2 \equiv S_K(\boldsymbol{q}). \tag{F.5}$$

Thus

$$Z_P = \mathcal{C}_P \beta^{-D/2} \int d\boldsymbol{r}_c \, d\boldsymbol{q} \; e^{-S_K(\boldsymbol{q})} \exp\left[-\frac{\beta}{P} \sum_j V(\boldsymbol{r}_c + \sqrt{\beta}\boldsymbol{q}_j) \right]. \tag{F.6}$$

When differentiating this expression, hold $\boldsymbol{r}_c$ and $\boldsymbol{q}_j$ fixed. The derivative of one potential term is

$$\begin{aligned} \frac{\partial}{\partial \beta} \left[\beta V(\boldsymbol{r}_c + \sqrt{\beta}\boldsymbol{q}_j) \right] &= V(\boldsymbol{r}_j) + \beta \nabla V(\boldsymbol{r}_j) \cdot \frac{\boldsymbol{q}_j}{2\sqrt{\beta}} \\ &= V(\boldsymbol{r}_j) + \frac{1}{2}(\boldsymbol{r}_j - \boldsymbol{r}_c) \cdot \nabla V(\boldsymbol{r}_j). \end{aligned} \tag{F.7}$$

Combining this with the derivative of $\beta^{-D/2}$ gives

$$-\frac{\partial \ln Z_P}{\partial \beta} = \frac{D}{2\beta} + \left\langle \frac{1}{P} \sum_j \left[V(\boldsymbol{r}_j) + \frac{1}{2}(\boldsymbol{r}_j - \boldsymbol{r}_c) \cdot \nabla V(\boldsymbol{r}_j) \right] \right\rangle_P, \tag{F.8}$$

which is Eq. 5.25. This is an identity for the discretized partition function under the stated assumptions. It does not remove the finite-P approximation to the quantum system; it changes the statistical estimator of the same E_P.

For a one-dimensional harmonic oscillator, $V(x) = m\omega^2 x^2/2$ and $V'(x) = m\omega^2 x$. The exact quantum energy

$$E = \frac{\omega}{2} \coth\left(\frac{\beta\omega}{2} \right) \tag{F.9}$$

provides a simple convergence test for both estimators. At the same finite P their means should agree within correlated statistical uncertainty, even though their variances differ.

F.2 Two-State Relaxation with Dephasing

This example specifies the assumptions behind the $\Delta^2 T_2$ relaxation scale. It is a resonant two-state model with pure dephasing in the localized basis, not a general thermal-bath formula. Use

$$\hat{H} = -\frac{\Delta}{2}\sigma^x, \qquad \frac{d\rho}{dt} = -i[\hat{H}, \rho] + \frac{1}{2T_2}(\sigma^z \rho \sigma^z - \rho). \tag{F.10}$$

In the absence of tunneling, the off-diagonal density-matrix elements then decay as e^{-t/T_2}. Write

$$\rho = \frac{1}{2}(\mathbb{I} + x\sigma^x + y\sigma^y + z\sigma^z), \qquad z = p_L - p_R. \tag{F.11}$$

Using the Pauli commutators gives

$$\dot{x} = -\frac{x}{T_2}, \qquad \dot{y} = \Delta z - \frac{y}{T_2}, \qquad \dot{z} = -\Delta y. \tag{F.12}$$

Eliminating y yields

$$\ddot{z} + \frac{1}{T_2}\dot{z} + \Delta^2 z = 0. \tag{F.13}$$

Without dephasing, and with $z(0) = 1$ and $y(0) = 0$, this gives $z(t) = \cos(\Delta t)$, so

$$p_R(t) = \frac{1 - z(t)}{2} = \sin^2(\Delta t/2). \tag{F.14}$$

With dephasing, the two characteristic exponents are

$$r_\pm = -\frac{1}{2T_2} \pm \sqrt{\frac{1}{4T_2^2} - \Delta^2}. \tag{F.15}$$

In the strongly dephased regime $\Delta T_2 \ll 1$, expansion of the slow exponent gives

$$r_+ = -\Delta^2 T_2 + O(\Delta^4 T_2^3), \qquad r_- = -T_2^{-1} + O(\Delta^2 T_2). \tag{F.16}$$

After the fast coherence transient, the imbalance obeys

$$\dot{z} \simeq -\Delta^2 T_2 z. \tag{F.17}$$

For a symmetric classical rate equation

$$\dot{p}_L = -kp_L + kp_R, \qquad \dot{p}_R = kp_L - kp_R, \tag{F.18}$$

one instead writes $\dot{z} = -2kz$. Hence the population-relaxation rate is $\Gamma_{\rm rel} = 2k \simeq \Delta^2 T_2$, and the directional hopping rate is $k \simeq \Delta^2 T_2/2$. This distinguishes the factor-of-two conventions while retaining the same gap scaling. Restoring $\hbar$ replaces Δ by $\Delta/\hbar$ in the dynamical equations.

F.3 An Explicit Quartic Instanton

Consider the dimensionless-coordinate potential used in Exercise 5.6.2,

$$V(x) = \lambda x^4 - x^2, \qquad \lambda > 0. \tag{F.19}$$

Its two minima lie at $x = \pm a$, with

$$a = \frac{1}{\sqrt{2\lambda}}, \qquad V_{\min} = -\frac{1}{4\lambda}, \qquad U(x) \equiv V(x) - V_{\min} = \lambda(x^2 - a^2)^2. \tag{F.20}$$

Subtracting $V_{\min}$ makes the action of a path sitting in either minimum vanish and gives a finite action to a connecting path on the infinite imaginary-time interval.

The Euclidean equation and its first integral are

$$m\ddot{x} = 4\lambda x(x^2 - a^2), \qquad \frac{m}{2}\dot{x}^2 = U(x). \tag{F.21}$$

For the path increasing from $-a$ to a,

$$\dot{x} = \sqrt{\frac{2\lambda}{m}}(a^2 - x^2). \tag{F.22}$$

Separating variables and integrating gives

$$\int \frac{dx}{a^2 - x^2} = \sqrt{\frac{2\lambda}{m}} \int d\tau, \tag{F.23}$$

$$\frac{1}{2a} \ln \frac{a + x}{a - x} = \sqrt{\frac{2\lambda}{m}} (\tau - \tau_0). \tag{F.24}$$

Solving for the path,

$$x_I(\tau) = a \tanh \left[a \sqrt{\frac{2\lambda}{m}} (\tau - \tau_0) \right] = a \tanh \left[\frac{\omega_0}{2} (\tau - \tau_0) \right], \tag{F.25}$$

where $\omega_0^2 = V''(a)/m = 8\lambda a^2/m = 4/m$ is the small-oscillation frequency squared in either well. The arbitrary center τ_0 reflects imaginary-time translation invariance. The anti-instanton follows by reversing the direction.

Using $m\dot{x}^2/2 = U(x)$, its action is

$$S_I = \int_{-\infty}^{\infty} d\tau \left[\frac{m}{2} \dot{x}_I^2 + U(x_I) \right] = \int_{-a}^{a} dx \sqrt{2mU(x)} \tag{F.26}$$

$$= \sqrt{2m\lambda} \int_{-a}^{a} (a^2 - x^2)\, dx = \frac{4}{3} \sqrt{2m\lambda}\, a^3 = \frac{2\sqrt{m}}{3\lambda}. \tag{F.27}$$

For a large barrier in the semiclassical sense, $S_I \gg 1$ in the units used here,

$$\Delta \simeq A(\lambda, m) e^{-2\sqrt{m}/(3\lambda)}. \tag{F.28}$$

The prefactor and corrections are not determined by this saddle calculation. Exact diagonalization is therefore the appropriate quantitative reference for the exercise. For a well-separated instanton–anti-instanton pair, the action approaches $2S_I$; finite separation and finite path length modify this approximation. An open path can contain one instanton, provided the boundary weights support both endpoints without adding a competing exponential penalty.